Master's Thesis

Estimating the Impact of Marine Biofouling on Ship Energy Efficiency Using Machine Learning

A Case Study of a Bulk Carrier

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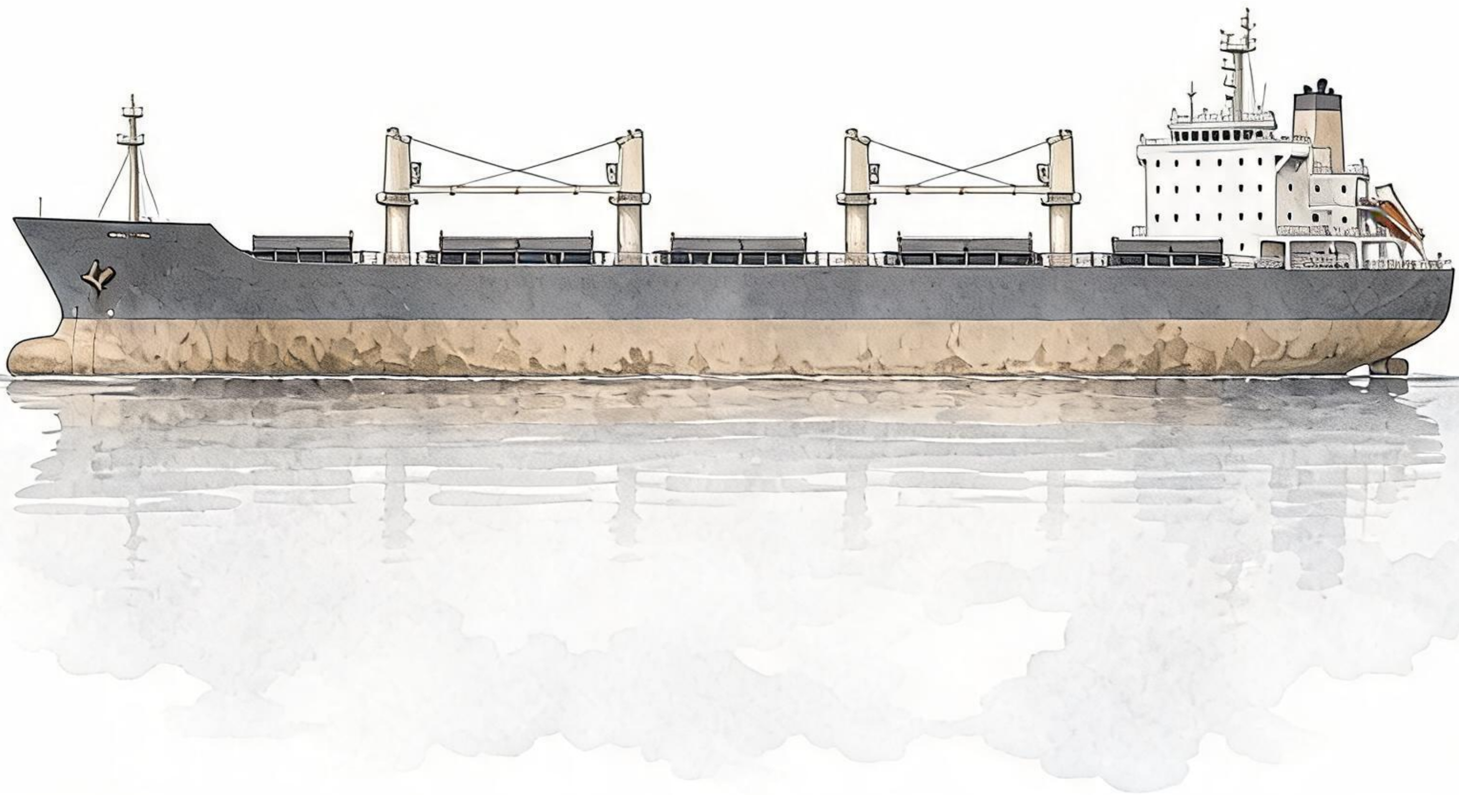

Abstract

The accumulation of marine organisms on submerged surfaces, also known as Marine biofouling, significantly reduces vessel energy efficiency. Yet its effect is difficult to quantify based on operational ship data because energy need is also determined by speed, cargo weight, and environmental conditions. This thesis investigates whether machine learning can provide a viable method for quantifying energy inefficiency caused by marine biofouling on a Panamax bulk carrier. Using one year of onboard sensor data, metocean forecasts, and noon reports, this thesis constructs a physics-informed preprocessing and modeling pipeline and uses it to compare the applicability of a simple Generalized Additive Model and a Multilayer Perceptron for this task. The results show that significant fouling accumulates on the vessel throughout 2024, increasing its energy need by about 31.6–36.3% six months after hull cleaning. The findings provide tentative evidence that physics-informed machine learning can serve as a practical vessel-specific alternative for quantifying energy inefficiency caused by marine biofouling when traditional naval engineering methods are difficult to apply. Furthermore, the results suggest that the performance gap between the simple Generalized Additive Model and the Multilayer Perceptron is too small to justify the added complexity and lack of interpretability of the latter.

Keywords: Marine Biofouling; Ship Performance; Machine Learning; Energy Efficiency; Explainable AI

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Contents

List of Abbreviations

Abbreviation	Meaning
ALE	Accumulated Local Effects
CFD	Computational Fluid Dynamics
CSC	Clean Shipping Coalition
DNV	Det Norske Veritas
DSC	Days Since Cleaning
DWT	Deadweight Tonnage
GAM	Generalized Additive Model
GCV	Generalized Cross-Validation
GenAI	Generative Artificial Intelligence
GPS	Global Positioning System
ICE	Individual Conditional Expectation
ICCT	International Council on Clean Transportation
IMO	International Maritime Organization
ISO	International Organization for Standardization
ITTC	International Towing Tank Conference
LIME	Local Interpretable Model-agnostic Explanations
MAD	Median Absolute Deviation
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MEPC	Marine Environment Protection Committee
ML	Machine Learning
MLP	Multilayer Perceptron
MMMCZCS	Mærsk-McKinney Møller Center for Zero Carbon Shipping
OLS	Ordinary Least Squares
PD	Partial Dependence
PE	Partial Effect
ReLU	Rectified Linear Unit
REML	Restricted Maximum Likelihood
RMSE	Root Mean Squared Error
RPM	Revolutions Per Minute
SFOC	Specific Fuel Oil Consumption
SGD	Stochastic Gradient Descent
SHAP	SHapley Additive exPlanations
SOG	Speed Over Ground
STW	Speed Through Water
UN	United Nations
UNCTAD	United Nations Conference on Trade and Development
UTC	Coordinated Universal Time
USD	United States dollar
VIF	Variance Inflation Factor
VLSFO	Very Low Sulfur Fuel Oil
WPR	World Population Review
XAI	Explainable Artificial Intelligence

1 Background

The shipping industry has an immense impact on the environment and economy, being responsible for 3% of anthropogenic carbon emissions and transporting 80% of goods worldwide (IMO, 2021; UNCTAD, 2021). Despite being a major contributor to both costs and pollution, vessel fuel consumption is often highly inefficient (IMO, 2021). A silent but significant source of inefficiency is marine organisms growing on the outside of the vessel, also known as marine biofouling (henceforth "fouling"). Fouling alone is estimated to cause USD 30bn additional yearly operating costs and 10% of greenhouse gas emissions in the world fleet, making its impact analogous in size to the entire government budget of Iceland or the greenhouse gas emissions of the Netherlands (CSC, 2011; Ship & Bunker, 2020; Worldometer, 2026; WPR, 2026).

Fouling develops gradually as organisms attach and grow on the hull and propeller, making them rougher and less hydrodynamic. The resulting "fouling effect" is a quantification of the inefficiency this creates, typically expressed as either the added propeller power required to maintain a given speed or the reduced speed at a given propeller power. The only way to avoid this is with hull coatings or by cleaning the hull with divers or drones at port stays (Ship & Bunker, 2020). Both options are expensive and time-consuming, and reliable measurement of fouling effects is therefore a key element of quantifying ship performance and optimizing maintenance schedules.

Yet quantifying fouling effects in practice is notoriously difficult because the fouling state cannot be measured directly, and a vessel's required propeller power is shaped simultaneously by a multitude of factors, such as speed, cargo weight, and environmental conditions. The most common ways to quantify fouling effects are with Computational Fluid Dynamics (CFD) models or by comparing measured power to a theoretical reference model based on standardized naval engineering frameworks, such as ISO 19030 (Cho et al., 2021; ISO, 2016; Song et al., 2019). However, recent advances in data availability and quality have made Machine Learning (ML)- based methodologies more relevant as they have become increasingly feasible. Therefore, this thesis aims to apply ML to operational data from a real vessel to estimate fouling effects and explore the potential of ML as an alternative to established methodologies for fouling effect estimation.

2 Research Approach

The following sections provide an overview of the thesis's objective and the approach used to achieve it. Section 2.1 states and motivates the research questions; Section 2.2 describes the applied philosophy of science; and Section 2.3 outlines the research design and scope. Lastly, Section 2.4 declares the extent to which generative AI has been used throughout the thesis.

2.1 Research Objective

Any methodology applied to estimate fouling effects in practice must be robust to the inherent noise and complexity that characterize operational ship data (Gupta et al., 2022; ISO, 2016; Y. Kim et al., 2025). The goal of this thesis is to assess whether ML models fit this description, and to what extent they can serve as an attractive alternative to traditional methods from the field of naval engineering, which often depend on strict filtering, correction, or reference-condition requirements (DNV, 2023; ISO, 2016). This assessment will be made by applying ML to estimate a vessel's fouling condition from its operational data and evaluating the relationships learned by the models against domain knowledge.

While ML models can offer strong predictive performance, their complexity and, at times, limited interpretability are often cited as making them harder to trust and adopt in practice (Murdoch et al., 2019; Rudin, 2019). This lack of interpretability is a challenge in this application, since predictive performance alone is insufficient to provide a useful estimate of fouling effects. A secondary contribution of this thesis is thus to compare the ability of black- and white-box ML models to estimate fouling effects, and assess which one strikes the best tradeoff between complexity and interpretability. Concretely, the thesis is guided by the following research question and sub-questions:

Can machine learning serve as a viable methodology for estimating the effect of marine biofouling on ship energy efficiency from operational data?

Sub-questions:

RQ1: *How does the energy inefficiency attributable to marine biofouling develop in the first 6 months after a cleaning event?*

RQ2: *Can machine learning be applied to reliably quantify the impact of marine biofouling on the energy efficiency of a vessel?*

RQ3: *Do black-box models outperform white-box models to an extent that justifies their complexity and relative lack of interpretability?*

2.2 Philosophy of Science

This thesis adopts a post-positivist perspective and assumes that an external reality exists independently of the researcher, even though that reality cannot be apprehended with complete certainty (Bell et al., 2022; Clark, 1998). This perspective implies that fouling is understood as a real physical phenomenon that produces real changes in ship performance. However, those effects can only be accessed indirectly through sensor measurements, modeling assumptions, and analytical choices. Consequently, the study seeks to approximate fouling-related performance effects as objectively as possible through transparent data processing, physics-informed modeling, and explicit evaluation of model behavior, without claiming that such estimates provide a perfectly certain or complete representation of reality (Bell et al., 2022; Clark, 1998). Representing this reality is particularly difficult in this domain because of statistical and modeling uncertainties, including idiosyncratic noise, generalization error, and structural model limitations.

This thesis follows a deductive approach to theory development. It begins from existing knowledge of ship propulsion physics and the known ways in which fouling accumulates, and these form the basis for the problem framing, variable selection, and modeling (Bell et al., 2022). Within this predefined framework, the study examines whether ML models can serve as a viable alternative for estimating fouling-related performance changes from operational data.

2.3 Research Design & Scope

This thesis adopts a bounded quantitative single-case case-study research design. A case-study strategy is deemed appropriate because ship data remains characterized by heterogeneous sources, formats, and sampling frequencies, complicating multi-vessel studies (Y. Kim et al., 2025; Tsarsitalidis et al., 2026). The study is grounded in operational data from a real vessel spanning one year, acquired through collaboration with the Mærsk-McKinney Møller Center for Zero Carbon Shipping (MMMCZCS). Accordingly, the purpose of the thesis is not to formulate a universal model for all vessels, but to assess the usefulness of ML-based methods for estimating fouling effects within the context of a real vessel.

The research design is also physics-informed, in that domain knowledge guides model development rather than treating the problem as a purely data-driven prediction task. In the literature, the field of ML has shown that such theory-informed modeling is often helpful when the modeling task involves a complex physical system, which is unarguably the case for ship propulsion (Murdoch et al., 2019; Rudin, 2019). Therefore, this thesis incorporates maritime knowledge, particularly in feature selection, model formulation, and evaluation. Models that, by design or through XAI techniques, allow for fouling effects to be assessed in isolation and related to physically meaningful behavior are therefore favored.

The intended form of generalization is therefore analytical rather than statistical. This thesis does not aim to establish how fouling affects the energy efficiency of vessels in general, as this is not feasible given the complexity of the task and availability and quality of data (Y. Kim et al., 2025). Instead, the study investigates whether ML can provide a practical alternative for estimating the accumulation and effect of fouling when traditional models are not applicable. The modeling is further restricted to open-water operating conditions at a relatively steady speed, where the propulsion state is more consistent and less affected by maneuvering and port stays(ISO, 2016). The analysis is limited to fouling as an isolated source of performance change, meaning that other influences, such as wind, waves, currents, and engine deterioration, are not examined in depth. In extension, the goal of the thesis is only to estimate the effect of fouling, and not to recommend an optimal maintenance schedule. Lastly, distinguishing between fouling of the hull and propeller is also outside the scope of this thesis.

Some domain knowledge and information about the vessel in question were acquired through personal communication with Naval Engineer Matthew Patey and Analytics Product Owner Frederik Lehn from MMMCZCS. This communication took place in person or via online meetings from November 1, 2025, to May 15th, 2026. Claims made based on these conversations are referenced explicitly as: (M. Patey / F. Lehn, Personal Communication, 2026).

2.4 Declaration of Generative AI Use

The author of this thesis has made use of GenAI in accordance with the guidelines of Copenhagen Business School (CBS, 2023). Various forms of autocorrect[1] were used for writing and programming. In addition, ChatGPT 5.4 and Claude Sonnet 4.6, accessed through Perplexity AI, were used to generate ideas, support conceptualization, and design the front page (see Appendix A for prompts). All AI-assisted outputs were critically evaluated and cross-referenced with primary sources, and GenAI was not used to generate any of the text in this thesis. The author takes full ownership and responsibility for the following content and considers it a genuine representation of their work, skills, and knowledge.

[1] Including Grammarly for checking grammar and spelling mistakes, Overleaf AI assist for formatting tables and figures in LaTex, and Github Copilot (integrated in VSCode) for correcting mistakes in Python code

3 Theoretical Framework

The following four sections present the theoretical foundation for this thesis. Firstly, Section 3.1 conceptualizes the quantification of ship energy efficiency. Secondly, Section 3.2 presents the known drivers of energy demand based on ship propulsion theory. Thirdly, Section 3.3 describes the known dynamics of the accumulation of marine organisms on submerged parts of the vessel. Lastly, Section 3.4 presents relevant concepts from the domain of Explainable Artificial Intelligence (XAI).

3.1 Ship Energy Efficiency

Ship performance is often quantified as energy efficiency, although it can encompass broader operational aspects (Gupta et al., 2022; ISO, 2016; ITTC, 2017). Since this thesis is centered on energy efficiency, the term "performance" will henceforth refer to the energy efficiency of the vessel. At its core, the energy efficiency of a vessel is defined by how much power is required to achieve a given speed, or conversely, how fast the vessel can travel given a fixed amount of power. This relationship is most commonly illustrated using a speed-power curve (see example in Figure 3.1).

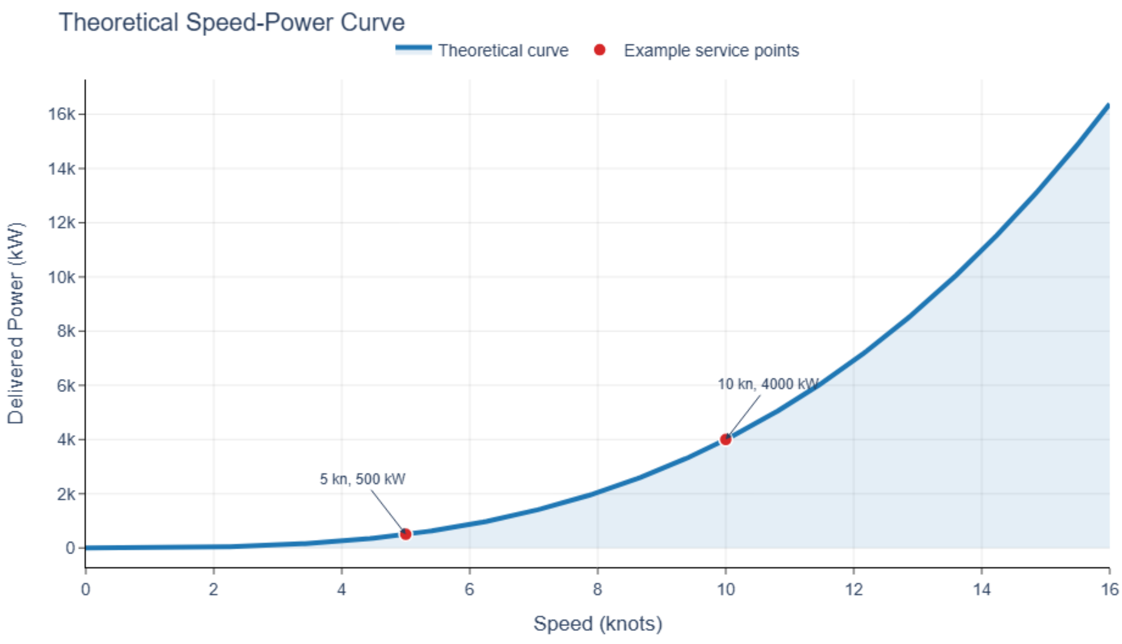


Figure 3.1: *Theoretical example of a speed-power curve. Note that doubling the speed increases the required power eightfold.*

When defining the speed-power curve, "speed" refers specifically to Speed Through Water (STW), which measures the movement of the vessel relative to the water mass itself, rather than Speed Over Ground (SOG), which is based on GPS measurements (Lewis, 1988). Similarly, when defining "power", the ideal metric is the power delivered directly on the propeller (ISO, 2015). However, because measuring the power delivered to the propeller is highly impractical, the propeller *shaft* power, which represents the power transmitted through the shaft just before reaching the propeller, is the standard proxy used in data analysis (ISO, 2016; Y. Kim et al., 2025). Henceforth, the word "power" thus refers to the power in the propeller shaft. While there is a slight mechanical transmission loss between the two, this discrepancy is minor and beyond the scope of this thesis.

The speed-power relationship has its roots in the work of English engineer William Froude, whose early studies laid the foundation for modern resistance and power analysis of ships (Froude, 1872, 1874). In its simplest form, the relationship states that the power required to propel a vessel increases approximately with the cube of its speed, so that doubling the speed from v to $2v$ requires about $2^3 = 8$ times as much power. This relationship follows from the common approximation that hydrodynamic resistance, R, is roughly proportional to the square of speed v^2. Because the power is given by $Power = Resistance \cdot v$, it follows that $Power \propto v^3$ (Lewis, 1988).

Under the common assumption that fouling adds an incremental resistance ΔR that varies less strongly with speed than the clean-hull resistance, the associated added power is $\Delta P = \Delta R\, v$, so the absolute added power in kW increases with speed. Because the baseline power requirement scales approximately as $P_{\text{clean}} \propto v^3$, the relative added power $\Delta P/P_{\text{clean}}$ decreases with speed, meaning that fouling can require more additional power in absolute terms while accounting for a smaller percentage increase. However, these relationships should be understood as theoretical rules of thumb rather than exact laws, and some deviation is entirely normal in practice due to differences in hull form, loading condition, sea state, and weather (DNV, 2023; ISO, 2016). In principle, this added power can come from a wide range of sources. Still, in this thesis, the term "added power" will refer specifically to the added power attributable to fouling.

A decrease in ship performance, such as that caused by fouling, can be understood as a change in the speed-power relationship, whereby more power is required to maintain the same speed, or equivalently, less speed is achieved at a given power (see Figure 3.2). Here, it becomes clear that speed loss and added power are two sides of the same coin, with the former referring to the horizontal difference between the two curves and the latter referring to the vertical difference. In practical applications, this relationship is typically represented by a more comprehensive model in which speed or power is expressed as a function of the other variable, along with operational and environmental factors (ISO, 2016; Laurie et al., 2021). When the purpose is to assess energy efficiency, the use of power, or the closest available proxy, as the target is often preferred to speed because it is more directly related to fuel consumption and is commonly used in both industry and academic studies (ISO, 2016; Lang et al., 2022; Valchev et al., 2022).

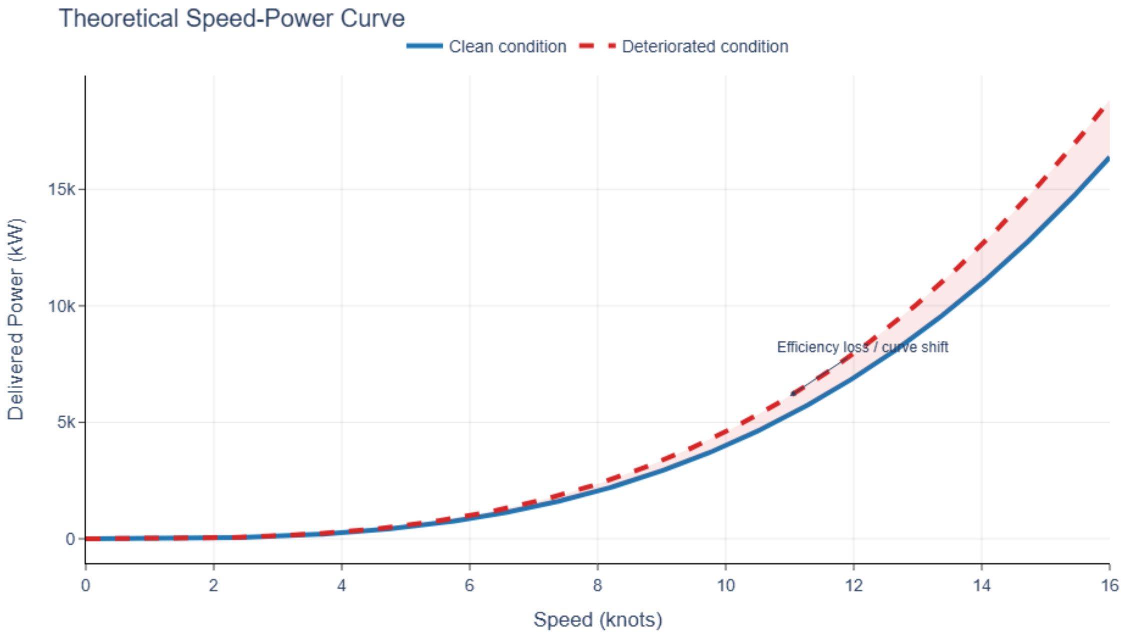


Figure 3.2: *Theoretical example of a change in the speed-power curve caused by marine biofouling. Source: Own creation.*

3.2 Determinants of Shaft Power

Although this thesis does not seek to develop a generalizable physical model of ship propulsion, a basic understanding of the underlying physics is still useful when specifying ML models for complex physical systems (Murdoch et al., 2019). In naval architecture, the power required to propel a vessel is commonly decomposed into added frictional, wave-making, and environmental resistance, which together capture the main ways in which operating conditions and the surrounding environment determine shaft power demand (Froude, 1872; ISO, 2015; Lewis, 1988). The propulsion system ultimately supplies this required power, so signals describing the operating state of the engine and propeller, such as fuel flow, propeller Revolutions Per Minute (RPM), and scavenger air pressure, are treated here not as primary drivers of power demand, but as mediating variables that reflect how the machinery responds to external resistance. In the available dataset, the variables measured by the torquemeter, transducer, control alarm monitoring system, and the RPM indicator should therefore be considered mediators rather than sources of power demand (see Appendix B.1).

Frictional resistance is the drag created by water rubbing along the propeller and submerged parts of the hull as the vessel moves forward, and is mainly driven by speed, water temperature, and fouling (Froude, 1872; ISO, 2015; Lewis, 1988). Among these, speed is the main driver. In contrast, the effects of fouling and water temperature are relatively small, which can, in some cases, complicate the task of distinguishing fouling from speed effects (ISO, 2015, 2016). Among the two speed measures, STW is preferred because friction depends on the motion of the vessel relative to the surrounding water rather than relative to the earth (ISO, 2015; ITTC, 2017). Water temperature is relevant because warm water is more viscous, which decreases friction along the hull and propeller (ISO, 2015; Lewis, 1988). Fouling increases frictional resistance by making the hull and propeller surfaces less hydrodynamic, so more power is required to maintain the same speed (ITTC, 2017; Lewis, 1988).

Wave-making resistance is the part of the vessel's drag that arises because energy is required to push water aside and generate waves around the hull as the vessel moves forward (Froude, 1874; Lewis, 1988). Wave-making resistance is mainly driven by speed, draft, trim, and water depth (ISO, 2015; Lewis, 1988). Draft refers to the depth of the vessel in the water. Thus, it is closely related to the amount of cargo the vessel is carrying and the amount of water it displaces as it moves, meaning that a higher draft increases the required power (Lewis, 1988). Because the draft is measured at multiple points along the hull, it is also useful to consider the "trim", defined as the difference between the draft at the back (aft) and front (fore) of the vessel (Lewis, 1988). Henceforth, the word "draft" will refer to the average of fore and aft draft unless otherwise specified. The specific combination of draft and trim at which the vessel is operating is also referred to as the loading condition. The depth of the water is relevant because shallow-water conditions alter the wave pattern around the vessel, increasing the power required to maintain the same speed. However, these effects are negligible in open sea (ISO, 2015; ITTC, 2017).

The added environmental resistance is the extra drag caused by external weather conditions, meaning the vessel needs more (less) power because the wind and waves actively work against (with) its motion (ISO, 2015; Lewis, 1988). Since the effect of both depends on the direction from which they are coming, it is useful to resolve them into *longitudinal* and *transverse* components, i.e., the parts acting along and across the vessel's axis. If x denotes the magnitude of the environmental quantity and θ_{rel} its relative angle to the vessel, these are calculated as

$$x_{\text{long}} = x\cos(\theta_{\text{rel}}),$$
$$x_{\text{trans}} = x\sin(\theta_{\text{rel}}).$$

Positive longitudinal values correspond to head conditions that oppose forward motion, while negative values correspond to following conditions that assist it. The impact of transversal values is considered more complex and less significant, but they are often included in ship performance models to have a complete representation of the environment (Fujiwara et al., 2005; Lewis, 1988). For waves, the relevant quantity is usually based on the *significant* wave height, which is a standard measure defined as the average of the highest one-third of waves (ISO, 2016). It is also common to distinguish between locally generated wind waves and "swell", where nearby wind conditions shape the former and the latter are longer, more regular waves generated by distant weather events (ISO, 2015; Lewis, 1988).

The sea current also affects the power need but plays a slightly different role compared to the other environmental variables. In theory, it is not relevant once STW is included, since propulsion power depends on the vessel's motion relative to the surrounding water, thereby implicitly accounting for the current. However, including a measure of current can still be useful in practice because STW measurements are based on speedlog sensors, which can be noisy, poorly calibrated, or covered by marine organisms (ISO, 2016). A local estimate of the current can be obtained from the difference between SOG and STW, which is more specific to the vessel's actual conditions but inherits any errors in the speed log. Alternatively, weather-provider current data can offer an estimate that is less locally precise but not affected by sensor inaccuracy (ISO, 2016). Therefore, current can be included in models not because current directly drives power after including STW, but because it may capture residual information about current and speedlog bias that STW alone does not represent.

It should be noted that the effect of each of these drivers tends to depend on the others. For example, fouling increases frictional resistance when submerged, so a higher draft leads to a greater adverse effect of fouling. The physics of these interactions is not fully understood, but treating each effect in complete isolation is undoubtedly a simplification (Lewis, 1988).

3.3 Marine Biofouling

Fouling is the gradual accumulation of organisms on submerged surfaces, such as the hull and propeller, typically starting with a thin microbial slime (biofilm) and, if left unchecked, developing into colonies of algae or larger marine organisms (Schultz, 2007; Townsin, 2003) (see examples in Figure 3.3). This process is cumulative, because the early slime layer makes it easier for later organisms to settle, so fouling tends to build over time rather than appear all at once (Townsin, 2003). It usually develops fastest when a vessel is idle or operating slowly, especially in warm, tropical, saline waters. Although the first thin layer biofilm can begin to form within hours to days, noticeable slime fouling is often not seen until after days to a couple of weeks under favorable conditions (Davidson et al., 2020; IMO, 2023). Calcareous fouling, i.e., organisms with hard shells such as barnacles, mussels, and tubeworms, tends to take multiple weeks to accumulate (Davidson et al., 2020; Schultz et al., 2011; Van Mooy et al., 2014).

Fouling can be mitigated mainly through antifouling or fouling-release coatings and periodic cleaning. Coatings do not completely stop fouling, but they can slow its accumulation and influence which organisms can attach to the surface (IMO, 2023; Townsin, 2003). When fouling builds up, it can be removed by cleaning the hull or propeller during port stops, which restores a smoother surface between maintenance intervals, or during dry-docking, where the vessel is taken out of the water for more extensive inspection, cleaning, and re-coating (IMO, 2023). Such events are referred to under the umbrella term "cleaning events"[2].

[2] In the dataset used in this thesis, only what type of cleaning event, which includes a polishing of the hull and propeller, is observed.

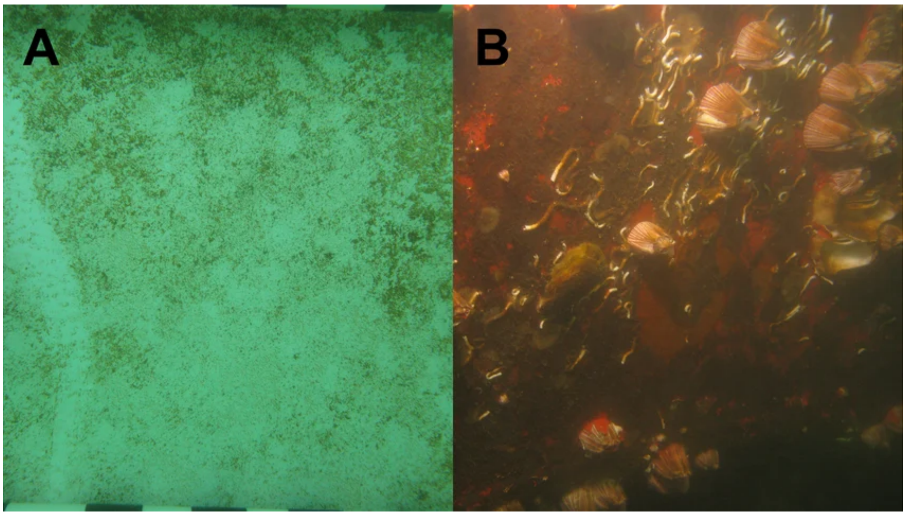


Figure 3.3: *Examples of biofilm (A) and calcareous fouling (B). Source: Georgiades et al. (2023)*

The relevant modeling problem is thus not simply to predict total power but to estimate how much of that power demand is attributable to fouling. Conceptually, this can be represented as $P_t = f(z_t, x_t) + \varepsilon_t$, where P_t is the power at time t, z_t is a time-dependent representation of the fouling state, x_t denotes the vector of observed operational and environmental variables that affect power demand, and ε_t captures the unmodeled variation and measurement noise. Since fouling tends to accumulate over time, the practical goal is to infer the growing added power attributable to fouling and, with the other variables in x_t fixed, to trace a fouling-effect curve over time for a given operating condition (Gupta et al., 2022; ISO, 2016; Laurie et al., 2021).

Modeling the effect of fouling is difficult because the power demand of a vessel is shaped by a large and interacting system of factors from which the fouling signal must be separated (ISO, 2016; Schultz, 2007). In practice, some of these inputs, especially the weather and sea-state variables, are only available after averaging, interpolation, or extrapolation over a certain spatial and temporal resolution, which introduces additional uncertainty before the model is even fitted (DNV, 2023; ISO, 2016). In addition, real operational data are noisy and subject to measurement error, outliers, and distribution changes over time, making it harder to identify a stable and interpretable fouling effect (Dalheim & Steen, 2020).

The fouling state of the hull can be treated as a latent variable, i.e., one that is not measured directly but inferred from the data (Murphy, 2012). In an ML context, this means that the task is primarily one of inference rather than prediction: the aim is not solely to predict power demand, but to infer an unobserved fouling state from noisy observations (Murphy, 2012; Shmueli, 2010). In practice, the best that can be done is to represent fouling using imperfect but measurable proxies, such as the time since the last cleaning event, while recognizing that these variables capture only part of the underlying fouling process (Murphy, 2012). This also creates an identification problem, because an observed increase in power cannot necessarily be attributed to fouling alone when several other factors may produce similar changes in the data (Allman et al., 2009; Lewbel, 2019).

3.4 Explainable Artificial Intelligence

Recent improvements in AI performance have come with an increase in model complexity, which has made many modern models highly effective, but also harder for humans to understand (LeCun et al., 2015; Linardatos et al., 2021). This trait is problematic when the goal extends beyond accurate prediction to encompass additional objectives, such as fairness, safety, or alignment with domain knowledge (Arrieta et al., 2020). This problem has laid the foundation for the field of XAI, within which such complex and opaque models are commonly referred to as "black-box" models (Arrieta et al., 2020). In contrast, "white-box" models can be defined as models that allow human inspection and interpretation by design (Arrieta et al., 2020). In this context, the "interpretability" of a model can thus be defined as its *"ability to explain or to present in understandable terms to a human"* (Doshi-Velez & Kim, 2017, p. 2).

Interpretability is needed when there is an *"incompleteness in formalizing the problem"* (Doshi-Velez & Kim, 2017, p. 3), which means that the objective used to train a model, such as minimizing the prediction error, does not fully capture all the objectives of the practitioner. For example, one may want a model to respect physical constraints, such as whether the learned relationships of a model are physically plausible, but an error-minimizing model can still fit spurious relationships that violate such expectations if doing so improves predictive accuracy (Doshi-Velez & Kim, 2017). While parts of this mismatch can sometimes be addressed by refining the model specification or its objective function, not all relevant criteria can be fully specified mathematically (Doshi-Velez & Kim, 2017). In such cases, interpretability helps humans assess whether the model satisfies important criteria outside the formal training objective (Doshi-Velez & Kim, 2017).

Black-box models are often preferred because they can deliver very high predictive accuracy, especially in high-dimensional tasks such as image recognition and natural language processing, where deep learning has substantially advanced the state of the art (LeCun et al., 2015; Linardatos et al., 2021). This advantage comes from their ability to learn complex nonlinear representations, allowing them to capture patterns that simpler linear or rule-based models cannot represent (LeCun et al., 2015; Linardatos et al., 2021). A second argument is flexibility: such models can often perform well with little explicit domain knowledge, even though incorporating domain knowledge will often still improve performance (Murdoch et al., 2019).

This interpretability problem can be addressed in two main ways: either by using a white-box model or by applying XAI techniques to make a black-box model more interpretable (Lipton, 2018). The latter is most commonly done post-hoc, i.e., after model fitting. These post-hoc methods are commonly divided into model-specific approaches, which rely on the internal structure of a particular model or model class, and model-agnostic approaches, which can be applied more broadly across different model types. Among the most common model-agnostic techniques are feature-attribution methods such as Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP), as well as effect-visualization tools such as Partial Dependence (PD), Individual Conditional Expectation (ICE), and Accumulated Local Effects (ALE), which help identify how individual variables influence model predictions (Apley & Zhu, 2020; Goldstein et al., 2015; Lundberg & Lee, 2017; Ribeiro et al., 2016).

The performance advantage for black-box models appears less pronounced for structured tabular problems with meaningful, human-understandable features, where interpretable models may perform similarly to black-box alternatives (Rudin, 2019). Fouling estimation arguably belongs to this category, since it is based on structured data where values correspond to real physical quantities with known relationships. For that reason, it is reasonable to ask whether a white-box model should be preferred or whether a black-box model combined with post-hoc XAI techniques offers a better compromise between predictive performance and interpretability.

4 Literature Review

The goal of the following literature review is to examine how previous scholars have solved the problem of estimating added power from fouling. Section 4.1 reviews the challenges of ship performance monitoring in practice. Section 4.2 then reviews the main approaches used to estimate fouling effects, with a distinction between traditional methods and ML-based methods. Finally, Section 4.3 considers XAI in adjacent domains, where similar problems arise when trying to model complex physical systems from noisy observational data.

4.1 Ship Performance Monitoring in Practice

In practice, ship performance must be assessed from noisy real-world data, which means that monitoring frameworks inevitably rely on simplifying assumptions and pragmatic pre-processing choices shaped by data availability and quality (Dalheim & Steen, 2020; Gkerekos et al., 2019). The available information is also typically heterogeneous, as it is assembled from multiple subsystems on board that are not always directly compatible in format, frequency, or reliability, and is often supplemented by other sources such as manually recorded noon reports and meteorological and oceanographic (Metocean) data (Dalheim & Steen, 2020; Gupta et al., 2023; Y. Kim et al., 2025). Although the literature does not offer a single agreed-upon preprocessing standard, there is now a substantial body of work on synchronization, filtering, validation, and aggregation of such mixed data sources, which makes it possible to construct a preprocessing pipeline for practical applications (Y. Kim et al., 2025).

As alluded to in Section 3.1, the theoretically ideal representation of a vessel's performance at a given moment in time would be a complete speed-power curve given the operating conditions. If such a curve were observable instantaneously, it could be compared with historical measurements or sea-trial data to derive the added power at a given speed (ISO, 2015). However, in practice, this is not feasible because a complete speed-power curve cannot be established at a single instant, as doing so would require the vessel to be observed at multiple speeds under otherwise unchanged conditions (DeKeyser et al., 2022; Gupta et al., 2021; ISO, 2015). As a result, nearly all practical and academic work is based on simplifications, typically by estimating how much power the vessel should theoretically require under an assumed optimal or reference condition and comparing that benchmark with the actual power observed in operation with implicit (Coraddu et al., 2019; Gupta et al., 2022) or explicit (DNV, 2023; ISO, 2016) corrections for operating conditions (Alexiou et al., 2022; Gupta et al., 2023). The formula used to estimate this theoretical power consumption is called a reference model.

In the literature, reference models are established in one of the following two ways. Firstly, the reference may be defined from physical knowledge of ship propulsion, combined with correction procedures that account for environmental and operational conditions. This approach is the foundation of standardized naval engineering frameworks such as ISO 19030, where observations are interpreted relative to a theoretical, physically informed baseline (ISO, 2016). Secondly, the reference may be estimated empirically from operational data collected during a period in which the vessel is assumed to be close to its optimal condition, for example, shortly after hull cleaning (Coraddu et al., 2019; DeKeyser et al., 2022).

Comparability is a central challenge in empirical approaches because observed power values are only comparable to those of a reference model under sufficiently similar operating and environmental conditions (Dalheim & Steen, 2020; DNV, 2023; Laurie et al., 2021). Since the speed-power relationship is influenced by many factors beyond the effect of interest, comparisons often rely on coarse standardized corrections to align observations with the reference conditions (ISO, 2015, 2016). Even with such corrections, some observations may still lie too far from the reference state to be treated as comparable, for example, under very harsh weather or port maneuvering, and are therefore commonly excluded (DNV, 2023; ISO, 2016). For this reason, it is generally considered good practice to filter operational data to focus on relatively steady cruising conditions in open waters. This filtering decision can be seen as a bias-variance trade-off, since stricter filtering improves comparability but reduces the amount of data available for modeling.

Physics-informed reference models provide a structured baseline, but their validity depends on how well correction procedures account for differences between reference conditions and actual operation. In practice, corrections for draft, environmental conditions, and other disturbances are imperfect, so the reference may still differ from the true clean-hull performance of the vessel (DeKeyser et al., 2022; ISO, 2015, 2016). By contrast, empirically estimated baselines are practical and highly vessel-specific, as they are derived from the vessel's operational data. Their main limitation is that the data used to define the baseline rarely represent a perfectly optimal condition, which means that the reference itself is imperfect (Coraddu et al., 2019; DeKeyser et al., 2022).

4.2 Fouling Effect Estimation

The literature on fouling effect estimation can broadly be divided into traditional methods, based on either naval engineering or CFD, and more recent ML-based methods. In addition to modeling logic, they differ substantially in their data requirements, interpretability, and practical suitability for operational monitoring of fouling effects.

4.2.1 Traditional Methods

There is a substantial body of literature on the accumulation of marine organisms on ships and the subsequent added power from fouling. The understanding of the former has mainly been achieved through controlled experiments with submerged metal panels, in which the number and types of organisms growing on the hull are monitored over a couple of weeks or months. The latter has traditionally been studied theoretically with CFD and empirically using ISO 19030 and similar frameworks. Research shows a high degree of heterogeneity in added power estimates, since both accumulation speed and impact on energy efficiency depend on a multitude of factors, including environmental conditions, hull shape, and operating speed.

Experimental immersion studies on metal panels indicate that fouling can begin to develop within days to weeks, although the rate is strongly dependent on season and local environment (Townsin, 2003; Van Mooy et al., 2014). Van Mooy et al. (2014) showed that measurable microbial biofilm accumulated on submerged test surfaces after as little as 10 days in summer, while winter experiments extended to 54 days. Davidson et al. (2020) similarly found that the accumulation of fouling increased rapidly during stationary periods, with limited fouling only during the first week to 10 days, but much heavier fouling after longer lay-up durations. The later stages showed a sharp increase between 10 and 28 days and heavy cover, including calcareous organisms, by 45 to 60 days under favorable conditions. However, studies of operating vessels have found that fouling accumulates significantly more slowly as the vessel moves, making it harder for organisms to stick to the hull (Townsin, 2003).

Experimental literature conducted on vessels in operation shows that the added power is strongly dependent on the severity of the fouling. For a naval surface ship sailing at 15 knots, Schultz (2007) estimates an increase in required shaft power of about 11% for a deteriorated coating or light slime, 54% for medium calcareous fouling, and up to 86% for heavy calcareous fouling (Schultz, 2007). A follow-up study by Schultz et al. (2011) found similar effects with 9% added power from light slime fouling, 18-31% for small calcareous fouling, and 76% for heavy calcareous fouling.

Several studies have also estimated fouling effects using naval engineering formulas as reference models, often using standardized performance monitoring frameworks (ISO, 2015, 2016). For example, Cho et al. (2021) found that the combined hull-cleaning and damage-recovery effect corresponded to a 11.7% reduction in the required power relative to the degraded pre-docking condition over three years, using an analysis based on ISO 19030. Meanwhile, Torii and Kifune (2023) found that fouling increased shaft power by about 20% in a slightly fouled condition and 35% in a seriously fouled condition, with worst cases reaching roughly 50% added power over the course of a year (Cho et al., 2021; Torii & Kifune, 2023).

Despite their popularity in industry, studies relying on standardized naval engineering frameworks generally suffer from three main limitations. Firstly, they rely on comparisons to reference conditions that may not be fully representative of actual operation. Even after correction, differences in draft, speed range, and environmental conditions can make the controlled baseline only partially comparable to the observed data, which weakens the interpretation of the estimated performance loss (Cho et al., 2021; DNV, 2023). Secondly, the preprocessing requirements used to improve comparability are often very strict, which can lead to the exclusion of upwards of 80% of data points (Laurie et al., 2021). This exclusion of data means that these studies tend to struggle to balance comparability with data availability (Dalheim & Steen, 2020; ISO, 2016). Finally, the identification problem means that the added power estimates inevitably contain some degree of speculation, since other factors such as hull deterioration or voyage heterogeneity also influence the difference between baseline and observed power (Cho et al., 2021; ISO, 2016; Torii & Kifune, 2023).

Another increasingly popular way of estimating fouling effects is CFD, in which a computer simulation of the hull and propeller is used to estimate the resulting changes in resistance and required power with different marine organisms attached (Farkas et al., 2020; Song et al., 2019). CFD studies show substantial fouling penalties that align in magnitude with empirical studies: Farkas et al. (2020) found that biofilm could increase power by 1.4% to 36.3% at constant speed, depending on the severity of fouling. In contrast, Song et al. (2019) reported increases in effective power of 18% to 60% at 24 knots and 22% to 73% at 19 knots for various levels of barnacle fouling. However, CFD is used mainly as a scenario-based prediction tool rather than as a routine monitoring method for an individual vessel, since it is typically used to analyze specified conditions on a model or full scale and remains considerably more demanding than empirical methods for repeated operational assessment (M. Kim et al., 2017; Liu et al., 2021).

A key limitation of CFD in this context is that it is poorly suited for operational monitoring, because it is not made to infer an unobserved latent fouling state from real-time data (M. Kim et al., 2017). In addition, CFD requires detailed geometric and physical inputs and substantial computational power, making them costly and difficult to apply repeatedly (M. Kim et al., 2017; Liu et al., 2021). Therefore, CFD is better suited to make theoretical benchmarks for expected fouling penalties.

4.2.2 Machine Learning-based Methods

ML, particularly supervised, is increasingly being used to monitor ship performance using operational data (Alexiou et al., 2022; Lang et al., 2022; Valchev et al., 2022). In the context of fouling, a growing body of research is using predictive power models to infer the added power required as hull and propeller condition deteriorate over time (Coraddu et al., 2019; Gupta et al., 2022; Laurie et al., 2021). Although this remains less common than industry standards or CFD-based approaches, it is emerging as a practical alternative, as some studies have found it to outperform traditional methods for fouling estimation (Coraddu et al., 2019; Valchev et al., 2022).

Tree-based models (decision trees, random forests, XGBoost) and neural network variants are popular model choices in recent literature and often achieve test Mean Absolute Percentage Errors (MAPE) around 1-3% (Lang et al., 2022; Laurie et al., 2021; Valchev et al., 2022). However, studies such as Gupta et al. (2022) and Bakka et al. (2022) indicate that physics-informed white-box models like Principal Components Regression and Generalized Additive Models can perform quite closely to black-box alternatives. The performance gap between white- and black-box models thus appears relatively small in this domain, supporting the notion of Rudin (2019) that white-box models can be a better choice when data is structured and represents physically meaningful relations. However, it should be noted that comparative literature in this area is still relatively scarce (H. S. Kim & Roh, 2024; Valchev et al., 2022).

Valchev et al. (2022) note that purely data-driven models can produce estimates that are not physically plausible in some situations, even when they perform well on average, implying that interpretability is a key consideration when assessing fouling effects (Shmueli, 2010; Valchev et al., 2022). This issue is not limited to black-box models; even naval engineering methods can yield physically inconsistent results due to their coarse assumptions and rigid formulas (Petridis & Vassilakopoulos, 2024). The logical conclusion from the literature is, therefore, that domain knowledge should be built into model design, feature engineering, and evaluation, regardless of model class (Gupta et al., 2022; Petridis & Vassilakopoulos, 2024; Valchev et al., 2022).

Coraddu et al. (2019) propose an approach that uses ML, instead of naval engineering standards, to create a reference model of clean-ship power demand. Here, a supervised model is trained on operational data from a vessel assumed to be in a clean condition (i.e., the first month after a cleaning event), and deviations between the predicted reference power and the observed power during later operation are interpreted as fouling-related performance loss. Petridis and Vassilakopoulos (2024) and Themelis et al. (2024) apply this framework and estimate added power of 20% and 20.5% after 300 days, respectively, during normal operation.

Gupta et al. (2022) criticize the reference-model approach because it is better at identifying whether performance has deteriorated than at quantifying how much of that deterioration is due to fouling, because the estimate depends strongly on how the presumed clean baseline period is defined. A more direct inferential alternative, therefore, fits a predictive model to shaft power over the full dataset and includes a time-dependent fouling proxy, such as time since the last cleaning event, so that the fouling effect is inferred jointly with the effects of speed, weather, and loading condition. Laurie et al. (2021) follow this logic and estimates that fouling increases required shaft power by 5.2% after 360 days. Themelis et al. (2024) also implement both approaches and, using the inferential version, estimate an average power increase of 11.3% after 300 days. Duan et al. (2024) estimate an even higher added power, with required power increasing by up to 30.5% after 777 days of operation.

Both of these ML-based approaches also suffer from the identification problem. None of them directly measures fouling, meaning that it cannot be distinguished from other time-dependent factors, such as engine deterioration, especially when using simple fouling proxies, such as time since the last cleaning event (Gupta et al., 2022; Laurie et al., 2021; Themelis et al., 2024). A commonly used method to mitigate this is to use a more detailed and domain-informed proxy, such as the *idle* time since last cleaning event (Bakka et al., 2022), the roughness factor of the hull (Gupta et al., 2022; Uzun et al., 2019), or a hybrid modeling approach (de Haas et al., 2023). Although there is a lack of studies comparing these proxies in a controlled manner, all of these proxies have been used to produce robust and physically consistent results (Bakka et al., 2022; Laurie et al., 2021).

A common critique of ML-based methodologies is that, although they eliminate the need for expensive sea trials[3], they instead require substantial amounts of high-quality operational data and careful preprocessing (Y. Kim et al., 2025). This requirement can be difficult to accommodate in practice, since such data are not always available, especially for older vessels (Valchev et al., 2022). In addition, ML models are generally more sensitive than naval engineering methods to poor data coverage, because they must learn the speed-power relationship from the patterns present in the training data rather than physical laws (Valchev et al., 2022). If, for example, the model is trained almost entirely on voyages at or near ballast draft, it has little basis for predicting power consumption in partially loaded conditions. Traditional methods are not immune to this problem, but are generally more likely to produce at least a rough and physically plausible estimate in sparsely covered regions (Valchev et al., 2022). For that reason, a sensible practical implication is to assess data coverage explicitly and remain aware of when the model extrapolates beyond the training data.

4.3 Explainable Artificial Intelligence in Adjacent Applications

Research applying XAI to fouling estimation remains limited. However, XAI methods have begun to appear in closely related maritime applications. For example, Wang et al. (2023) use SHAP values to interpret black-box fuel-consumption models and assess how individual features contribute to the predicted fuel consumption of ships. Relevant methodologies have also been applied to similarly complex physical systems with noisy observational data and latent degradation processes, such as wind turbine monitoring and weather systems (Astolfi et al., 2023; Letzgus & Müller, 2024)

SHAP and LIME values are popular methodological choices in this regard, because they make black-box models easier to inspect by quantifying which input variables drive a prediction. Although they cannot quantify effects (such as fouling), they can improve interpretability, support model checking, and increase confidence that a model is relying on physically meaningful patterns rather than spurious correlations. Letzgus and Müller (2024) and Astolfi et al. (2023) both apply SHAP values to assess whether ML models of wind turbine power-curve models have learned physically plausible patterns to support more informed model selection. These studies show that ML models can learn which features are physically important (Astolfi et al., 2023; Letzgus & Müller, 2024). However, they do not themselves provide an estimate of how changes in power output can be causally attributed to a deterioration process.

Another common post-hoc method is PD, which summarizes how a model's prediction changes as a

[3] Controlled trial runs of the ship with a clean hull in calm conditions

feature varies, while the remaining features are held constant (Friedman, 2001). In contrast to SHAP, PDs are mainly useful for showing the overall shape of a learned relationship, e.g., whether it appears linear, monotonic, or more complex, and they therefore support interpretability by making model behavior easier to compare with domain expectations. Zhang et al. (2021) provide a relevant example in environmental studies, in which random forest models and PD functions are used to examine how climate and ecosystem characteristics relate to the occurrence and impact of overshoot droughts. In their study, they use PD to plot the relationship between drought occurrence and annual temperature, biodiversity, aridity, and land cover. While PD is a better tool than SHAP values for isolating single-feature effects, PD can become misleading when features are correlated (Apley & Zhu, 2020).

When stronger feature interactions are expected, a sensible alternative to PD is ALE, which has the same overall goal of describing how predictions change with a feature, but does so using local differences calculated within the observed data distribution rather than averaging over all possible feature combinations[4] (Apley & Zhu, 2020). ALE is less sensitive to the unrealistic combinations that can distort PDs when predictors are correlated, and therefore, is often more suitable for structured physical systems where variables are strongly interdependent. Kantz et al. (2024) support the case for ALE with a robustness study of industrial process modeling, where effect-based methods such as ALE more closely recovered ground-truth relationships than additive methods such as SHAP and LIME, especially when the underlying model captured the process well. Related applications also appear in environmental science, where ALE has been used to interpret models of soil-moisture droughts and broader bioclimatic response patterns (Huang et al., 2023; Yi et al., 2026).

4.4 Summary & Research Gaps

The literature on fouling estimation is most mature within CFD and industry-standard naval engineering methods, while ML applications remain more limited and recent. Across study types, the reported magnitudes show a high degree of heterogeneity, likely because the added power is very context-dependent. However, it can be concluded that the added power can become rather substantial, ranging from single-digit percentage increases under light slime conditions to 70-80% under heavy calcareous fouling. However, traditional approaches have important weaknesses. Industry standard methods face a clear trade-off between comparability and data retention, and often depend on costly or imperfect sea-trial baselines. At the same time, CFD requires substantial modeling effort and computational cost and is better suited to theoretical benchmarking than to continuous monitoring of an operating vessel.

ML-based methods have appeared as a promising pragmatic alternative for modeling fouling development from operational data, especially when standardized baselines are difficult to apply consistently. However, the literature remains too sparse to consider ML-based fouling estimation a mature method-

[4] Explained in further detail in Section 5.7.2

ology, and further validation is needed across vessels, datasets, and modeling pipeline designs. In particular, the question of whether black-box models actually offer a meaningful advantage over more interpretable alternatives remains unresolved in this domain.

A similar gap exists for XAI. Although SHAP, PD, and ALE have proven valuable in adjacent domains for inspecting learned relationships and identifying key drivers, their use in marine fouling studies remains very limited. No studies are found that use XAI specifically to isolate fouling as a source of added power. However, adjacent applications suggest that such methods may be useful for separating latent effects in complex physical systems. This makes ML, especially, explainable, physics-informed AI a relevant and promising area for further research.

5 Methodology

The following sections present each step of the modeling pipeline and document the choices that are made before applying the models. Section 5.1 gives a high-level overview of the dataset, while Section 5.2 describes the preprocessing steps in detail. It should be noted that 5.2 is mostly relevant to reproducing the results, but less so to interpretation and discussion. Section 5.3 describes the approach to feature selection and the final feature set. Sections 5.4, 5.5, and 5.6 presents the fundamental theory of the models, as well as the training approach. Section 5.7 describes the tools and metrics used for evaluation. Finally, Section 5.8 presents the tool stack, including software and libraries. Note that exploratory data analysis is documented in Section 6.1, since it builds on concepts and decisions presented throughout the methodology section.

5.1 Dataset

The analysis is based on a dataset provided as part of the "Real Vessel Data Challenge" organized by MMMCZCS, which is a workshop for fouling effect estimation based on operational data for academics and industry (Tsarsitalidis et al., 2026). The dataset comes from a 76,000 Deadweight Tonnage (DWT) bulk carrier, a large cargo ship designed to transport unpackaged commodities such as grain, coal, or iron ore. It is therefore representative of a common long-haul, fuel-intensive vessel type (Tsarsitalidis et al., 2026). The vessel is anonymized, which implies that neither its operational history before 2024 nor its GPS location is disclosed (Tsarsitalidis et al., 2026). The vessel is 225 meters long and has a beam of 32 meters. Its main engine has a maximum continuous rating of 9,500 kW at 105 rpm, which are all characteristic of Panamax ships in the 60,000 to 80,000 DWT range (Tsarsitalidis et al., 2026).

The dataset comprises three data streams: operational monitoring data from onboard sensors, externally provided metocean data, and daily noon reports, which span January 1, 2024, through December 31, 2024 (Tsarsitalidis et al., 2026). 17 variables were measured by onboard sensors at 15-second intervals, while 15 hourly metocean variables in the dataset stem from external providers. The hourly metocean variables are forecasts, which are generally considered less accurate than nowcasts or hindcasts. 14 variables were collected manually by the crew in the noon reports at 12:00 PM each day in the vessel's time zone. Since the vessel's location is unknown, the exact timing of the noon report data cannot be determined. An overview of all available variables is provided in Appendix B.

Additional contextual information is provided for the vessel, but several important details remain undisclosed. Data from sea trials, it should be noted that these were conducted before a propeller upgrade that is expected to have reduced the vessel's power requirement by approximately 3-5% on average (M. Patey, Personal Communication, 2026). In addition, the results are provided as an image (see Appendix C), and the underlying data used to estimate the curves are not available. Four standard loading conditions, under which the vessel is designed to operate, are provided for reference with the sea trials (see Table 5.1). Data owners further confirmed that two cleaning events, each including hull cleaning and propeller polishing, took place in late January and late July, respectively, although the exact dates are unknown (M. Patey & F. Lehn, Personal Communication, 2026).

Table 5.1: *Standard loading conditions, with corresponding drafts and trim, used for reference with sea trials. Fore refers to the draft at the front of the ship, while Aft refers to the draft at the back of the ship.*

Condition	Fore [m]	Aft [m]
Design	12.2	12.2
Scantling	14.25	14.25
Ballast	5.0	7.8
Heavy Ballast	4.4	9.4

5.2 Preprocessing

The objective of preprocessing is to transform the relevant data from raw noon reports, sensor streams, and metocean data into a final tabular dataset that is as free as possible from disturbances and reaches an appropriate bias-variance tradeoff (Dalheim & Steen, 2020). In this thesis, preprocessing is carried out in five steps, each of which will be explained in further detail in the following sections. Section 5.2.1 covers a preliminary feature selection step that removes variables that are clearly irrelevant, redundant, or technically unreliable. Section 5.2.2 covers the synchronization of observations in time between the different onboard subsystems and external data sources. Section 5.2.3 covers the identification and removal of wrong or otherwise undesirable values such as outliers, dropouts, or spikes. Section 5.2.4 covers the validation of remaining measurements against domain expectations and cross-sensor relationships. Section 5.2.5 covers the aggregation of the validated high-frequency data to a common target sampling interval of 15 minutes. The preprocessing steps and their placement within the full modeling pipeline are shown in Figure 5.1.

5.2.1 Preliminary Feature Removal

As a first step, variables that are clearly unreliable or entirely irrelevant are removed from the dataset. While Dalheim and Steen (2020) and Gupta et al. (2023) recommend removing all variables not required for modeling as a first step, a deliberate choice is made to keep a broader set of variables and only remove those that are clearly unreliable. This approach allows additional variables to be used for validation or supplementary analyses, even if they are not used as features in modeling.

Seven sensor signals are removed entirely. The transducer water-depth variable is excluded because its readings are unreliable and its effect is only notable in shallow-water conditions that typically coincide with maneuvering (Dalheim & Steen, 2020; Laurie et al., 2021). One turbocharger signal and the four draft sensor signals are dropped because they contain no observations. Instead, draft data is taken from the noon reports.

For metocean data, wave height and sea water velocity signals are provided by both providers, but

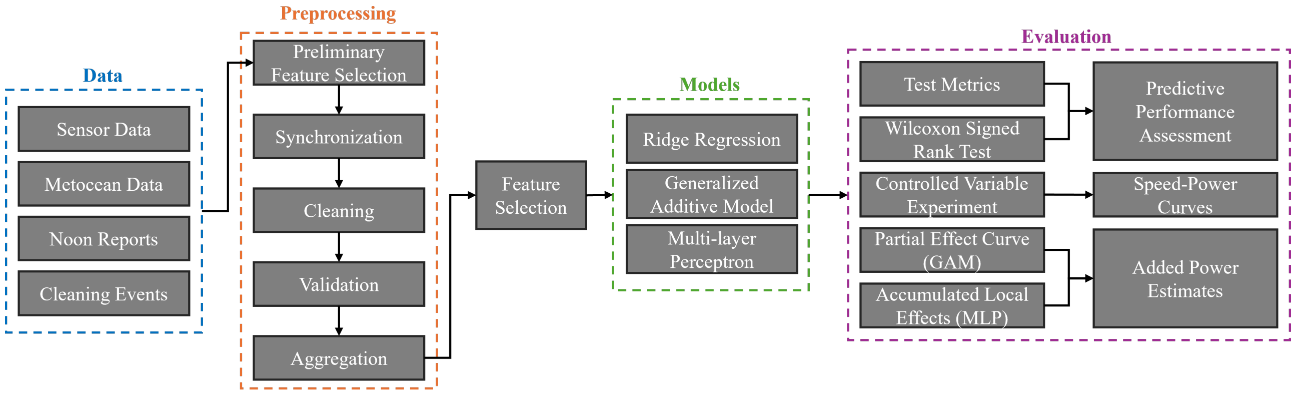


Figure 5.1: *Overview of the full preprocessing and modeling pipeline. Source: Own Creation*

with significantly different values. For example, the significant wave heights of Provider MB and Provider S show only moderate agreement, with a Pearson correlation coefficient r = 0.727. Because wave height is known to be associated with an increase in required shaft power (DNV, 2023; ISO, 2016), the provider that produces the strongest correlation between significant wave height and shaft power is treated as the most reliable. A preliminary analysis identifies Provider MB as superior in this respect (see Appendix D). For internal consistency, Provider MB is also selected as the source of seawater velocity signals.

For the noon reports, all variables except draft are removed, since these manually reported quantities are considered coarser and less reliable proxies of the same physical properties that are already measured at higher resolution and accuracy by the onboard sensors and metocean providers.

5.2.2 Synchronization

To enable joint analysis of signals from different subsystems, all time series are synchronized to a common time reference, following the synchronization procedures described by Dalheim and Steen (2020). In their framework, time "jumps" are first identified before constructing continuous time intervals where all variables are aligned onto a common time reference, in this case Coordinated Universal Time (UTC). Draft values from noon reports are assumed to be made in UTC, although the original entries are recorded in local time in an unknown timezone.

The space between two measurements is considered a time jump if it exceeds 1.5 times the intended sampling interval, i.e., 15 seconds for sensor data and 1 hour for metocean data. For example, if two measurements of shaft power (sensor measurement) are made with 45 seconds in between, those 45 seconds are marked as a time jump because it is longer than $1.5 * 15 = 22.5$ seconds (see illustrative example in Figure 5.2). Boundaries between clean segments are defined so that a new segment starts whenever at least one signal exhibits a time jump, and no time jumps are present for any signal inside the segment. Only segments with a minimum duration of 2 hours are kept. This procedure yields 519 continuous time segments, mostly shorter than one day, which together cover approximately 349 days of operation or 96% of the raw data (Histogram of segment lengths in Appendix E).

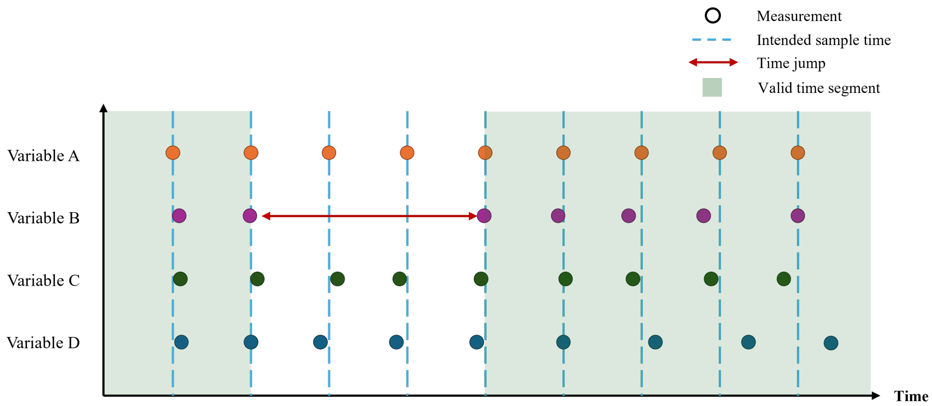


Figure 5.2: Illustrative example of a dataset where some of the data is discarded due to a time jump. Note that only the stretches where all variables are continuously measured are used, meaning that values for variable A, C, and D are discarded during the time jump for variable B. Source: Own creation

Within each continuous segment, an ideal timeline is then imposed by populating the interval with regularly spaced timestamps at the nominal sampling rates, i.e., every 15 seconds for onboard sensor data and every 1 hour for metocean data. Visual inspection of measurement times (example in Appendix F) indicates that observations tend to lie very close to these grids, and only 0.04% of sensor measurements and 0.5% of metocean data points deviate by more than 1.5 times the intended sampling interval. To obtain values on the common grid while preserving representativeness, each variable is linearly interpolated between the two closest observations to an intended sampling point (see illustrative example in Figure 5.3). For each continuous interval, a merged table is created that combines sensor and metocean data on the same time axis, with missing values represented as missing for intermediate timestamps in the metocean data. After synchronization, the 519 segments are combined into a single table containing 2,013,093 observations.

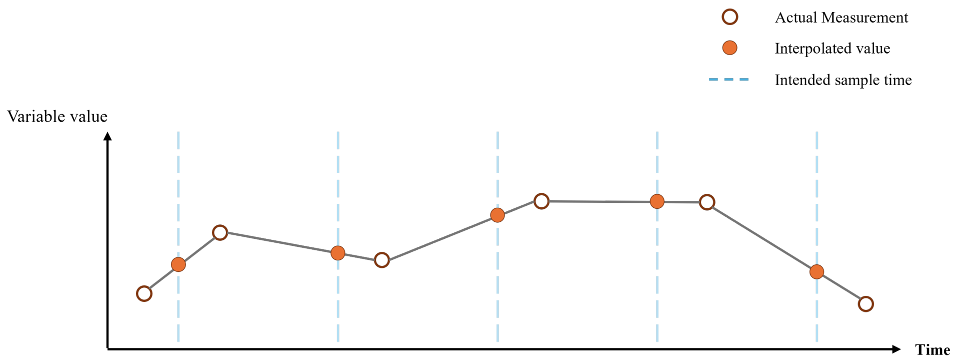


Figure 5.3: Illustrative example of linear interpolation procedure for a single variable to the intended sampling times. Source: Own creation

5.2.3 Cleaning

A cleaning philosophy similar to that of Laurie et al. (2021) is applied to obtain data that approximate steady-state operation in open waters by removing observations that resemble port maneuvering or clearly invalid measurements. The former is removed because including it would introduce unnecessary variance and make the results less useful to practitioners who are primarily interested in steady-state operations in open waters (Dalheim & Steen, 2020; ISO, 2016; Laurie et al., 2021). However, some conventional limits used in naval architecture, particularly on metocean variables such as wave heights, are relaxed because ML models can accommodate higher variance and partially understood physics better than the rigid physics-based formulas usually applied in these fields (ISO, 2016; Laurie et al., 2021). Cleaning is performed considering all sensor variables, although most are not used in modeling. This decision is based on the conservative assumption that inaccuracies in one subsystem may indicate sensor failure or inaccuracy in another (Dalheim & Steen, 2020).

A sensible first step in the cleaning process is to remove measurements likely to represent sensor dropouts rather than genuine measurements (Dalheim & Steen, 2020; Gupta et al., 2023). Firstly, inconsistent combinations of propeller RPM and shaft power are flagged when the implied shaft power deviates more than 2% from the value calculated from the torque and rotational speed using:

$$\text{Power} = \text{Torque} \times 2\pi \times \text{Propeller RPM}$$

as this is interpreted as a possible indication of torquemeter malfunction (ISO, 2016). Secondly, all water temperature values of exactly 6 or 2 degrees are also flagged as dropouts, since they appeared suspiciously often (11.5% of pre-filtering n) and often in constant stretches after instantaneous shifts from temperatures well above 20 degrees (see Figure 5.4). In addition, filters are implemented to address inconsistencies among propeller RPM, engine RPMs, and shaft power, negative values for speed, propeller and engine RPM, and any angular values outside the range $[0, 360]$.

Following Dalheim and Steen (2020), a spike filter is also applied to a selected set of high-frequency sensor variables, namely scavenging air pressure, fuel oil inlet mass flow, as well as propeller shaft torque, thrust force, and power. For each variable, a rolling median over 20 observations is used as the local reference level, corresponding to a 5-minute window at the 15-second sampling interval. A rolling median absolute deviation (MAD). is then computed in the same window, and an observation is flagged as a spike when its absolute deviation from the rolling median exceeds five times the local MAD (see Appendix G for example). For short spike runs (< 10 consecutive observations), the nearest valid value is used for imputation, while longer runs are considered missing values.

Values that are either spikes or dropouts are considered missing. The default in this domain is to remove entire rows with at least one missing value, as imputation can be difficult and often introduces

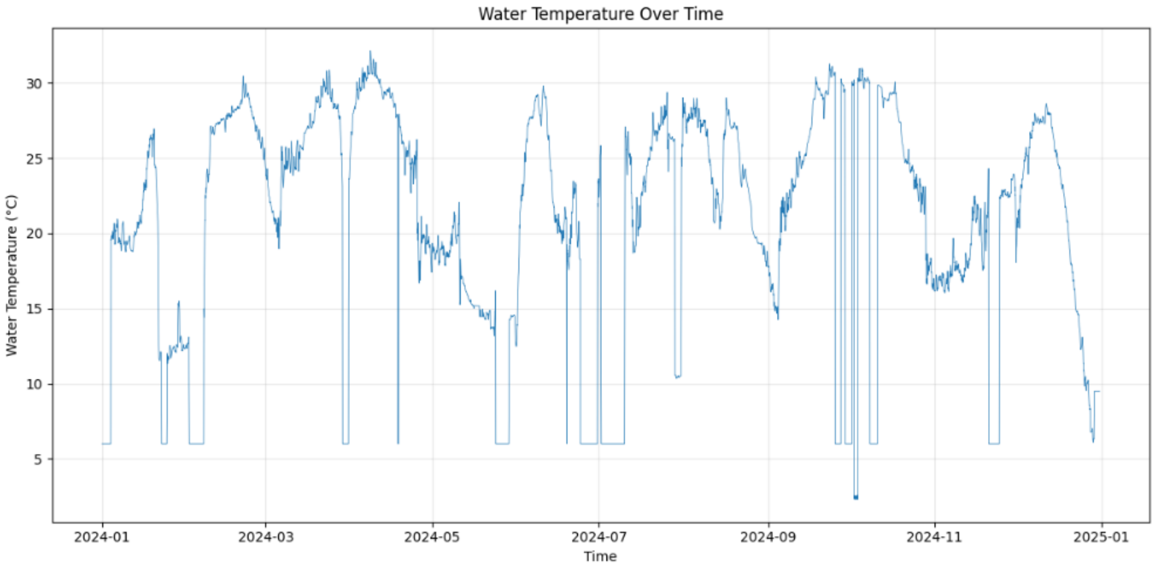


Figure 5.4: *Raw series of water temperature values measured by the ship sensor. Measurements of 6 and 2 degrees are considered invalid and treated as missing values.*

spurious relationships that are not consistent with physics (Dalheim & Steen, 2020). The data is filtered under the conservative assumption that all sensor variables are required, resulting in the removal of 380,767 rows (18.9% of the total) due to missing or invalid values, most of which are removed because of missing shaft power data. This includes segment boundaries, where missing values naturally occur because values cannot be interpolated.

In addition to quality-based filters, two further actions are taken to ensure the dataset excludes periods of port waiting or maneuvering. Firstly, a rolling standard deviation filter is applied to STW, SOG, and heading angle as recommended by DNV (2023). Thresholds for speed are deliberately doubled from 0.194 knots to 0.388 knots to allow slightly more variance in the training data, since the DNV thresholds are set with naval engineering methods in mind and appear very restrictive relative to the data at hand (see Appendix H). Secondly, rule-based filtering is applied to remove observations with low values for speed (< 9 knots), propeller RPM (< 0, i.e., reversing), and shaft power (< 1000 kW). All filtering operations are summarized in Table 5.2. In total, 50.5% of the raw data is removed, resulting in 1,017,126 observations.

Table 5.2: *Summary of filtering operations*

Filter	**Rule(s)**	**n removed**	**n**
Raw synchronized data	–	–	2,013,093
Segment boundaries	NaN values at segment start/end	1,038	2,012,055
Required vars NaN	Any variable is NaN, repeated or dropout	374,021	1,638,034
Spikes	Any variable is marked as a spike	5,708	1,632,326
Rolling stds	Rolling std above threshold in speed / heading	350,971	1,281,355
Ship maneuvering	Speed <4 kn **or** Propeller RPM <0 **or** Shaft power <1000 kW	264,229	1,017,126
Filtered data	–	–	1,017,126

5.2.4 Validation

Because operational data are noisy and assembled from multiple sources, some basic validation is needed to ensure that the main variables have sensible magnitudes and are mutually consistent (Dalheim & Steen, 2020). Therefore, three plausibility checks are performed, covering speed, power, and wind.

As shown in Figure 5.5, the relationship between speed and power is vaguely consistent with the expected cubic trend, but with substantial residual variation. Relative to the sea-trial curves (Appendix C), the power values are significantly higher, which is expected since the vessel is not in calm conditions. At 13 knots, for instance, the fitted line implies a power requirement of roughly 5500 kW, whereas the sea trials indicate about 3800–4500 kW depending on draft. The moderate goodness of fit ($R^2 = 0.53$) is also expected, since speed alone cannot explain the full variation in shaft power when weather, draft, and fouling remain uncontrolled. The relationship between speed and power thus appears valid.

Shaft power can be validated by comparing it with the main engine fuel load, with which it is expected to show an approximately linear relationship. Industry guidance suggests that the resulting correlation should typically be in the range of 0.9 to 0.95 (M. Patey, Personal Communication, 2026). However, in the present data, the fuel-load-power relationship initially appears as two distinct clouds rather than a single trend, indicating a systematic inconsistency in the shaft-power signal rather than ordinary operating scatter (see Figure 5.6). A plausible explanation is that the power is calculated from the torque and rpm measured by the torquemeter, and that the torquemeter was recalibrated during the observation period. MMMCZCS confirms that this is the case and that the calibration most likely took place on July 1st (M. Patey, Personal Communication, 2026).

Since the data owners could not disclose the exact recalibration method, a data-driven approximation is applied by fitting a shared-slope linear model to the pre- and post-calibration observations,

$$P_i = \beta_0 + \beta_1 L_i + \beta_2 I_i^{\text{post}} + \varepsilon_i, \tag{1}$$

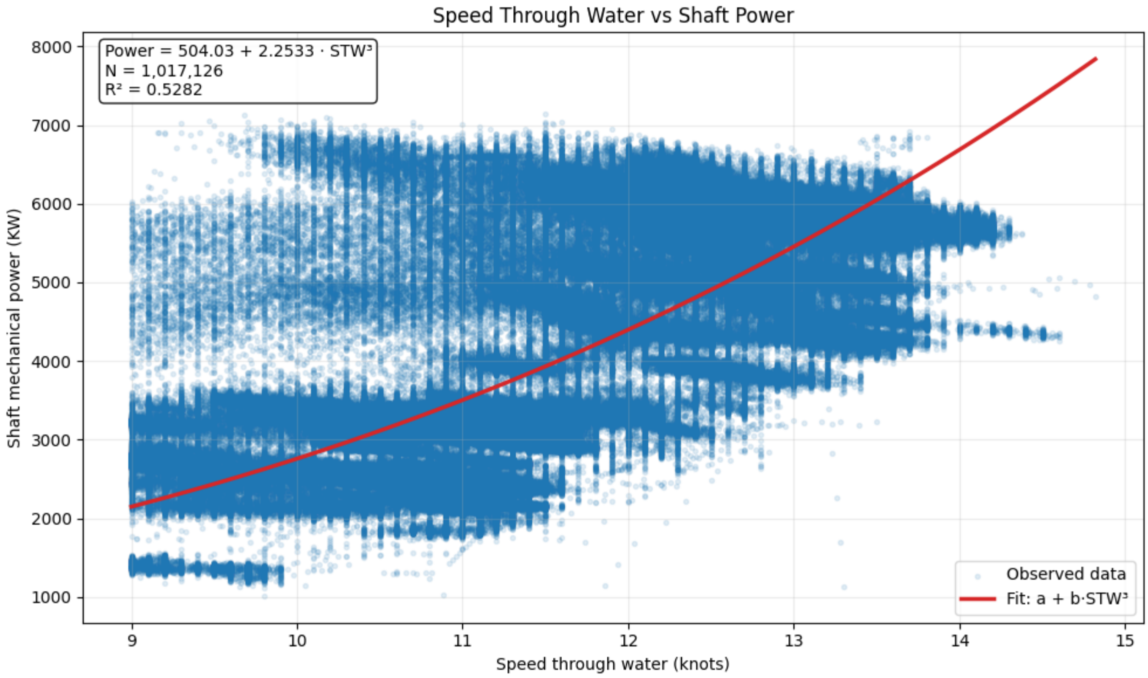


Figure 5.5: *Scatterplot of STW and Shaft Power, with a fitted cubic trend of the form* $P = a + b \cdot STW^3$.

where P_i is the measured shaft power in kW for observation i, L_i is the main engine fuel load for observation i, I_i^{post} is a binary variable equal to 1 if observation i occurs after the assumed calibration date and 0 otherwise, β_0 is the intercept for the pre-calibration period, β_1 is the common slope relating fuel load to shaft power, β_2 is the intercept shift between the post- and pre-calibration periods, and ε_i is the residual error term. This specification constrains the two periods to share the same slope while allowing different intercepts, thereby modeling the discrepancy between the two point clouds as an additive offset (Figure 5.6).

The estimated common slope is 218.07, the pre-calibration intercept is -4965.07 kW, and the post-calibration intercept is -6865.85 kW, implying an offset of

$$\Delta = \beta_2 = -1900.78 \text{ kW}.$$

The pre-calibration observations are then shifted onto the post-calibration scale according to

$$P_i^{\text{corr}} = P_i + \Delta, \qquad \text{for } I_i^{\text{pre}} = 1,$$

which corresponds to subtracting 1900.78 kW from the pre-calibration shaft-power values. After this adjustment, the fuel-load-power relationship appeared substantially more coherent, with a Pearson correlation of 0.9446 between main engine fuel load and the corrected shaft power label (Figure 5.7).

This correction should not be considered as a true physical recalibration of the torquemeter. Rather,

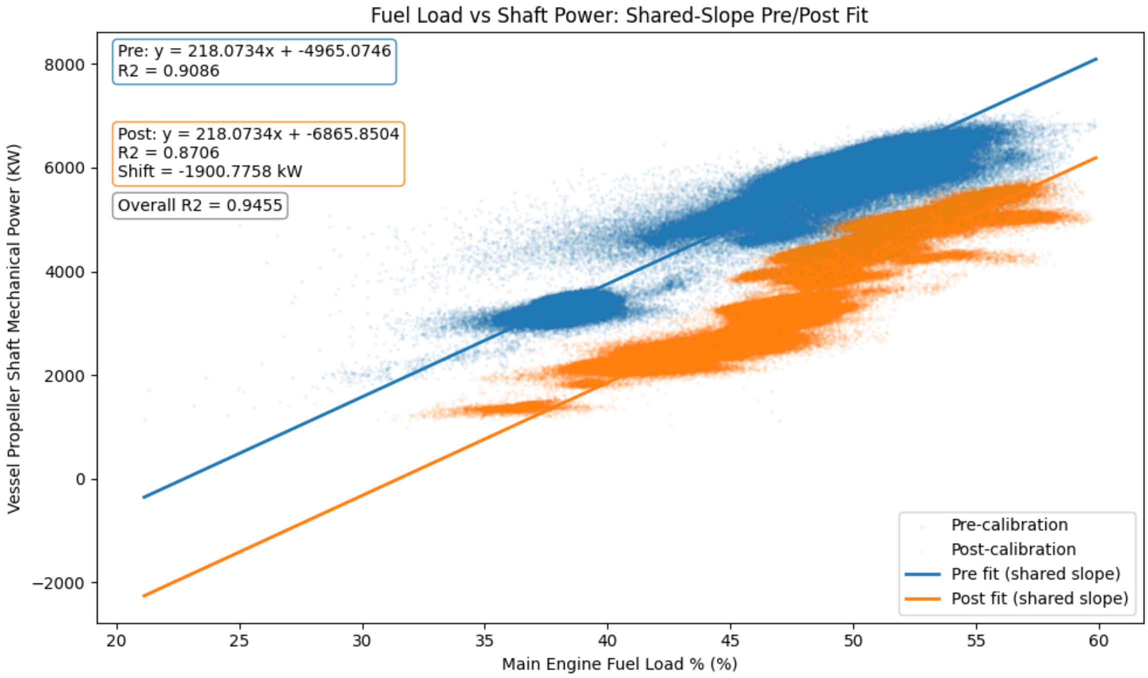


Figure 5.6: *Scatter plot of main engine fuel load and propeller shaft power before and after the calibration event on July 1st. Lines illustrate the common-slope model applied to each subset, and the space between the lines is the estimated offset of -1900.78 kW.*

it is a pragmatic correction introduced to make the pre- and post-calibration data comparable. The approach assumes that the calibration effect can be represented as a constant additive offset in shaft power across the operating range, which is a simplification. As mentioned in section 5.2.3, shaft power depends on both torque and rotational speed, which means that any change in the torque measurement chain does not necessarily translate into the same power difference at all RPM levels.

A final plausibility check of the wind signals is performed by comparing onboard-derived longitudinal true wind with the corresponding provider-derived longitudinal wind. The onboard estimate is obtained from the anemometer's relative wind measurement, together with vessel motion, and the comparison is used to assess whether the provider values are broadly consistent with onboard observations and to determine which source appears more reliable.

Figure 5.8 shows a clear positive relationship between the two estimates, indicating that they capture the same overall wind pattern despite some scatter and some outlying observations. These outliers are more plausibly attributed to disturbances in the onboard anemometer signal than to errors in the provider data, as the onboard measurement is susceptible to local flow distortion due to the vessel's shape and the sensor's surroundings (M. Patey, Personal Communication, 2026).

This interpretation is supported by Figure 5.9, where the wind derived onboard appears to be more dispersed than the wind derived from the provider. While the broader spread could also be attributed

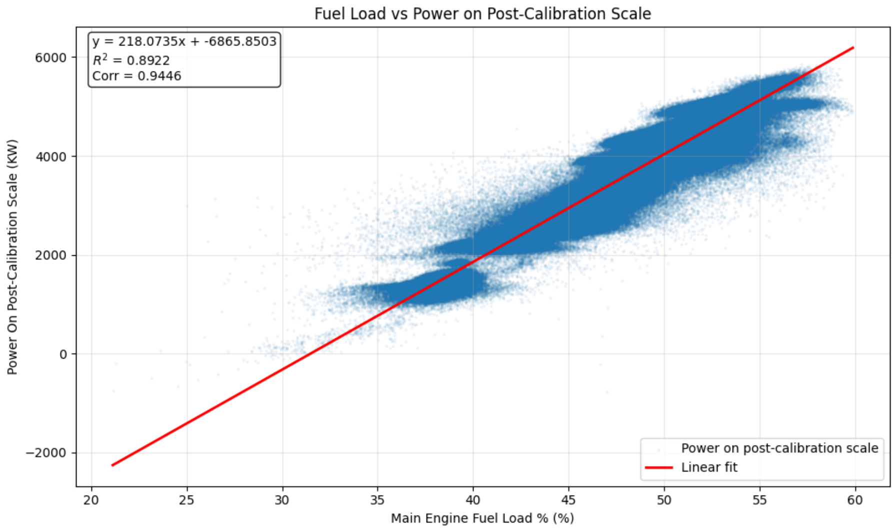


Figure 5.7: *Scatter plot of main engine fuel load against corrected propeller shaft power after removal of the post-calibration offset.*

to the weather providers aggregating over a longer time window[5], it could also be at least partially attributed to the on-board sensor being noisier. Therefore, the weather provider variable is chosen for modeling.

5.2.5 Aggregation

The raw sensor data is recorded at a 15-second sampling frequency, producing a high volume of observations that capture many short-term fluctuations. This frequency is too high to model the relationship between shaft power and slowly evolving factors such as fouling, loading conditions, and weather (Dalheim & Steen, 2020; ISO, 2016). The data is therefore aggregated with fixed windows to better approximate stable operating states while preserving the variance and data volume needed for training of ML models (Dalheim & Steen, 2020). Based on industry standards and prior academic studies, a 15-minute aggregation interval is deemed appropriate (Gupta et al., 2023; ISO, 2016; Laurie et al., 2021).

Aggregation is performed within the synchronized time segments rather than across the full time series to avoid extrapolation (Dalheim & Steen, 2020). Concretely, the sensor data within each segment is partitioned into non-overlapping 15-minute windows, and all observations falling within a given window are summarized into a single row. A window is retained only if it has at least 90% of the expected number of rows (60). Most variables are aggregated using the arithmetic mean, which is appropriate for continuous quantities representing instantaneous states. For directional variables such as wave direction, wind direction, and vessel heading, the circular mean is used, defined as:

[5] Exact aggregation window, if any, used by the metocean providers was not disclosed

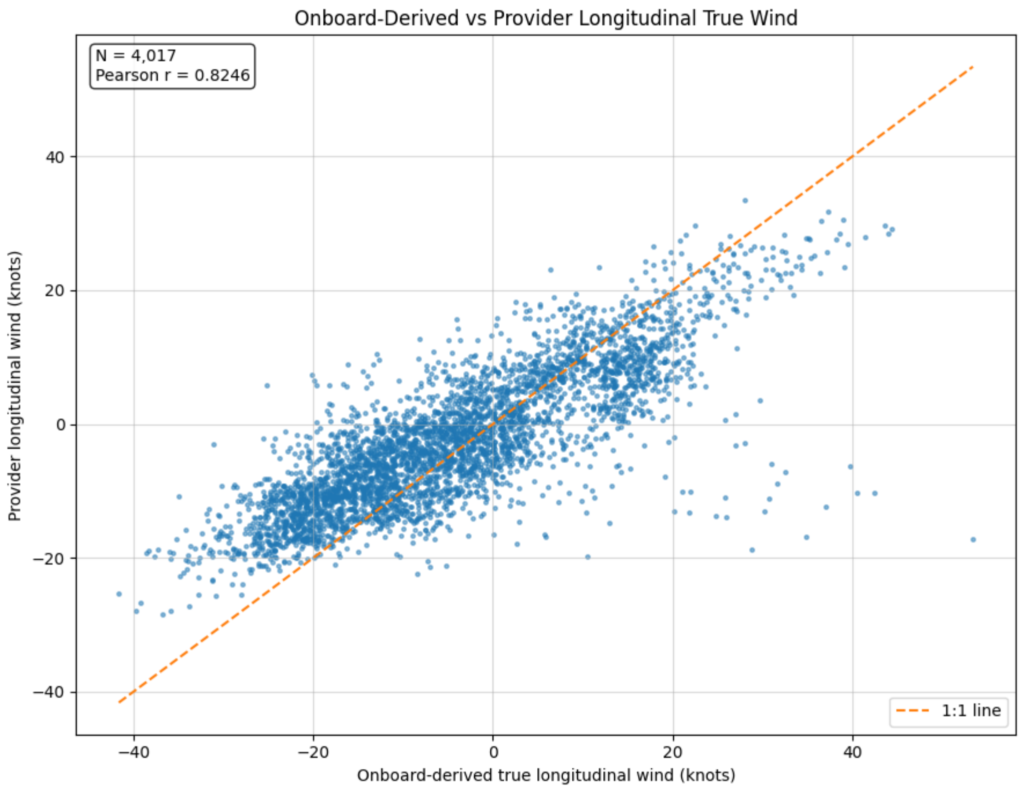


Figure 5.8: *Onboard-derived longitudinal true wind vs. provider-derived longitudinal true wind. Note that N is relatively low since the metocean provider data is hourly.*

$$\bar{\theta} = \operatorname{arctan2}\Big(\overline{\sin(\theta_i)},\ \overline{\cos(\theta_i)}\Big) \pmod{360^\circ}$$

where the mean is taken component-wise over all observations in the window. For cumulative counter variables that record a running total rather than an instantaneous value, the relevant quantity per window is the total increase accumulated during that interval. This is computed as the sum of all positive successive differences within the window.

Because metocean data and noon report draft values are recorded at lower frequencies than the onboard sensors, they cannot be aggregated directly alongside the sensor variables. Instead, these sources are joined to the aggregated sensor table after aggregation. As recommended by DNV (2023), the metocean variables are matched to each window using the most recent available forecast observation within a backward tolerance of four hours. Similarly, draft values from the noon reports are matched using a 24-hour tolerance.

As evident in Figure 5.10, the aggregation achieves its intended purpose: the 15-minute windows substantially reduce the noise present in the raw sensor signal while preserving the underlying trend (Figure 5.10a). For metocean variables, the values are more step-wise in nature due to the forward fill (Figure 5.10b).

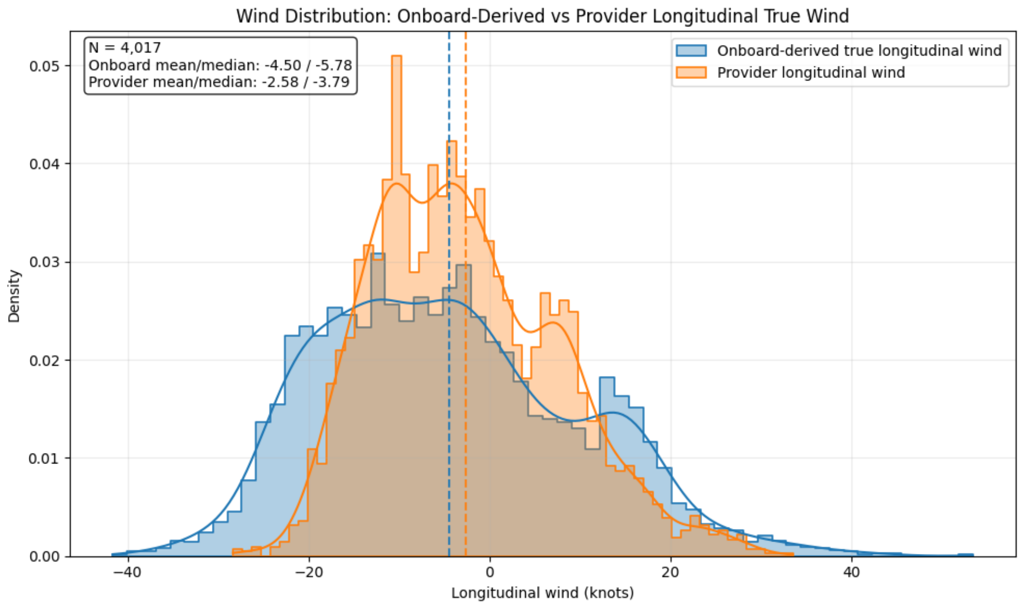
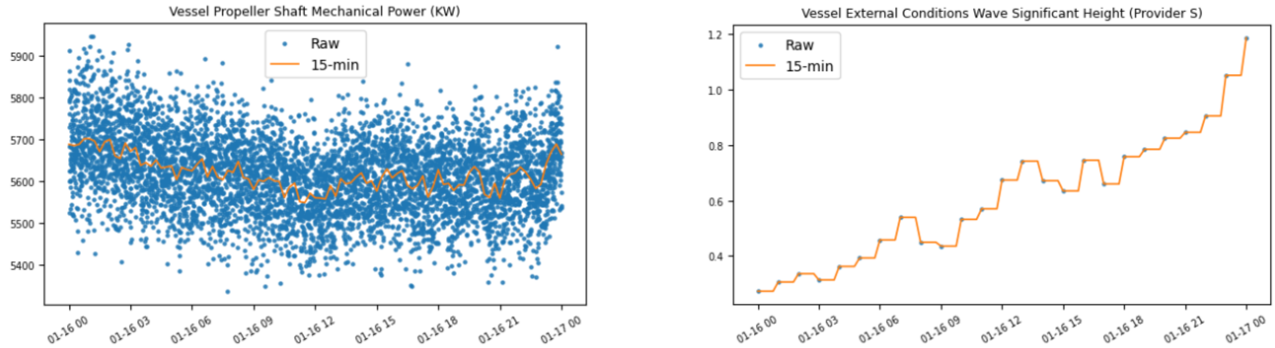


Figure 5.9: *Distributional comparison of onboard-derived and provider-derived longitudinal true wind.*



(a) Shaft power (sensor variable)

(b) Wave height (metocean variable)

Figure 5.10: *Example of 15-minute aggregation over 24 hours for two variables. The raw observations are shown in blue, and the 15-minute window means are shown in orange.*

Lastly, data points with unknown time since last cleaning are dropped. Since the exact cleaning dates are not provided, the two cleaning events are inferred from the SOG time series, assuming that hull cleaning must occur when the vessel is idle for at least a day. Visual inspection of the January and July SOG series reveals two clear port stays with a prolonged SOG of exactly 0 in the second half of the months, occurring on January 23rd to 24th and July 29th to 30th, respectively (Appendix K). The cleaning events are therefore assumed to have taken place on the last day of each port stay, January 24th and July 30th. Consequently, all observations before Jan 24th contain no value for time since last cleaning and are therefore dropped. The final aggregated dataset thus comprises 15,061 observations.

5.3 Feature Selection

Features are selected based on the theoretical arguments outlined in section 3.2, to achieve a complete representation of the operating conditions of the vessel without introducing mediating effects. To represent environmental conditions, wind, waves, and swell are each decomposed into longitudinal and transverse components relative to the vessel's heading and included as separate features. In addition, wave period, sea water temperature, and longitudinal current (estimated as $SOG - STW$) are included.

To represent the vessel's operating mode, STW, average draft, draft trim, and days since last cleaning event (DSC) are included to capture the main non-environmental drivers of shaft power. Specifically, speed is the primary determinant of resistance, while draft and trim reflect loading condition and hull attitude, both of which influence frictional and wave-making resistance. DSC is selected as the fouling proxy because it provides a simple and interpretable measure that has also been used in previous fouling studies (Gupta et al., 2022; Laurie et al., 2021). However, it is acknowledged that constructing a more comprehensive proxy might yield a model more consistent with domain knowledge.

To ensure that the effect of fouling can be reliably distinguished from other features, multicollinearity among features is assessed using pairwise correlations and Variance Inflation Factors (VIFs). VIF measures how strongly a feature is explained by the other features and, in turn, how much this inflates the variance of its estimated linear regression coefficient (Marquardt, 1970). This approach is somewhat limited in that it only accounts for linear relationships. However, concurvity, i.e., nonlinear dependence, between features can still be problematic, notably when interpreting generalized additive models (Hastie & Tibshirani, 1986). However, this is considered an acceptable approximation.

Several features could reasonably be suspected of exhibiting multicollinearity. Wind speed and wave height, for example, are physically linked, since surface winds generate ocean waves, typically with a delay of a few hours (ISO, 2016). Because swell and locally wind-generated waves are not always reliably distinguished in metocean data, it could also be suspected that the two signals correlate. Similarly, speed and draft are likely to be correlated with the broader environmental context, since the crew may consciously adjust the vessel's speed or ballast in response to prevailing conditions and loading requirements (M. Patey, Personal Communication, 2026).

These expectations are confirmed in the data (see Table 5.3). Speed and metocean variables exhibit the highest VIFs, with STW, wave components, and water temperature all exceeding VIFs of 30. STW and draft are also somewhat explained by the remaining features. This collinearity is not itself a concern, since the goal is not to isolate the causal effect of any individual metocean variable but only to build a capable predictive model and infer the effect of fouling. All of the aforementioned features are, therefore, retained based on their known physical relevance.

__Table 5.3:__ Final list of features for the ML models.

Feature name	**Unit**	**VIF**	**Formula / explanation**
STW	knots	82.5	Longitudinal speed of the hull relative to the surrounding water mass.
Transversal wave	m	43.3	Computed as $x \sin(\theta_{\mathrm{rel}})$. Captures waves crossing the longitudinal direction of the ship.
Sea water temperature	°C	41.2	Sea temperature from the weather provider.
Longitudinal wave	m	34.7	Computed as $x \cos(\theta_{\mathrm{rel}})$, where x is wave magnitude and θ_{rel} is the ship-relative wave angle. Positive values represent head conditions.
Average draft	m	32.4	Calculated as $\frac{\mathrm{Draft}_{\mathrm{fore}}+\mathrm{Draft}_{\mathrm{aft}}}{2}$.
Transversal swell	m	31.1	Computed as $x \sin(\theta_{\mathrm{rel}})$. Captures swell crossing the longitudinal direction of the ship.
Longitudinal swell	m	24.5	Computed as $x \cos(\theta_{\mathrm{rel}})$, where x is swell magnitude and θ_{rel} is the ship-relative swell angle. Positive values represent head conditions.
Wave period	seconds	6.7	Mean wave period from the weather provider.
Transversal wind	knots	5.8	Computed as $x \sin(\theta_{\mathrm{rel}})$. Captures crosswind conditions relative to the vessel axis.
Days since last cleaning	days	5.7	Number of days elapsed since the most recent recorded cleaning event. Used as a proxy for fouling accumulation.
Longitudinal wind	knots	5.2	Computed as $x \cos(\theta_{\mathrm{rel}})$, where x is wind magnitude and θ_{rel} is the ship-relative wind angle. Positive values represent headwind conditions.
Draft trim	m	4.7	Calculated as $\mathrm{Draft}_{\mathrm{aft}} - \mathrm{Draft}_{\mathrm{fore}}$.
Longitudinal current	knots	1.9	Estimated as SOG − STW.

Although it is moderate in comparison to the other features, a VIF of 5.7 for DSC is arguably the most concerning. From the correlation matrix (Appendix I), it seems that the main reasons are correlations with Avg draft (corr = -0.65), water temperature (corr = -0.30), and current (corr = 0.25). Although removing all three does improve the VIF (5.7 vs 3.9), this is hard to defend from a theoretical standpoint, since all three are known to drive shaft power, as discussed in Section 3.2. Thus, removing any of them would almost certainly introduce an omitted variable bias that would arguably be as bad for model validity. However, this collinearity should be taken into account when interpreting the results. This is particularly true for draft, as the vessel tends to sail with higher loads immediately after cleaning events (see Figure 5.11).

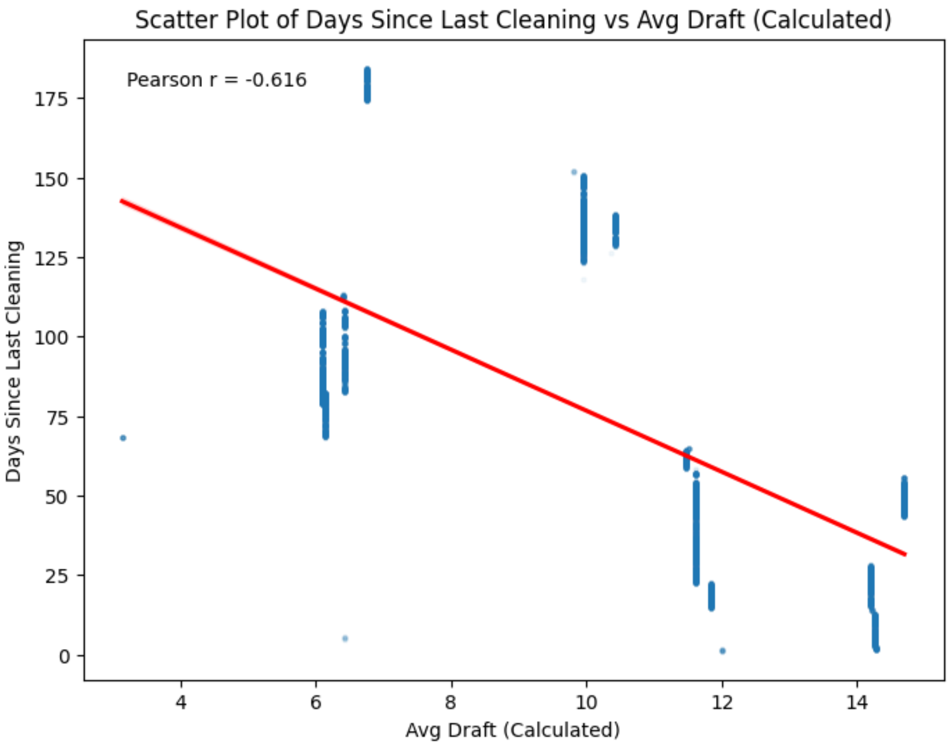


Figure 5.11: *Scatterplot between draft and DSC.*

A more rule-based feature selection strategy, for example, by iteratively removing variables and testing predictive performance, could also have been applied. Such an approach is not pursued because subsequent models will be evaluated not only by out-of-sample accuracy but also by alignment with domain knowledge. A variable may therefore be worth retaining even if its marginal predictive contribution is small, provided that it helps separate fouling from other drivers of shaft power or represents a known resistance mechanism that is theoretically relevant.

5.4 Modelling Framework

As discussed in Section 2.3, the purpose of this study is not to develop a generalizable ship performance model, but to assess the ability of ML to estimate the added power from fouling for a specific vessel.

The task is therefore treated primarily as one of inference rather than prediction: three models are formulated to predict shaft power from the features specified in Table 5.3, and the learned relationships are subsequently interpreted to infer added power from fouling. This study thus builds upon the work of Gupta et al. (2022) and Laurie et al. (2021), where shaft power is modeled as a function of the operating state of the vessel and fouling is inferred directly without the need for a separately constructed reference model.

To support the interpretation, a brief exploratory analysis is conducted to characterize the vessel's operational conditions during the observation period. The purpose of this is not to draw conclusions in advance, but to provide context on how the vessel operated in terms of speed, loading condition, and environmental exposure, and thereby assess how favorable the observed conditions were for fouling development.

5.5 Models

Three models are analyzed: a Ridge Regression, a Generalized Additive Model (GAM), and a Multilayer Perceptron (MLP). The former is included as a simple benchmark model. In contrast, the latter two are included as a comparison between an interpretable nonlinear white-box model and a more flexible black-box model. This comparison makes it possible to assess both whether a performance gap exists between the two and whether any such gain is large enough to justify reduced interpretability and the use of post-hoc XAI methods. The following three sections outline the mathematical formulation and key properties of each model.

5.5.1 Ridge Regression

As a supplement to the comparison of black- and white-box models, a ridge regression is fitted to the data as a simple performance baseline. Ridge regression is a regularized alternative to the ordinary least squares (OLS) regression, with the only difference between the two being the estimation criterion. In both OLS and Ridge, the response variable is modeled as a linear combination of the input features. Formally, this is expressed as:

$$y_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i,$$

where y_i denotes the response for observation i, and $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^\top$ denotes the corresponding feature vector. This model specification assumes that the conditional mean of the response is linear in the predictors and additive, in the sense that each predictor contributes through a separate term unless interaction or nonlinear terms are explicitly introduced. OLS estimates the intercept β_0 and coefficient vector $\boldsymbol{\beta}$ by minimizing the residual sum of squares:

$$\min_{\beta_0, \boldsymbol{\beta}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta} \right)^2 .$$

However, OLS can become unstable when predictors are noisy or strongly correlated. In such settings, this may lead to large estimated coefficients, which can hurt the model's generalization. This issue is particularly relevant in this thesis, where some features contain overlapping information. Ridge regression addresses this issue by retaining the linear structure of the model while adding a penalty on the magnitude of the coefficients (Hoerl & Kennard, 1970b). The ridge estimator is thus obtained by solving

$$\min_{\beta_0, \boldsymbol{\beta}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2,$$

where $\lambda \geq 0$ is a hyperparameter controlling the amount of shrinkage. When $\lambda = 0$, the formulation reduces to OLS, whereas larger values of λ shrink the coefficients more strongly toward zero. Importantly, ridge regression does not perform hard variable selection. Instead, it penalizes large weights leading to a model that is less sensitive to noise and collinearity (Hoerl & Kennard, 1970a, 1970b). In practice, the regularization strength λ is selected from the training data, typically by cross-validation, so that the final model balances goodness of fit against coefficient shrinkage (Géron, 2019, Section 4). Since the role of the ridge regression is solely to act as a benchmark for predictive performance, the physical interpretation of its coefficients will not be considered.

5.5.2 Generalized Additive Model

A GAM is a model in which the conditional mean of the target variable is expressed as a sum of smooth functions of the features (Hastie & Tibshirani, 1986). GAMs are often positioned as a white-box compromise between fully linear models, such as the ridge regression, and highly flexible models, such as neural networks, because they allow complex nonlinear feature effects while retaining an interpretable functional form (Chang et al., 2021; Hastie & Tibshirani, 1986). The underlying idea is somewhat similar to that of basis transformations in linear regression, where features undergo a simple transformation, such as a log or cubic, but with more flexibility to fit more complex or "wiggly" curves. A standard GAM formulation without interactions can be written as:

$$\eta(\mathbf{x}) = g(\mathbb{E}[Y \mid \mathbf{x}]) = \beta_0 + f_1(x_1) + \cdots + f_p(x_p).$$

Where Y denotes the target variable and $\mathbf{x} = (x_1, \ldots, x_p)$ denotes the vector of p features. The quantity $\eta(\mathbf{x})$ is the additive predictor, defined as an intercept, β_0, plus a sum of feature-specific contributions, $f_j(x_j)$. The function $g(\cdot)$ is the link function that maps the conditional mean $\mathbb{E}[Y \mid \mathbf{x}]$ to the additive predictor. In the Gaussian case with identity link, which will be used in this thesis, this implies that $Y \mid \mathbf{x} \sim \mathcal{N}(\mu(\mathbf{x}), \sigma^2)$, where $\mu(\mathbf{x}) = \beta_0 + \sum_{j=1}^{p} f_j(x_j)$ and σ^2 is the residual variance estimated from the training data (Hastie & Tibshirani, 1986). GAMs are estimated by representing each smooth f_j with a finite basis expansion and fitting the resulting coefficients by penalized likelihood. In the Gaussian identity-link case, this becomes penalized least squares. The amount of smoothness is controlled by one or more smoothing parameters, which are typically selected from the data using criteria such as Restricted Maximum Likelihood (REML) or the generalized cross-validation (GCV) (Wood, 2006, 2017). Consequently, the Gaussian GAM with an identity link function can be formally defined as:

$$Y = \beta_0 + f_1(x_1) + \cdots + f_p(x_p) + \varepsilon.$$

In the Gaussian identity-link formulation, this rests on the assumptions that the conditional mean of shaft power is additive in the included predictors, that observations are conditionally independent, and that residual errors have mean zero and approximately constant variance, with normality required only for likelihood-based inference. Each $f_j(x_j)$ is an unknown univariate smooth function that captures a potentially nonlinear effect of the feature x_j. Each term is represented through a basis expansion.

$$f_j(x_j) = \sum_{k=1}^{K_j} \theta_{jk}\, b_{jk}(x_j),$$

where $b_{jk}(\cdot)$ are fixed basis functions and θ_{jk} are coefficients estimated from data (Hastie & Tibshirani, 1986). If mandated by domain knowledge, shape constraints can be imposed on a specific $f_j(x_j)$, such as monotonicity (Pya & Wood, 2015).

In general, GAMs are estimated by penalized likelihood, where smoothness is enforced through a quadratic penalty on the spline coefficients (Wood, 2006). In the Gaussian case with identity link, this reduces to a penalized least-squares problem. For a GAM with only a single smooth represented using a spline basis with a difference penalty, this can be written as:

$$\min_{\boldsymbol{\theta}} \left\{ \sum_{i=1}^{n} \left(y_i - \sum_{k=1}^{K} \theta_k b_k(x_i) \right)^2 + \lambda \sum_{k=d+1}^{K} (\Delta^d \theta_k)^2 \right\},$$

where $\Delta^d \theta_k$ denotes the d-th order finite difference of adjacent spline coefficients and λ controls the amount of smoothing (Eilers & Marx, 1996).

In this formulation, each $b_{jk}(\cdot)$ is a B-spline basis function defined by a spline degree q (often cubic, $q = 3$) and a knot sequence (most commonly uniform/equidistant) spanning the domain of x. Each B-spline is a piecewise polynomial between knots, joined so that the function is smooth at the knots, and it has local support (it is non-zero only on a small interval of x). For a spline degree q, each B-spline consist of $q + 1$ polynomial pieces of degree q across $q + 2$ consecutive knots (Eilers & Marx, 1996). Consequently, at any given x_j, at most $q + 1$ B-splines are non-zero, so evaluating $f_j(x_j) = \sum_{k=1}^{K_j} \theta_{jk}\, b_{jk}(x_j)$ involves only a small local subset of coefficients (Eilers & Marx, 1996).

The number of B-spline basis functions is therefore a hyperparameter that sets the basis dimension and, therefore, the maximum flexibility of the smooth, where increasing it increases the model's capacity to fit local variation. Provided the number of splines is large enough, the smoothing parameter λ typically plays the dominant role in the bias–variance trade-off by controlling how strongly coefficient differences are penalized (Eilers & Marx, 1996).

GAMs are well-suited for this domain because they are flexible enough to model complex physical relationships while retaining interpretability. The assumption of additivity allows for the isolation of marginal effects from a single x_j by plotting only the corresponding term $f_j(x_j)$, also called the partial effect (PE) plot (Chang et al., 2021). Figure 5.12 shows example PE plots from a study on food scarcity, where each panel displays the estimated smooth contribution of a single predictor to the fitted response.

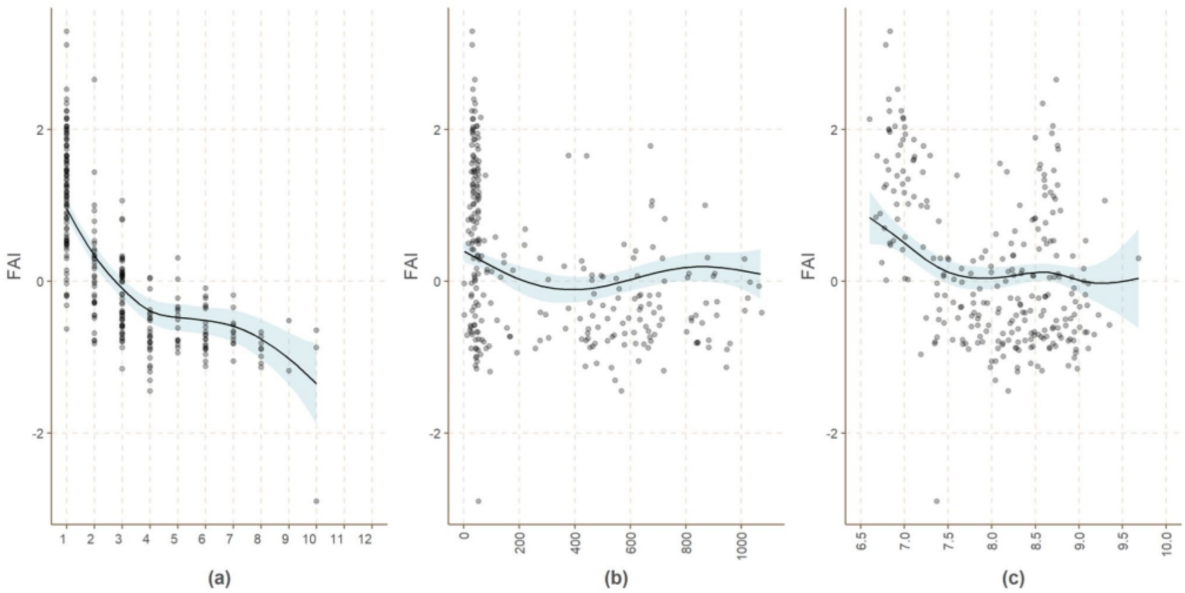


Figure 5.12: *Example PE plots from a research paper on food access. The curves represent the effects of road proximity, precipitation, and cereal prices (x-axes) on households' food access in Ethiopia (y-axes). Source: Campomanes et al. (2024)*

The interpretation of GAMs does come with some caveats, two of which are particularly relevant in this domain. Firstly, differently specified GAMs with similar accuracies can imply different, and sometimes contradicting, relationships for specific features (Chang et al., 2021). The evaluation of GAMs should therefore always include a qualitative assessment of the PEs and whether the relationships behave as expected. Secondly, GAMs tend to make overly confident predictions in sparse data regions, and one should therefore always take data coverage into account when interpreting near the boundaries of the data space (Chang et al., 2021).

It should be noted that GAMs can model interactions using tensor products. However, including them means the shape of the PE curve is not global. Since this global curve is a cornerstone of the analysis, tensor products are intentionally omitted.

5.5.3 Multi-Layer Perceptron

An MLP is included as a flexible nonlinear regression model capable of representing complex relationships thata purely linear baseline may insufficiently capture, such as ridge regression, or strictly additive models, such as the purely additive GAM described in the previous section (Géron, 2019, Section 10). MLPs are commonly treated as black-box models because their internal mapping from inputs to predictions is generally not directly interpretable for humans without additional post-hoc analysis (Arrieta et al., 2020). Among the many possible architectures, the relatively simple MLP is chosen because the goal is to study model behavior rather than to maximize predictive performance.

An MLP is a type of feed-forward neural network, meaning that it transforms inputs into predictions by passing them through a sequence of layers. Each layer takes the outputs from the previous layer as input and transforms them using a set of learned weights and biases (Géron, 2019, Section 10). A layer consists of multiple units or "neurons", each of which produces an output by multiplying all its inputs by a weight and adding a bias, before passing the result through a nonlinear activation function. In the simplest dense layer, this transformation can thus be formally defined as $z = Wx + b$, where x is the input vector of features, W is a weight matrix, and b is a bias vector, after which the nonlinear activation function is applied as a transformation of the output. In a dense, or fully connected, layer, each unit receives input from all units in the preceding layer (Géron, 2019, Section 10). Figure 5.13 illustrates this general structure and the transformations carried out in each layer.

When specifying the architecture of an MLP, four important design choices must be made. First, one must choose the number of hidden layers, which determines how many successive transformations the input undergoes before reaching the output. Second, one must choose the activation function used in the hidden layers (see Appendix L for common functions). Third, one must choose the size (i.e., the number of neurons) of each hidden layer. Finally, one must choose how the units are connected across layers, where the standard MLP uses dense, or fully connected, layers in which each unit receives input from every unit in the preceding layer (Géron, 2019, Section 10).

The weights and biases in the network are not obtained from a closed-form expression, as in ordinary least squares. Still, they are instead learned through iterative minimization of a regression loss, most commonly squared error (Géron, 2019, Section 11). If θ denotes the full collection of all weights, and $\hat{y}_i(\theta)$ denotes the prediction for observation i, the objective is to minimize the loss function, most commonly the squared error plus L2 regularization. Formally, the loss is thus given by:

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \big(y_i - \hat{y}_i(\theta)\big)^2 + \alpha \|\theta\|_2^2.$$

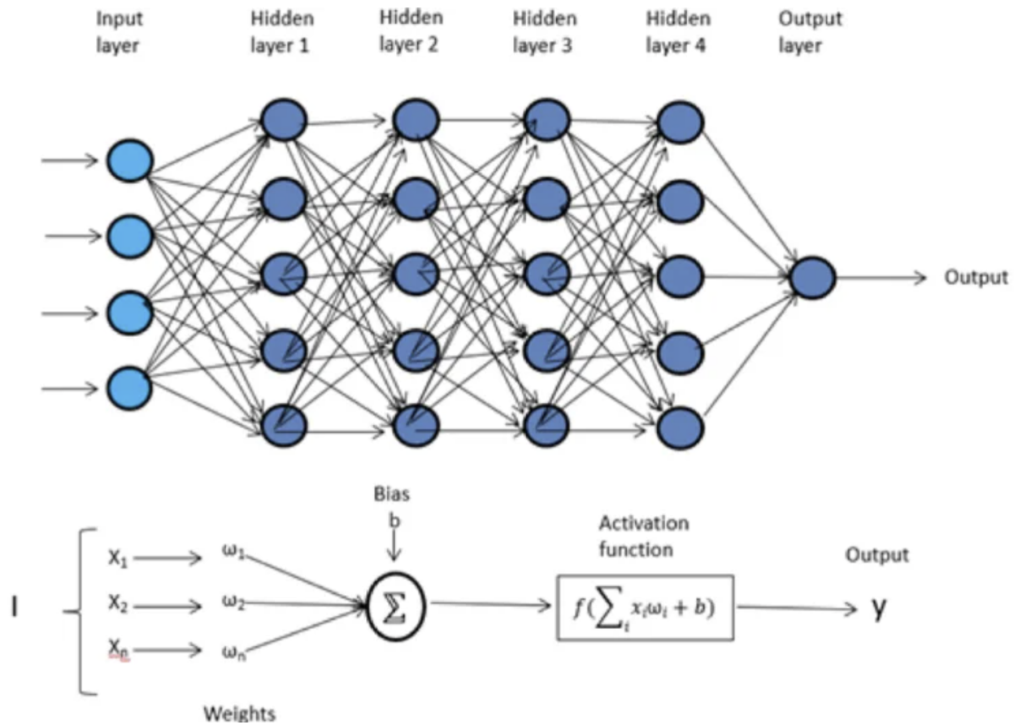


Figure 5.13: *Generic example of an MLP Architecture. The upper half shows the overall architecture with connections between neurons, while the lower half shows a single neuron. Source: Gill et al. (2022).*

α is an optional regularizing parameter that penalizes large weights, which mitigates the tendency of the MLP to overfit to noise in the training data.

The training is carried out by initializing θ at arbitrary values and repeatedly adjusting them in the direction that reduces the loss. This adjustment is done by computing the gradient of the loss function with respect to the parameters, $\nabla_\theta \mathcal{L}(\theta)$, which indicates how the loss changes for small changes in each weight and bias. The parameters are then updated iteratively until the loss no longer improves materially, or until a stopping criterion is reached.

A famous and foundational algorithm for updating the weights is Stochastic Gradient Descent (SGD). In SGD, the update at iteration t is based on the gradient evaluated on a single observation or a small mini-batch rather than the full dataset. Writing $g_t = \nabla_\theta \mathcal{L}_t(\theta_t)$ for the gradient based on the current batch, the parameter update is

$$\theta_{t+1} = \theta_t - \eta g_t,$$

where $\eta > 0$ is the learning rate. The learning rate controls the step size of each update: if it is too large, training may become unstable, while if it is too small, learning may be unnecessarily slow.

SGD suffers from two main weaknesses. First, it is highly sensitive to the choice of η. Second, SGD uses the same global step size for all parameters, even though different parameters may have gradients of very different magnitudes and therefore require different update sizes. For these reasons, Adaptive

Moment Estimation (Adam) has become a popular alternative in practice (Géron, 2019, Section 11). Adam addresses the first weakness by replacing the raw current gradient with a smoothed average of recent gradients, which makes the update direction less noisy. It addresses the second weakness by maintaining a separate running average of squared gradients, allowing the algorithm to scale the update size independently for each parameter. Since these moving averages are initialized at zero, Adam also applies a bias correction in the early iterations to avoid systematically underestimated updates. The resulting update rule is therefore typically more stable, less sensitive to gradient scale, and less dependent on careful tuning than plain SGD (Géron, 2019, Section 11).

As with the architectural specification, training an MLP also involves several hyperparameter choices. The learning rate determines the magnitude of each parameter update. It therefore affects both the speed and stability of convergence, while α controls the degree to which large weights are penalized, thereby helping to limit overfitting. Batch size determines how many observations are used in each parameter update. In addition, one must decide how long training is allowed to continue, most directly through the maximum number of iterations or epochs, but often also through early stopping, whereby training is terminated when performance no longer improves sufficiently (Géron, 2019, Section 11). When early stopping is used, a validation fraction must be set aside from the training data to monitor out-of-sample performance during training, and the stopping decision is then governed by additional criteria, such as the improvement tolerance and the number of consecutive iterations without meaningful improvement allowed before training is stopped.

5.6 Training Strategy

The dataset is divided into a training set and a test set using an 80/20 random split. A chronological split is deliberately avoided, as it would prevent the models from observing the full range of voyage types, loading conditions, and environmental exposures present in the data. Random splitting implicitly assumes independence among observations, which is admittedly a simplification given the data's temporal structure. However, this is considered acceptable because the primary goal is inference rather than out-of-sample forecasting, and the risk of temporal leakage is limited given that the fouling proxy is the only explicitly time-dependent feature in the model. All features are standardized before model fitting.

The regularization parameter λ of ridge regression is selected via a grid search over $\{0.01, 0.1, 1, 10, 100\}$ using K-fold cross-validation on the training set, with RMSE as the selection criterion (Géron, 2019, Section 14). For the GAM, a single global smoothing penalty λ is tuned over a logarithmically spaced grid of 20 values in the range $[10^{-6}, 10^{3}]$ using grid search, which selects the penalty that minimizes GCV score (Servén & Brummitt, 2018). Note that a single shared λ is applied across all spline terms rather than a separate penalty per term.

For the MLP, hyperparameters are determined through manual heuristic tuning. While not as rigorous as a formal grid search over the hyperparameters, this is deemed acceptable since maximizing predictive performance is not the central goal of the analysis. Once the best configuration for each model has been identified, the predictive performance of the three models is compared on the test set.

5.7 Evaluation

Since simply minimizing the predictive error is not guaranteed to yield a model suitable for inference, the models will be evaluated based on both *formalized* criteria, which are strictly mathematically defined, and *unformalized* criteria, which are qualitative assessments grounded in domain knowledge (Doshi-Velez & Kim, 2017; Shmueli, 2010).

The main formalized criterion is predictive performance, which is assessed using three metrics: root mean squared error (RMSE), mean absolute percentage error (MAPE), and mean absolute error (MAE), defined as

$$\begin{aligned} \mathrm{RMSE} &= \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \\ \mathrm{MAPE} &= \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|, \\ \mathrm{MAE} &= \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right|, \end{aligned}$$

where n is the number of observations, y_i is the observed shaft power for observation i, and $\hat{y}_i$ is the corresponding model prediction. RMSE serves as the primary criterion for model comparison because it penalizes large errors more heavily than MAE and is expressed in the same units as the target variable. MAPE and MAE are reported alongside RMSE to provide a more interpretable characterization of average error magnitude. Together, these metrics are used to assess whether a meaningful performance gap exists between the GAM and the MLP, and whether, if present, that gap is large enough to justify the additional complexity and reduced direct interpretability of the latter.

Beyond overall predictive performance, each model is estimated twice: once with the DSC feature and once without. Following the approach of DeKeyser et al. (2022) and Laurie et al. (2021), the difference in predictive performance between the two versions is used here to evaluate whether the fouling proxy carries meaningful information. If including the fouling proxy yields a significant improvement in predictive performance, this is considered evidence that the proxy captures a real signal beyond what speed, loading condition, and environmental variables can account for. Whether the difference is statistically significant will be determined by the Wilcoxon signed-rank test (elaborated in Section

5.7.1).

The unformalized evaluation criteria concern whether the fouling-related effects learned by each model are physically plausible and align with domain knowledge. Concretely, this means assessing whether the estimated added power associated with the DSC feature increases over time in a direction, shape, and approximate magnitude that is consistent with what is known about fouling accumulation and its effect on hull resistance. For the GAM, this is assessed directly through PE curves, which isolate the marginal contribution of each feature to the predicted shaft power as described in Section 5.5.2. For the MLP, the same assessment is conducted using ALE curves. This model-agnostic post-hoc method estimates how the model's predictions change on average as a given feature varies, while accounting for feature correlations (Apley & Zhu, 2020). Section 5.7.2 explains the ALE methodology in more detail. In both cases, an absolute added power can be inferred by comparing the power predictions at DSC=0 (or earliest available) after cleaning with those at later stages. To assess the learned added power across the speed-power curve, both models will also undergo a controlled-variable experiment, as performed by Duan et al. (2024) and Laurie et al. (2021). This approach is elaborated in Section 5.7.3.

5.7.1 Wilcoxon Signed Rank Test

To assess whether including the DSC feature improves predictive performance, a one-sided Wilcoxon signed-rank test is performed on the paired prediction errors from models with and without DSC as a feature. Because both models are evaluated on the same test observations, the comparison is treated as paired, and it is examined whether the observed error reductions are sufficiently systematic to be unlikely under the null hypothesis of no improvement (Wilcoxon, 1945).

For each observation $i = 1, \ldots, n$, the prediction error is defined as the absolute error,

$$e_{i,\text{base}} = \left|y_i - \hat{y}_{i,\text{base}}\right|, \qquad e_{i,\text{DSC}} = \left|y_i - \hat{y}_{i,\text{DSC}}\right|,$$

where y_i denotes the observed shaft power, $\hat{y}_{i,\text{base}}$ denotes the prediction from the model without DSC, and $\hat{y}_{i,\text{DSC}}$ denotes the prediction from the model including DSC. The paired difference is then defined as

$$d_i = e_{i,\text{base}} - e_{i,\text{DSC}},$$

such that $d_i > 0$ indicates lower error for the model including DSC. The hypotheses are formulated as

$$H_0: \text{ the distribution of the paired differences is centered at 0,}$$

against the one-sided alternative

$$H_1: \text{ the paired differences are greater than } 0,$$

which corresponds to lower prediction errors for the model including DSC (Wilcoxon, 1945).

To compute the test statistic, all cases with $d_i = 0$ are omitted, and the remaining absolute differences $|d_i|$ are ranked from smallest to largest. The original sign of each difference is then attached to its rank. For the one-sided alternative considered here, the test statistic is defined as the sum of the ranks associated with positive differences,

$$W^+ = \sum_{i:\, d_i > 0} R_i,$$

where R_i denotes the rank of $|d_i|$. A large value of W^+ indicates that the model including DSC tends to produce lower paired errors than the baseline model. Under the null hypothesis, the positive-rank sum follows a known reference distribution, which allows the p-value to be calculated. Under H_0, the signed-rank statistic follows a known null distribution, which allows the one-sided p-value to be calculated as

$$p = P(W \geq w_{\text{obs}} \mid H_0).$$

This is the probability of observing a statistic at least as favorable as w_{obs} if the paired differences are symmetric about zero. For small samples, the p-value is obtained from the exact null distribution of W, whereas for larger samples (such as this case) it is based on an asymptotic standard normal approximation. At the $\alpha = 0.05$ level, H_0 is rejected when $p < 0.05$, indicating that inclusion of DSC yields systematically lower absolute prediction errors.

5.7.2 Accumulated Local Effects

When the main goal of a model is explanation, a natural goal is to construct a reliable average approximation of how a single feature X_j influences the model output $f(\mathbf{x})$. In black-box models, common approaches include PD plots (Friedman, 2001), which marginalize over the distribution of the remaining features, and marginal plots, which use local conditional averages but can mix the effects of correlated predictors (Apley & Zhu, 2020). The standard remedy is to remove correlated features before analysis, but this is not feasible in this context, since removing vessel speed or DSC would directly undermine the analysis. As an alternative, Apley and Zhu (2020) propose ALE plots, which estimate feature effects by averaging small, local changes in the model output within narrow intervals of X_j, thereby restricting all model evaluations to regions of the predictor space that are well-supported by data.

More formally, let $f(\mathbf{x})$ denote the fitted prediction function, and let $\mathbf{x}_{-j}$ denote all predictors except x_j. Assuming f is differentiable in x_j, the uncentered ALE main effect for feature X_j can be written as

$$g_{j,\mathrm{ALE}}(x_j) = \int_{x_{\min,j}}^{x_j} \mathbb{E}\left[\frac{\partial f(z, \mathbf{X}_{-j})}{\partial z} \mid X_j = z\right] dz,$$

where $x_{\min,j}$ is a lower bound of the support of X_j (Apley & Zhu, 2020) and z is dummy integration variable representing a specific value of X_j at which the local sensitivity is evaluated. The quantity inside the integral can thus be interpreted as the average local sensitivity of the prediction to X_j at z, conditional on the data actually observed near z. The plotted ALE curve is then centered to have a mean of zero,

$$f_{j,\mathrm{ALE}}(x_j) = g_{j,\mathrm{ALE}}(x_j) - \mathbb{E}\left[g_{j,\mathrm{ALE}}(X_j)\right],$$

so that it is interpreted relative to the average prediction level rather than as an absolute value (Apley & Zhu, 2020).

In practice, ALE is estimated using finite differences over small intervals rather than a smooth curve. For a given feature, the observed range is partitioned into K intervals, typically based on empirical quantiles. This gives narrower bins in dense regions of the feature space and wider bins where data are sparse, which helps maintain local support (Apley & Zhu, 2020). The choice of binning introduces a bias-variance tradeoff: too few observations per bin lead to unstable local effect estimates, whereas overly wide bins may smooth away important local structure. In practice, this is therefore controlled through the number of bins, or equivalently, the minimum amount of data required within each interval (Apley & Zhu, 2020).

For each interval $[x_{i,\min}, x_{i,\max}]$, the subset of observations with feature values inside that interval is then identified. For every such observation, the feature of interest is replaced once by $x_{i,\min}$ and once by $x_{i,\max}$, while all remaining features are left at their observed values. The local effect in interval i is then estimated as the average prediction difference

$$\widehat{\mathrm{effect}}_i = \frac{1}{|S_i|} \sum_{j \in S_i} \left[f(x_j^+) - f(x_j^-)\right],$$

where S_i denotes the observations falling in interval i, and x_j^+ and x_j^- denote the same observation with the feature replaced by the upper and lower interval boundary, respectively (Apley & Zhu, 2020). The ALE curve is then obtained by accumulating these local effects from the lowest interval upwards, such that the ALE value at the upper edge of an interval equals the ALE value at the lower edge plus the estimated local effect for that interval. Finally, the accumulated values are centered by subtracting their mean, so that the resulting curve is interpreted relative to the average prediction level rather than as an absolute prediction.

The practical output of the ALE procedure is a sequence of accumulated effect values across the range of a feature, which can be visualized as a curve (see Figure 5.14 for example). After centering, the vertical axis is interpreted relative to the average model prediction, so positive values indicate that the feature is associated with a higher-than-average predicted response. In comparison, negative values indicate a lower predicted response.

For example, if an ALE curve for DSC is approximately −100 kW at day 0, this means that observations near this value are associated with a predicted shaft power about 100 kW below the average prediction. If the ALE value is around 400 kW at day 150, this indicates that the model has learned the vessel uses approximately 500 kW more power after not being cleaned for 150 days. It is important to note that such statements should be interpreted as the models *learned* effect, rather than as direct causal effects (Apley & Zhu, 2020).

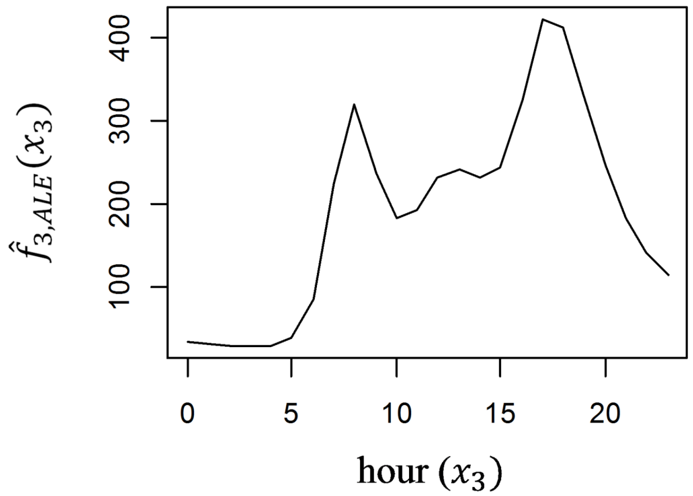


__Figure 5.14:__ Example of an ALE curve applied to a dataset on bike rentals. The curve shows the effect of the hour of day on predicted bike rentals in Washington, D.C. from a neural network model. The curve shows two clear peaks, corresponding to morning and evening rush hours. Source: Apley and Zhu (2020).

5.7.3 Controlled Variable Experiment

To examine how added power varies with speed under different fouling conditions, both models are used to generate predicted power values across the vessel's normal operating speed range of 9–14 knots. For each set of predictions, all remaining input variables are held constant at their median, except draft, which is varied across ballast, design, and scantling drafts. A cubic function is then fitted to the predicted values and treated as an implied speed–power curve, which can be repeated for different levels of fouling. At a given speed, the difference between the estimated speed-power curve on day 0 and on that day can be interpreted as the added power for that specific speed and day since cleaning. This allows for a more granular analysis of the fouling penalty at different speeds, rather than just a point estimate.

5.8 Software & Implementation

The analysis is implemented in Python 3.12.10 using pandas for data cleaning and formatting, matplotlib and seaborn for visualization, scikit-learn for ridge regression and the MLP model, and pyGAM for the GAM model. Code was executed either on a Lenovo Yoga PC with an Intel Core Ultra 7 processor and 32 GB of RAM, or, when additional computational resources were needed, on a virtual machine with 16 vCPUs based on Intel Xeon Gold 6130 processors and 96 GB of RAM, provided by Ucloud through CBS.

6 Results

The following five sections present the results of the analysis. Section 6.1 provides exploratory insights into the dataset, including whether the vessel encountered conditions favorable for fouling development. Section 6.2 presents the final model specifications and compares their predictive performance. Sections 6.3 and 6.4 report the fouling penalties estimated by the models and use their predictions to estimate speed-power curves for different levels of fouling. Section 6.5 examines residual diagnostics, while Section 6.6 assesses whether the main findings are robust to selected changes in the GAM specification and preprocessing choices.

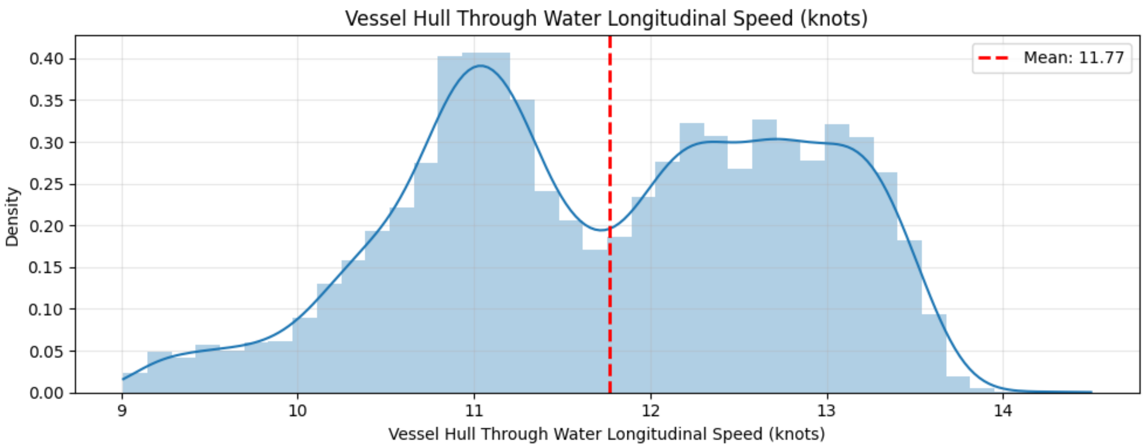
6.1 Exploratory Data Analysis

Table 6.1 shows descriptive statistics for shaft power and all features. Regarding speed, the vessel operated at a mean STW of 11.8 knots, with a standard deviation of 1.08 knots. The bimodal shape of the distribution (see Figure 6.1a) suggests two distinct operating modes, with modes around 11 knots and 13.3 knots. Compared to speed, the shaft power distribution is noticeably less smooth. This could suggest that the vessel operated at fixed power consumptions for extended periods (see Figure 6.1b), which is common for a vessel operating at autopilot at open sea. This is significantly lower than the speeds observed during sea trials, where the vessel was tested from 11.5 to 18 knots (Appendix C). This discrepancy is expected, as sea trials are conducted at or near full engine load. In contrast, commercial operation of a bulk carrier in this size range typically involves "slow-steaming", i.e., operating at 60-80% of design speed to reduce fuel costs (ISO, 2016). These speeds are relatively slow compared to what has been tested in previous literature on fouling estimation (mostly above 15 knots), meaning that the added power from fouling, expressed as a percentage, should be slightly higher (Schultz, 2007; Schultz et al., 2011; Townsin, 2003).

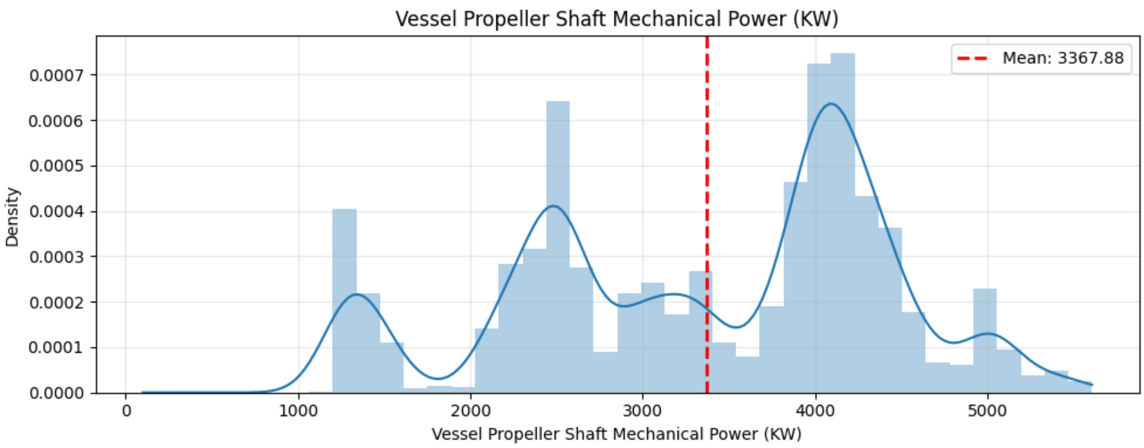
Table 6.1: *Descriptive statistics of the modelling dataset after preprocessing and filtering to open-water sailing conditions. All environmental force variables are calculated from provider weather data as described in Section 3.2.*

Variable	n	**Mean**	**Std**	**Min**	**25%**	**50%**	**75%**	**Max**
Wave period (s)	15,061	3.91	1.60	0.72	2.59	3.84	5.08	9.96
Long. wave force (m height)	15,061	0.28	1.43	-5.13	-0.71	0.51	1.37	3.92
Long. wind force (m/s)	15,061	1.45	5.25	-18.97	-1.94	2.03	5.47	14.61
Long. swell force (m height)	15,061	0.22	1.17	-3.74	-0.56	0.37	1.01	3.40
Trans. swell force (m height)	15,061	0.02	1.23	-4.26	-0.89	-0.03	0.94	3.00
Trans. wind force (m/s)	15,061	0.32	5.13	-12.27	-3.45	-0.07	4.25	16.87
Trans. wave force (m height)	15,061	0.05	1.49	-4.26	-1.13	-0.04	1.32	4.17
Avg. draft (m)	15,061	9.94	3.08	3.15	6.42	10.43	11.62	14.70
SOG − STW (kn)	15,061	-0.45	0.68	-4.09	-0.81	-0.37	-0.03	4.04
Days since last cleaning	15,061	77.3	48.1	1.3	35.0	78.8	107.5	184.4
STW (kn)	15,061	11.77	1.08	9.01	10.94	11.80	12.70	14.50
Shaft power (kW)	15,061	3,367.88	1,067	1000	2,496	3,704	4,157	5,606

As shown in Table 6.1 and the histograms in Appendix J, the vessel experienced mildly adverse envi-



(a) *STW*



(b) *Shaft Power*

Figure 6.1: *Histograms for speed and shaft power ($n = 15{,}061$).*

ronmental conditions on average, which is typical for bulk carriers operating on major deep-sea trade routes where extreme weather events are deliberately avoided. The longitudinal force components show small positive means, indicating a slight tendency toward head conditions during the observation period. Concretely, the mean force components observed are equivalent to the vessel facing a 0.28m wind wave, a 0.22m swell wave, and 1.45 m/s wind gusts directly opposing its heading direction on average. The longitudinal wind force shows the largest spread (std 5.25 m/s, range −18.97 to 14.61), while wave and swell forces are more tightly distributed with standard deviations below 1.5 m.

Distributional coverage is generally good for all variables (see Appendix J for histograms). The three exceptions are DSC, trim and draft, all of which have notable gaps that require closer attention. DSC shows noticeable gaps in its distribution (see Figure 6.2), which is likely a consequence of the filtration of port stays and low-speed operations. As expected, this means that the model's behavior at certain values, especially beyond 150 days, is supported by relatively few training observations, which should be taken into account when interpreting.

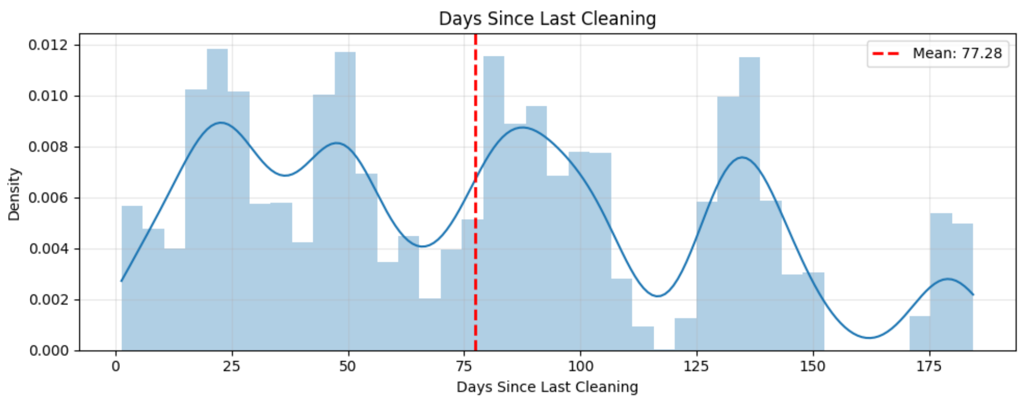


Figure 6.2: *Distribution of DSC ($n = 15{,}061$).*

The loading condition of the vessel (draft and trim) is effectively discrete, with the dataset only containing 24 unique combinations of the two (see Figure 6.3). This is because the draft rarely changes during a voyage. Apart from 82 observations at scantling draft, it never operated under its exact standard loading conditions. Therefore, the loading condition most frequently observed within a ±1m average draft and ±1m trim of each standard condition is chosen as a proxy for further analysis (see Appendix N for exact values). This avoids forcing the model to extrapolate to predict loading conditions not present in the data when analyzing different loading scenarios. Since it never operated near the heavy ballast condition, that condition is disregarded in the following analyses. Although the draft feature is retained as a numeric input rather than bucketed, any interpretation should respect its effectively discrete support, since treating it as a continuous variable would force the model to extrapolate between operating conditions not present in the training data.

Figure 6.4 indicates that the vessel operated in conditions favorable to fouling. The water temperature

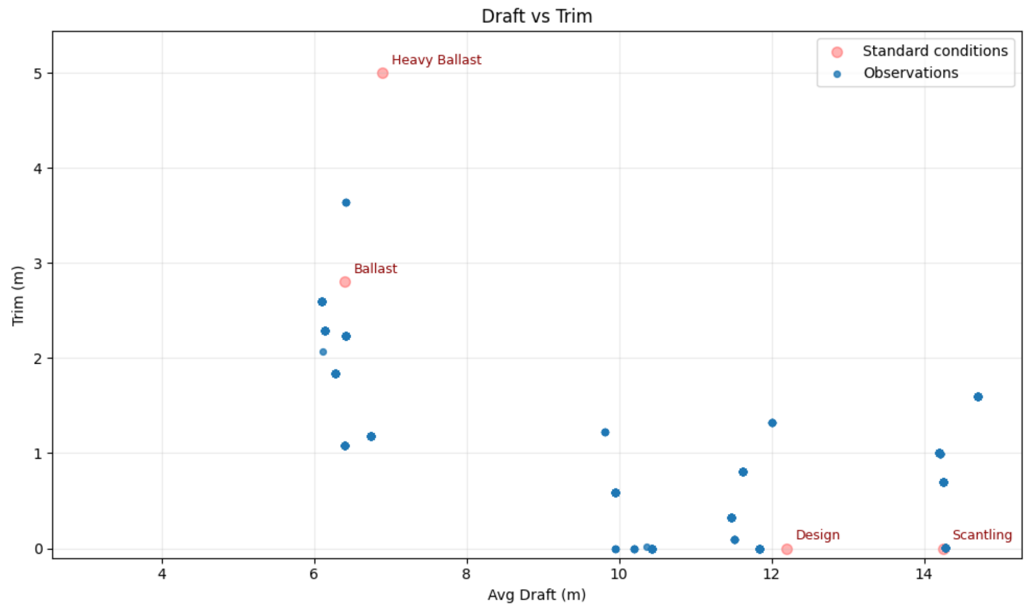


Figure 6.3: *Observed Loading Conditions. Blue dots indicate observed loading conditions, while red dots represent the standard loading conditions of the vessel. ($n = 15{,}061$).*

was often above 20,°C and a significant share of the observations occurred below 4 knots. 5.1% of the observations, corresponding to approximately 18 days, satisfy both these constraints, and therefore considerable amounts of fouling should be expected throughout the year (Schultz, 2007; Townsin, 2003).

The most prominent cluster around 9-14 knots and 20-30 °C likely reflects the vessel's predominant open-water sailing conditions, while the high-density regions near zero speed across all temperatures likely correspond to port stays. Several smaller dark clusters are visible at very low speeds, which is where the conditions for fouling are optimal. The darkest hexagons each represent approximately 10,000 observations of 15 seconds, or approximately 1.7 days, which in itself is sufficient for initial biofilm formation (Schultz, 2007; Townsin, 2003).

6.2 Model Comparison & Performances

This section presents the final specifications of the models and compares their predictive performance on the test set. It also evaluates whether the GAM and MLP yield meaningful gains over ridge regression, whether DSC improves predictive accuracy, and whether the models exhibit signs of overfitting.

Ridge regression is included as a simple linear benchmark, with its role limited to predictive performance rather than interpretation of individual coefficients. The regularization parameter λ is tuned using grid search over $\lambda \in \{0.01, 0.1, 1, 10, 100\}$, optimizing RMSE. The resulting final model is fitted with $\lambda = 1$.

The GAM is fitted with an identity link function, assuming normally distributed residuals. Each term

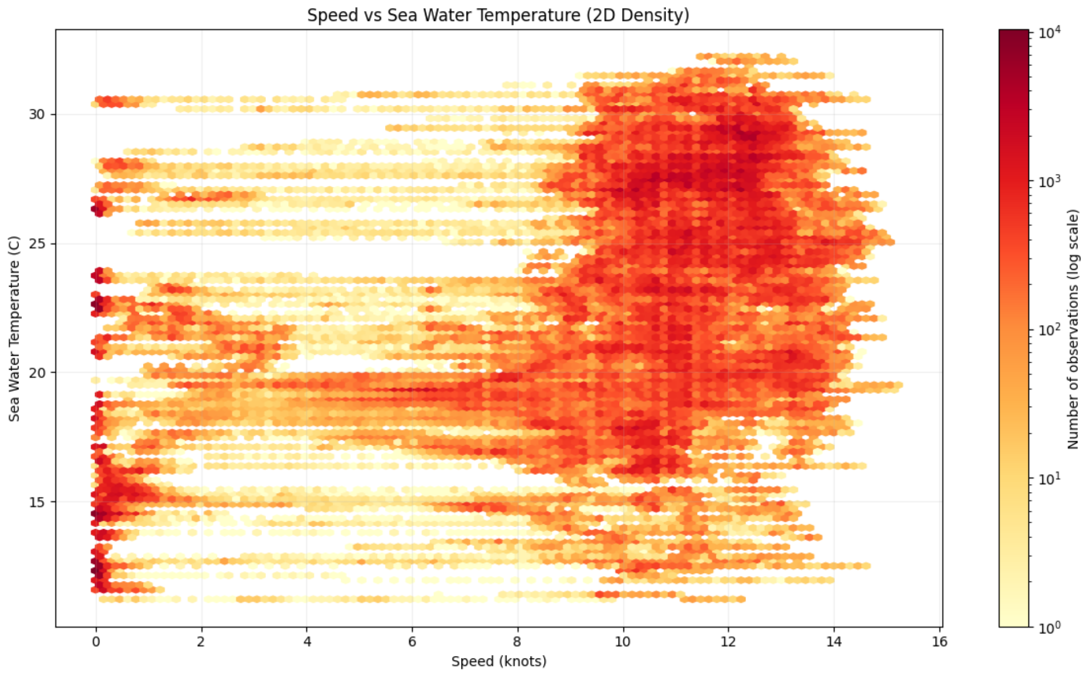


Figure 6.4: *Heatmap of the vessel's speed and surrounding sea water temperature across the full observation period, based on unaggregated data and before filtering for speed. Each hexagonal bin represents the number of observations recorded at the corresponding combination of speed and seawater temperature, with darker colors indicating more observations (log scale).*

is represented by 10 cubic B-splines with uniformly spaced knots, which is generally a sufficiently flexible choice when the underlying relationships are expected to be relatively smooth and physically bounded (Hastie & Tibshirani, 1986). Using a grid search over a 15-point logarithmically spaced grid of smoothing parameters, $\lambda \in [10^{-8}, 10^{2}]$, a global smoothing parameter of $\lambda = 5.1795 \times 10^{-8}$ is identified as optimal and applied to all terms. The applied constraints and effective degrees of freedom are summarized in Table 6.2. To keep the model as interpretable as possible, no tensor products are included.

Table 6.2: *GAM feature constraints and effective degrees of freedom (EDoF). EDoF is the effective complexity used by each smooth term after penalization, with larger values indicating a more flexible fitted term.*

Feature name	Constraint	EDoF
Avg Draft	Monotonic Increasing	8.9
Days Since Last Cleaning	Monotonic Increasing	3.0
STW	Monotonic Increasing	6.7
Sea Water Temperature	Monotonic Decreasing	3.1
Wave Period	None	8.6
Current	None	9.0
Draft Trim	None	4.5
Longitudinal Wave Force	Monotonic Increasing	5.3
Longitudinal Wind Force	Monotonic Increasing	4.8
Longitudinal Swell Force	Monotonic Increasing	2.1
Transversal Wave Force	None	8.1
Transversal Wind Force	None	8.1
Transversal Swell Force	None	7.8

The final MLP architecture is selected through a manual heuristic search. It consists of two hidden layers with 128 and 64 neurons, respectively, using Rectified Linear Unit (ReLU) activation functions and a single continuous output neuron that predicts shaft power. The model is trained using a 10% internal validation split, Adam optimizer with a learning rate of 0.001, $\alpha = 0.01$, and mean squared error as the loss function. To prevent overfitting, early stopping is applied with a maximum of 1000 training epochs and a patience of 5 epochs. Figure 6.5 visualizes the architecture of the model with DSC included. Note that some weights are reduced to nearly 0, presumably due to the relatively high value of α.

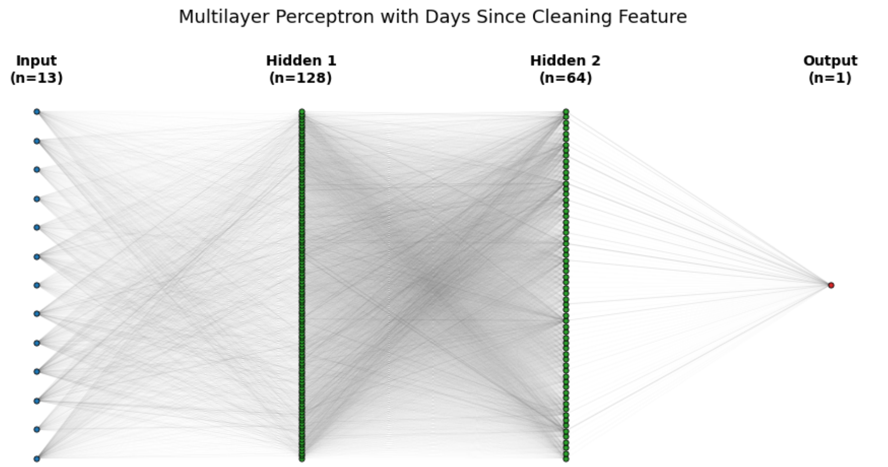


Figure 6.5: *Architecture of the MLP including DSC. Each blue dot represents an input variable, and each green dot represents a neuron. The thickness of lines indicates the absolute value of weights. See Appendix O for the architecture of MLP excluding DSC and Appendix P for training curves.*

Table 6.3 summarizes the performance metrics of the fitted models. MAPEs reach single digits for both the nonlinear models, which is relatively good compared to traditional empirical methods, which often lie around 5% (Laurie et al., 2021; Lu et al., 2015). Both nonlinear models outperform the ridge benchmark by a substantial margin. Ridge regression yields test RMSEs of 458.2 and 473.3 (including and excluding DSC), whereas the GAM reduces these to 286.5 and 297.7, respectively. The MLP performs best overall, with test RMSEs of 138.3 and 150.8. The only models showing minor signs of overfitting are the MLPs, which show train/test gaps of 5.6% and 2.6% with and without fouling, respectively. This gap is tolerated because the training curves (Appendix P) show that the validation loss continuously improved throughout training, and early stopping with a relatively low patience of 5 prevented further training once the validation loss stopped improving.

The inclusion of the DSC feature improves the performance in all three models. The gain is most pronounced for the MLP, while the GAM also benefits, albeit to a lesser extent. In all three model classes, the one-sided Wilcoxon signed-rank test rejects the null hypothesis of no improvement. It is

Table 6.3: *Training and test performance metrics for models with and without fouling features. Train-test gap is the difference between RMSE on the training and test sets, with positive values indicating worse performance on the test set. Lower is better for all metrics.*

Model	DSC	Train RMSE (kW)	Test RMSE (kW)	Train–Test Gap (%)	Train MAPE (%)	Test MAPE (%)	Train MAE (kW)	Test MAE (kW)	Wilcoxon p-value
Ridge	Incl.	455.2	458.2	0.7	13.46	13.47	342.6	345.9	$p < 0.0001$
Ridge	Excl.	472.7	473.3	0.1	13.94	13.95	360.1	362.2	
GAM	Incl.	286.2	286.5	0.1	7.48	7.34	212.2	213.6	$p < 0.0001$
GAM	Excl.	295.6	297.7	0.7	7.49	7.42	216.5	219.1	
MLP	Incl.	130.9	138.3	5.6	3.15	3.16	92.5	96.6	$p < 0.0001$
MLP	Excl.	147.0	150.8	2.6	3.59	3.45	103.9	105.4	

therefore concluded that the inclusion of DSC leads to a statistically significant reduction in paired absolute prediction errors. Together with the lower values of RMSE, MAPE, and MAE, this suggests that including the DSC figure improves the predictive performance of the models.

6.3 Estimated Fouling Effects

Figure 6.6 shows the PE curve for DSC in the fitted GAM. It suggests limited fouling during the first 2.5 months after cleaning, followed by a more pronounced increase. Assuming the vessel's state on day 2 is representative of an unfouled condition, the estimated added power attributable to DSC reaches 327.8 kW at DSC=150 and 1224.1 kW at DSC=180. Relative to the average shaft power in the dataset (3368 kW), this corresponds to approximately 9.7% and 36.3% higher power demand at DSC=150 and DSC=180, respectively. Since the observations are to some extent dependent and the QQ plots show signs of non-normal residuals (see Appendix U), the confidence intervals are likely deceivingly narrow and are therefore disregarded.

Figure 6.7 presents the ALE curve for DSC based on the predictions of the MLP on the training set. It suggests a pattern similar to the GAM, with only limited fouling development during the first 2.5 months after cleaning, followed by a relatively steep increase. Assuming that the state of the vessel at DSC=7 is representative of an unfouled condition, the estimated increase in shaft power is 539.3 kW at DSC=150 (16.0% increase relative to average power) and 1065.6 kW at DSC=179 (31.6% increase relative to average power). It is worth noting that, although the sharp increase in the last 30 days persists, the difference between the DSC=150 day and DSC=180 points is significantly smaller than what is implied by the GAM (525.7 kW vs 896.3 kW).

While the GAM is forced to fit the same additive curve across speeds, ALE curves can be made separately for different subsets of the data. Partitioning by speed allows inspection of whether the learned patterns of the MLP meet the expectation that the absolute added power increases while the relative added power decreases with speed for a given level of fouling. Splitting the dataset in half at the median STW (11.8 kn) and computing separate ALEs for each half yields the curves shown in Figure 6.8. The high-speed curve tells a similar story to the overall curves, with limited fouling for the first 2.5 months, followed by a relatively steep increase. However, the low-speed curve shows a

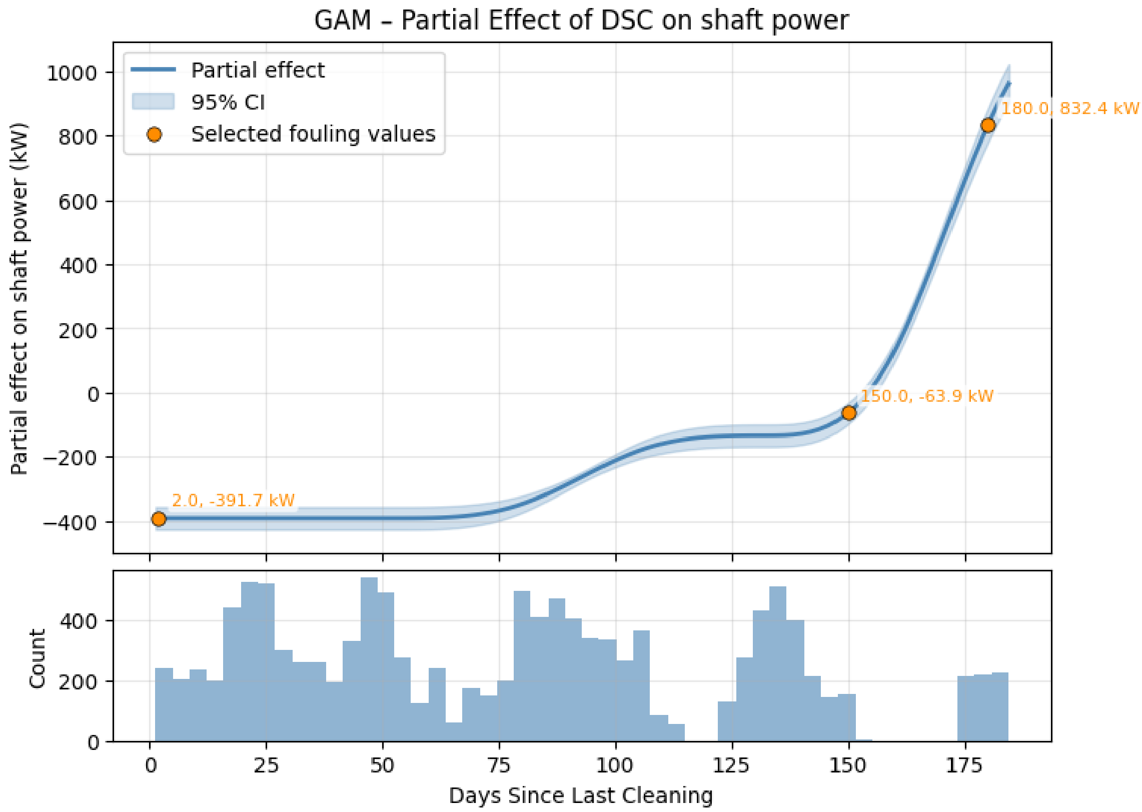


Figure 6.6: *PE curve of DSC impact on predicted shaft power for the fitted GAM. Orange dots show DSC's impact on the predicted shaft power at selected values with all other variables held constant. The histogram indicates the data support along the curve.*

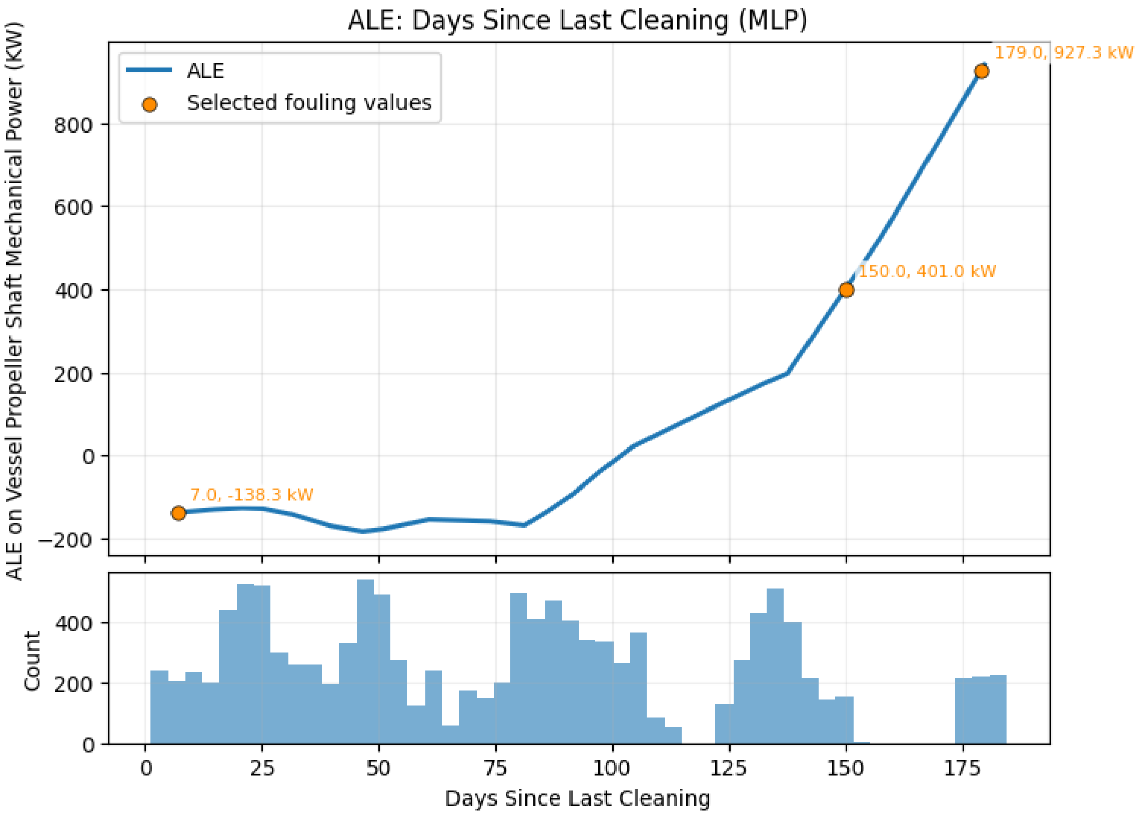
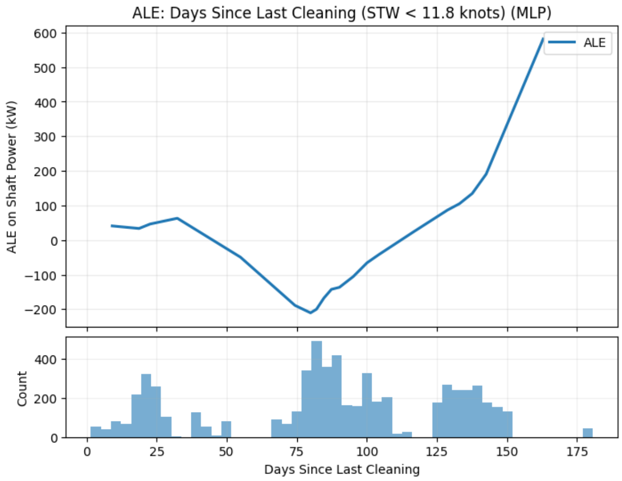
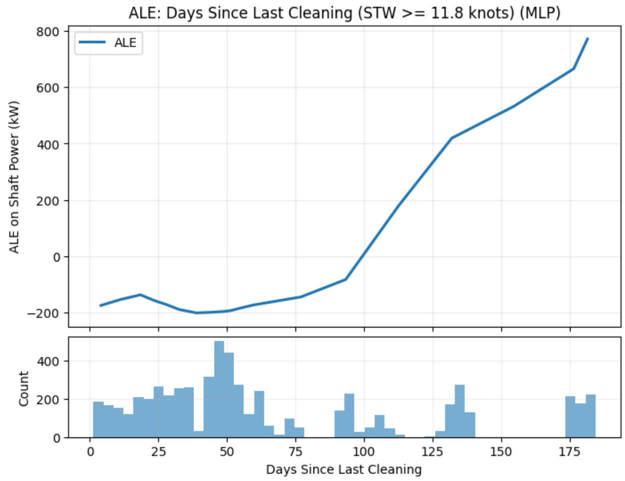


Figure 6.7: *ALE curve of DSC impact on predicted shaft power for the fitted MLP. Orange dots show the accumulated local effect at selected fouling levels. The histogram indicates the distribution of DSC observations along the curve.*

shape that is very inconsistent with expectations, with a pronounced dip around 75 days. It is also unable to produce an estimate for DSC=180 due to data sparsity.



(a) (STW < 11.8 kn).



(b) (STW ≥ 11.8 kn).

Figure 6.8: *ALE of DSC impact on predicted shaft power, constructed separately for observations below and above the median speed.*

Looking at the power estimates for each scenario, it appears that the MLP has learned that the added power as a percentage of total power increases with speed, estimating 9.8% and 13.4% after 150 days, respectively, compared to the scenario means of 2981 and 5709 kW (see Table 6.4). Meanwhile, the GAM implies the opposite since the additivity assumption forces it to apply the same absolute added power regardless of speed. Consequently, the added power as a percentage of total power decreases with speed in its predictions. The flexibility of the MLP thus allows it to fit a relationship that is contrary to the expectation that the added power as a percentage of total power would decrease with speed, as mentioned in Section 3.1.

Table 6.4: *Absolute and relative power penalties for the two models in high- and low-speed scenarios. Percentage increases are calculated relative to mean for the subsample (2981.0 kW for low speed and 5709.3 kW for high speed).*

Model	Speed	DSC = 150	DSC = 180
GAM	Low	327.8 kW (11.0%)	1224.1 kW (41.1%)
	High	327.8 kW (5.7%)	1224.1 kW (21.4%)
MLP	Low	290.6 kW (9.8%)	N/A
	High	763.9 kW (13.4%)	1032.5 kW (18.1%)

To put these estimates into perspective, Figure 6.9 shows the estimated added fuel for the period covered by the aggregated dataset (Jan 24 to Dec 31, 2024). The curve is constructed by attributing the appropriate share of the fuel use for each row in the aggregated dataset based on the added power for that period, as estimated by the PE and ALE curves, respectively (see detailed methodology in Appendix W). Assuming a constant fuel price equal to the Global 4 Ports[6] average for 2024 of 626

[6]The Global 4 Ports average is the average fuel price across Singapore, Rotterdam, Fujairah, and Houston ports.

USD/mt for Very Low Sulfur Fuel Oil (VLSFO) (Ship & Bunker, 2026), and a constant specific fuel oil consumption (SFOC)[7] over each time interval equal to the observed average, the estimated added fuel costs over the period were 90,990 USD according to the MLP and 105,737 USD according to the GAM. These estimates are conservative because the dataset excludes observations below 9 knots. For reference, in-water hull cleaning for a commercial vessel of this size is typically reported to cost around 15,000–20,000 USD (We4Sea, 2021).

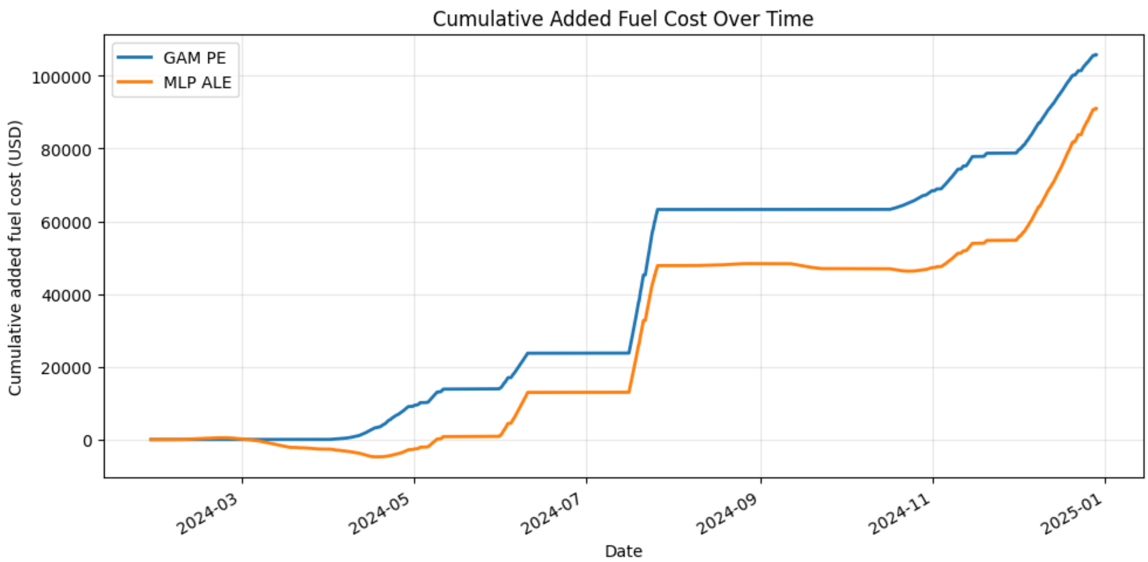


Figure 6.9: *Estimated cumulative added fuel costs over time. Methodology elaborated in Appendix W.*

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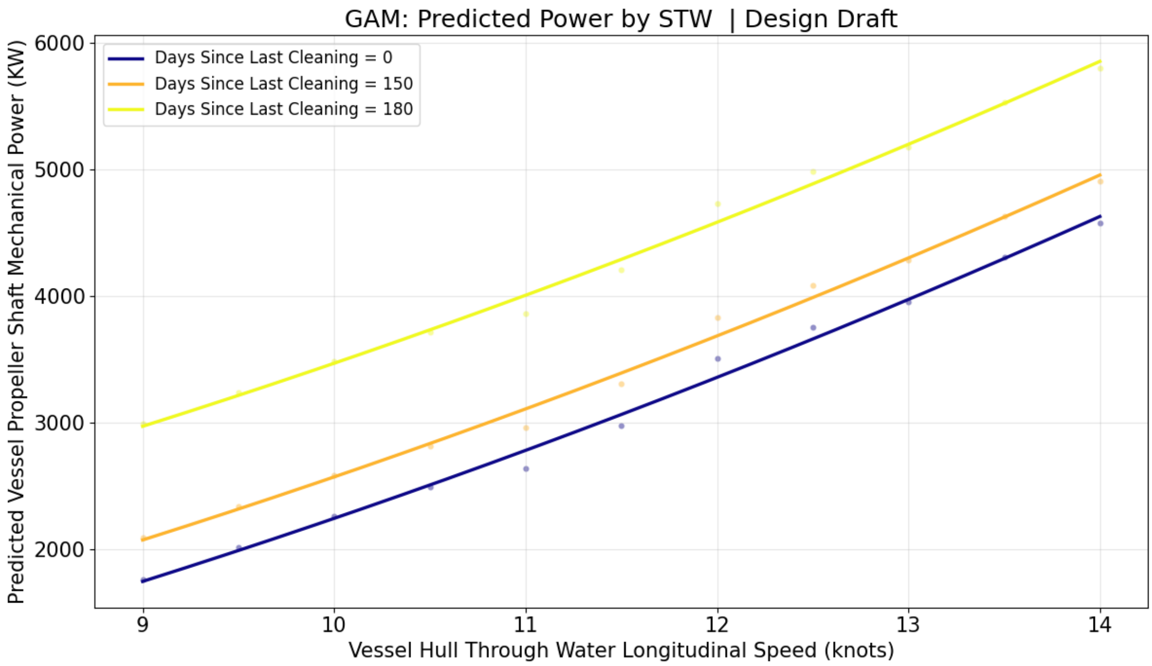
6.4 Speed-Power Curves

The estimated speed–power curves, constructed using the controlled variable experiment explained in Section 5.7.3 at design draft, are shown in Figure 6.10. In both models, the predicted shaft power increases with speed, as expected, but the curvature is more strongly pronounced for the MLP, whereas the GAM produces a nearly linear curve. The ordering of the curves is as expected, with higher DSC values corresponding to higher predicted power at a given speed.

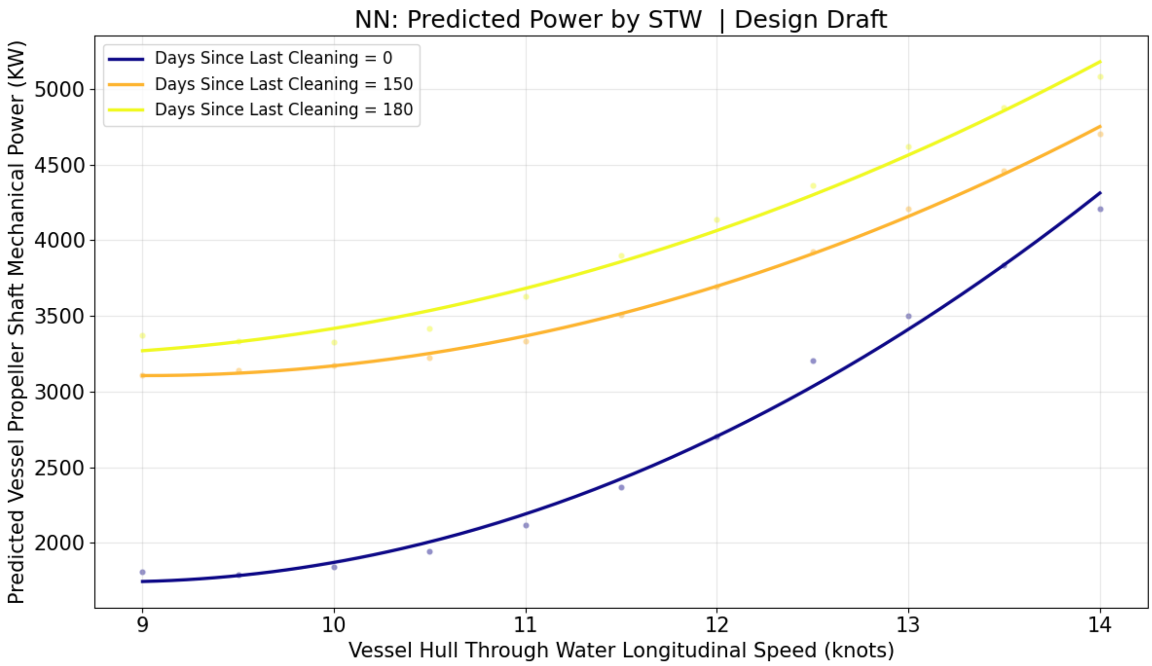
For the GAM, the spacing between curves is constant across speed by design. By contrast, the MLP curves are more flexible in shape, as the vertical distance between them decreases with speed, implying a smaller absolute increase in power at higher speeds. Contrary to the high-/low-speed ALE curves, this suggests that the MLP has learned that added power decreases, both in absolute and relative terms, as speed increases. This pattern is still inconsistent with expectations, but in a different manner than what is observed when binning into high and low speeds.

Comparing these curves with the sea trial curves in Appendix C allows the assessment of whether the

[7]Amount of fuel consumed to produce one kilowatt-hour of power output.



(a) GAM



(b) MLP

Figure 6.10: *Estimated speed–power curves at design draft for different fouling levels based on predictions made by each model.*

magnitude of the predicted shaft power is reasonable. While the curves are not fully comparable due to differences in draft and propeller retrofitting, they can still serve as a sanity check. Both models produce clean-hull baselines (DSC=0) that appear flatter and lower in level than expected for design draft. At 11.5 knots, the sea-trial reference is approximately 3100 kW, compared with 2978 kW for the GAM and 2365 kW for the MLP. while at 14 knots, the sea-trial reference is approximately 5900 kW, compared with 4575 kW for the GAM and 4206 kW for the MLP. The GAM baseline is thus closer to the sea-trial magnitude than the MLP baseline. However, both remain below the reference values, particularly at the upper end of the speed range. Once again, the MLP is further from the expectation despite its flexibility. It is also worth noting that the power estimates are more in line with the sea trials at 11.5 kn, which is close to the dataset median of 11.8 kn, than at 14 kn, where the data support is very low. This difference could suggest that some of the inaccuracy is due to extrapolation.

Table 6.5 shows that the estimated added power due to fouling is generally of a similar order of magnitude to the estimates of the PE and ALE curves. However, the MLP tends to produce higher estimates, particularly at the lower end of the speed range. For the GAM, the relative added power ranges from 7.2% to 18.6% at DSC=150 and from 26.8% to 69.5% after 180 days, while the corresponding ranges for the MLP are 11.8% to 72.5% and 20.8% to 86.1%, respectively.

Table 6.5: *Estimated added power due to hull fouling relative to the respective 0-day baseline at design draft (avg draft = 11.62 m, trim = 0.8 m).*

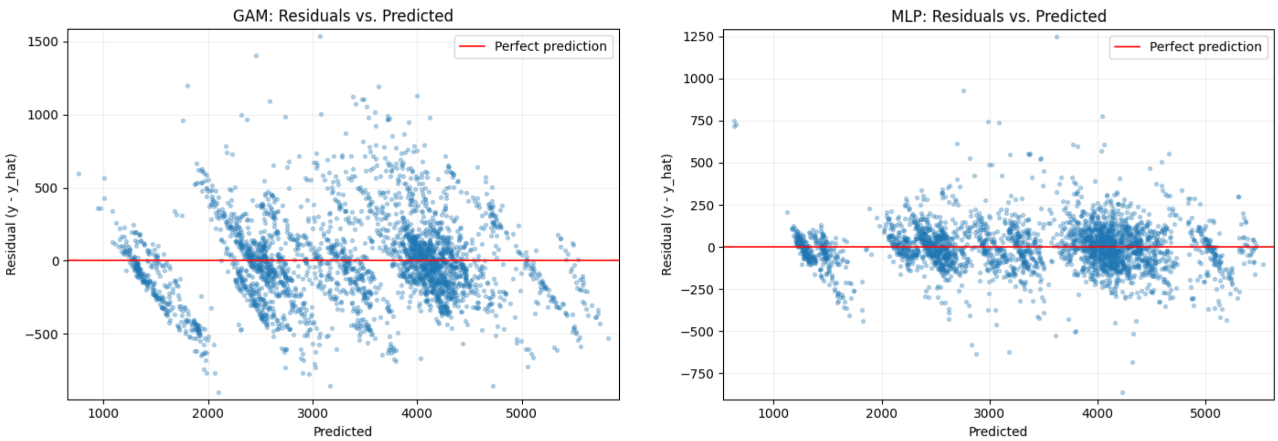
	GAM		**MLP**	
Speed	**150 days**	**180 days**	**150 days**	**180 days**
9 kn	327.8 kW (18.6%)	1224.1 kW (69.5%)	1298.9 kW (71.7%)	1560.5 kW (86.1%)
10 kn	327.8 kW (14.5%)	1224.1 kW (54.2%)	1334.5 kW (72.5%)	1484.9 kW (80.7%)
11 kn	327.8 kW (12.4%)	1224.1 kW (46.5%)	1214.9 kW (57.3%)	1507.6 kW (71.1%)
12 kn	327.8 kW (9.3%)	1224.1 kW (34.9%)	991.5 kW (36.7%)	1433.2 kW (53.0%)
13 kn	327.8 kW (8.3%)	1224.1 kW (31.0%)	710.9 kW (20.3%)	1123.2 kW (32.1%)
14 kn	327.8 kW (7.2%)	1224.1 kW (26.8%)	496.8 kW (11.8%)	873.3 kW (20.8%)

In general, the estimated speed–power curves appear only partially plausible and should be interpreted with caution. The observed inconsistencies are likely because the models make predictions based on feature combinations not represented in the training data. The heat maps in Appendix Q indicate that, even when considering only speed and DSC across draft conditions, all estimated speed-power curves, except the DSC=0 curve for scantling draft, are based entirely on extrapolation. This may also explain the contradictory speed-power curve for the ballast draft estimated by the MLP, where the curves cross, and the 0-day curve does not increase monotonically over the full speed range (see Appendix R). This issue is likely even more pronounced when considering the full set of metocean features, since each of them constitutes yet another dimension for the models to extrapolate into.

6.5 Residual Diagnostics

To assess whether some patterns remain unmodeled, three residual diagnostic checks are conducted for both models. First, residuals are plotted against predicted values to assess whether they are centered around zero across the prediction range and whether any systematic patterns are present. Second, residuals are plotted over time to assess whether model errors exhibit time drift or period-specific clustering. Third, residuals are examined across the feature ranges, with scatterplots and residual-feature correlations, to identify any dependence between the residuals and individual features.

In the residual-versus-predicted plots (Figure 6.11), both models appear to capture the overall relationship fairly well, with residuals generally centered around zero in most of the prediction range. However, the residuals do not appear to be completely random. Instead, they form visible diagonal clusters. One plausible explanation is the discrete nature of the operational data. As established in Section 6.1, the vessel appears to spend significantly more time operating at certain speeds and loading conditions. This discreteness can naturally lead to clustered observed shaft power values and give rise to diagonal bands in residual plots, meaning that this pattern alone should not be interpreted as evidence that the models are invalid.



(a) GAM *(b) MLP*

Figure 6.11: *Residuals vs. predicted values for the GAM and MLP.*

Figure 6.12 shows the residuals by model over time. The MLP residuals appear reasonably centered around zero throughout the year, with only limited evidence of time drift and mainly a few short episodes of increased spread, e.g., late spring and August, indicating that the model is struggling a bit more with some periods than others. Meanwhile, the GAM residuals show slightly clearer time-local clustering, including stretches with predominantly positive residuals around August and more negative residuals in parts of November and late December. This difference suggests that the MLP captures the broad relationships slightly more effectively. However, the time dependence of the residuals is not alarming for either model. It is worth noting that there is no growing under-prediction of shaft power between cleaning events, which, if present, would suggest the existence of an unmodeled fouling effect.

For both models, the residuals are generally centered around zero across the feature ranges and show no clear signs of severe heteroskedasticity (see Appendix S for scatterplots). The residual-feature correlations are also very low (< 0.1, see Appendix T). The main exception is the relationship between DSC and the GAM residuals, where some periods show a slightly larger spread, suggesting that part of the fouling effect, or its interactions with other variables, remains unmodeled. This is expected since DSC is only an imperfect proxy for the latent fouling state, and the GAM does not include tensor products. Notably, this issue is not present for the MLP.

6.6 Robustness Checks

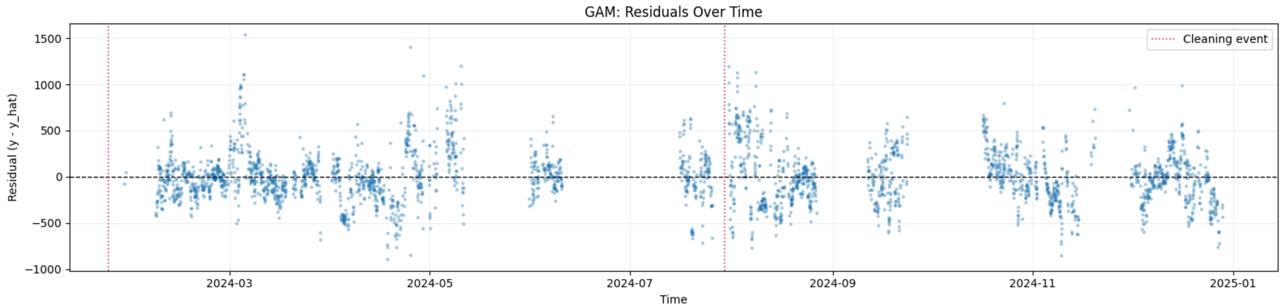
This subsection conducts two robustness checks to assess whether the main findings depend on specific modeling and preprocessing choices. Specifically, it tests the sensitivity of the estimated fouling effect to the monotonicity constraint on DSC in the GAM and to the chosen rolling standard deviation threshold for the speed variables.

Re-estimating the GAM without the monotonically increasing constraint on DSC improves predictive performance relative to the constrained GAM, with test RMSE decreasing from 286.5 to 274.8 kW, and test MAPE from 7.34% to 7.08% (see Table 6.6). Despite what one could have suspected, this specification showed no signs of overfitting when looking at the test metrics, with the model performing slightly better on the test set than the training set. These results suggest that DSC still contributes useful predictive information, and that its contribution becomes even stronger when the fouling term is allowed to vary freely. However, the improved accuracy could also suggest that the unconstrained model is using its added flexibility to fit time-related variation in the data that is not necessarily due to fouling.

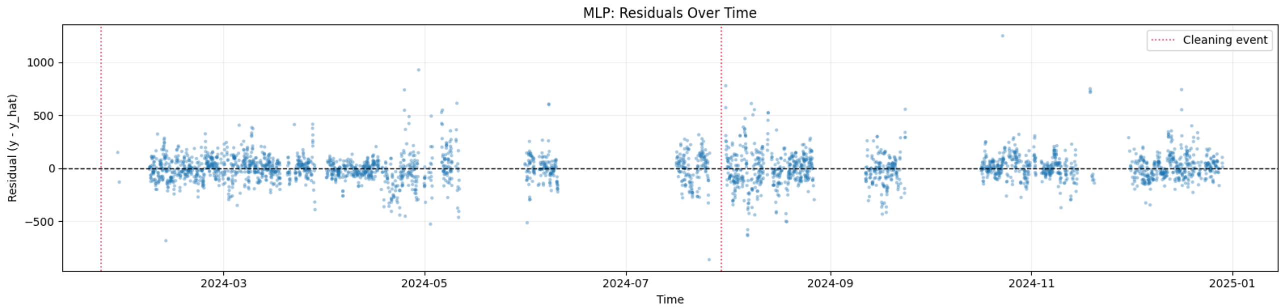
Table 6.6: *Training and test performance metrics for GAM model specifications used in the sensitivity analysis. Train-test gap is the difference between RMSE on the training and test set. Positive values indicate worse performance on the test set. Lower is better for all metrics.*

Model	DSC	Train RMSE	Test RMSE	Train-test Gap	Train MAPE	Test MAPE	Train MAE (kW)	Test MAE (kW)
GAM	Excl.	295.6	297.7	0.7%	7.49%	7.42%	216.5	219.1
GAM	Incl.	286.2	286.5	0.1%	7.48%	7.34%	212.2	213.6
GAM	Incl. (no mono.)	278.2	274.8	−1.2%	7.30%	7.08%	205.9	205.2

This suspicion is supported when looking at the PE curve (Figure 6.13), which no longer follows a sensible shape. Instead of showing a gradual increase in required shaft power over time, the curve changes direction several times and rises sharply at high DSC values. This shape is inconsistent with a credible fouling process, suggesting that some of the variance captured by the DSC feature is unrelated to fouling. Therefore, the unconstrained specification does not provide a meaningful basis for estimating the added power from fouling, even though it shows a better predictive performance.



(a) GAM



(b) MLP

Figure 6.12: *Residual diagnostics over time for GAM and MLP models, respectively.*

Throughout the preprocessing of the data, several choices impact the bias-variance tradeoff, including removing low-speed observations, retaining features with high VIFs, and loosening the thresholds for rolling standard deviations of speed. The choices intentionally favor higher variance than the recommendations for engineering-based formulas, on the expectation that ML models are better suited to handle it. Therefore, a second robustness check is included to assess whether the findings persist under stricter preprocessing, thereby reducing variance. A wide range of steps can be chosen to achieve lower variance, but tightening the rolling standard deviation thresholds from 0.388 knots to the DNV-recommended 0.194 knots is chosen since it follows an explicit industry guideline (DNV, 2023).

Table 6.7 shows that the stricter threshold improves test performance for all three model classes. This improvement suggests that it makes the prediction task easier overall by reducing short-term variability and thereby improving the consistency of the speed-power relationship. For Ridge and GAM, the inclusion of DSC remains clearly beneficial, as the Wilcoxon signed-rank test still indicates a statistically significant reduction in paired absolute prediction errors ($p < 0.0001$). By contrast, this pattern no longer holds for the MLP, where the DSC version does not improve relative to the earlier specification and no longer outperforms the model without DSC. One caveat is that the MLP hyperparameters are not re-tuned after the filtering change, which makes this comparison somewhat less fair and may partly explain why the MLP does not improve with DSC.

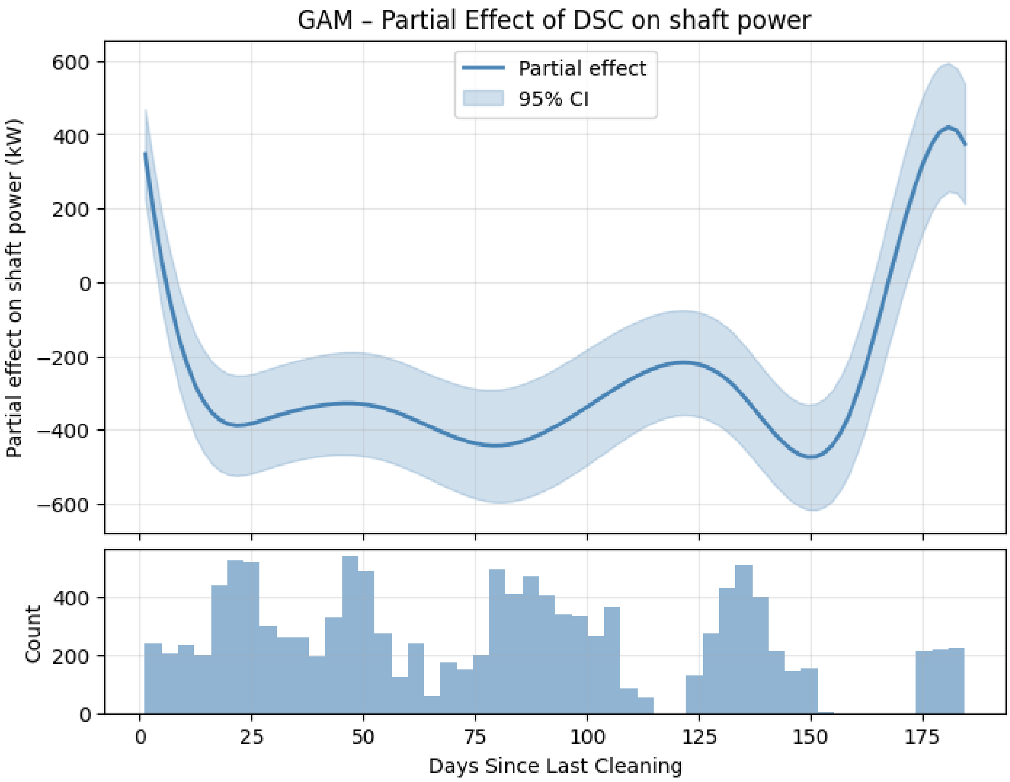


Figure 6.13: *GAM PE estimate without shape constraints. The Histogram indicates data coverage along the curve.*

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Table 6.7: *Training and test performance metrics for models with and without fouling features. Train-test gap is the difference in RMSE on the training and test sets, with higher values indicating worse performance on the test set. Lower is better for all metrics. Δ Test RMSE is the change in test RMSE relative to the original analysis (0.388-knot filtering), with negative values indicating an improvement.*

Model	DSC	Train RMSE (kW)	Test RMSE (kW)	Train–Test Gap (%)	Δ Test RMSE (kW)	Train MAPE (%)	Test MAPE (%)	Train MAE (kW)	Test MAE (kW)	Wilcoxon p-value
Ridge	Incl.	437.7	431.8	−1.3	−26.4	12.99	12.84	329.1	326.8	$p < 0.0001$
Ridge	Excl.	458.7	454.8	−0.9	−18.5	13.51	13.38	348.6	345.7	
GAM	Incl.	262.6	259.7	−1.1	−26.8	6.74	6.61	195.5	192.7	$p < 0.0001$
GAM	Excl.	268.5	268.0	−0.2	−29.7	6.75	6.68	198.6	197.2	
MLP	Incl.	118.7	130.7	10.1	−7.6	2.77	2.88	83.9	89.9	$p = 0.99$
MLP	Excl.	121.1	130.2	7.5	−20.6	2.90	2.86	82.9	85.9	

The estimated fouling-effect curves remain sensible in shape, but with some subtle, yet significant, differences (see Figures 6.14 and 6.15). The GAM PE retains a shape that is similar to the previous version. In contrast, the MLP effect becomes more clearly U-shaped, which is difficult to reconcile with the expected relationship. The implied fouling penalties are also lower than in the original specification. For the GAM, the estimated added power is 211.1 kW at DSC=150 days and 877.3 kW at DSC=180, corresponding to 6.3% and 26.0% of the average shaft power of 3368 kW (9.7% and 36.3% in the original specification). For the MLP, the corresponding estimates are 182.0 kW after DSC=150 and 528.5 kW at DSC=180, corresponding to 5.4% and 15.7% of the average shaft power (16.0% and 31.6% in the original specification).

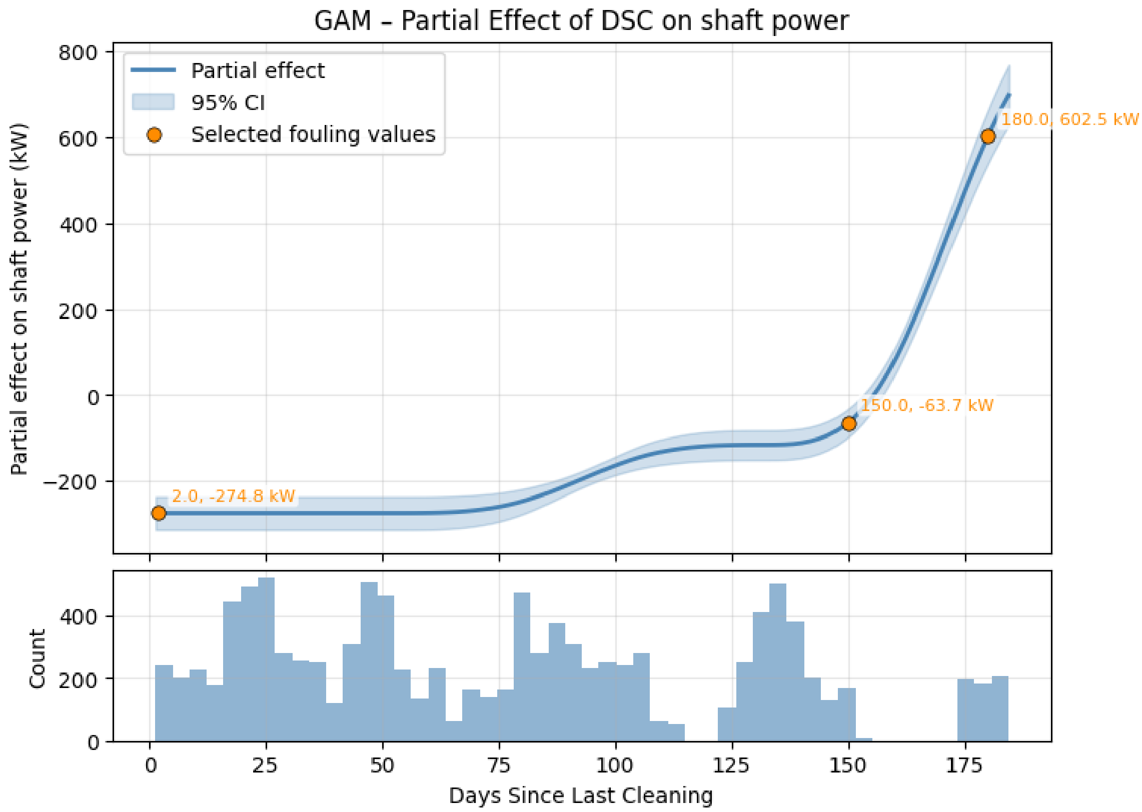

__Figure 6.14:__ PE curve for impact of DSC on shaft power predictions for the GAM using the DNV-recommended threshold for rolling std for STW and SOG. Orange dots show DSC's impact on the predicted shaft power at selected values with all other variables held constant.

The results from using a stricter filtering do not overturn the main analysis, but the changes are large enough to cast some doubt on the robustness of the results reported in Sections 6.2 and 6.3. A plausible explanation is that part of the previously estimated fouling effect is attributable to unmodeled variance rather than to fouling itself, which could lead to an overestimation of the added power. This interpretation is consistent with the observation that all models improve under stricter filtering, while the estimated fouling penalties decrease.

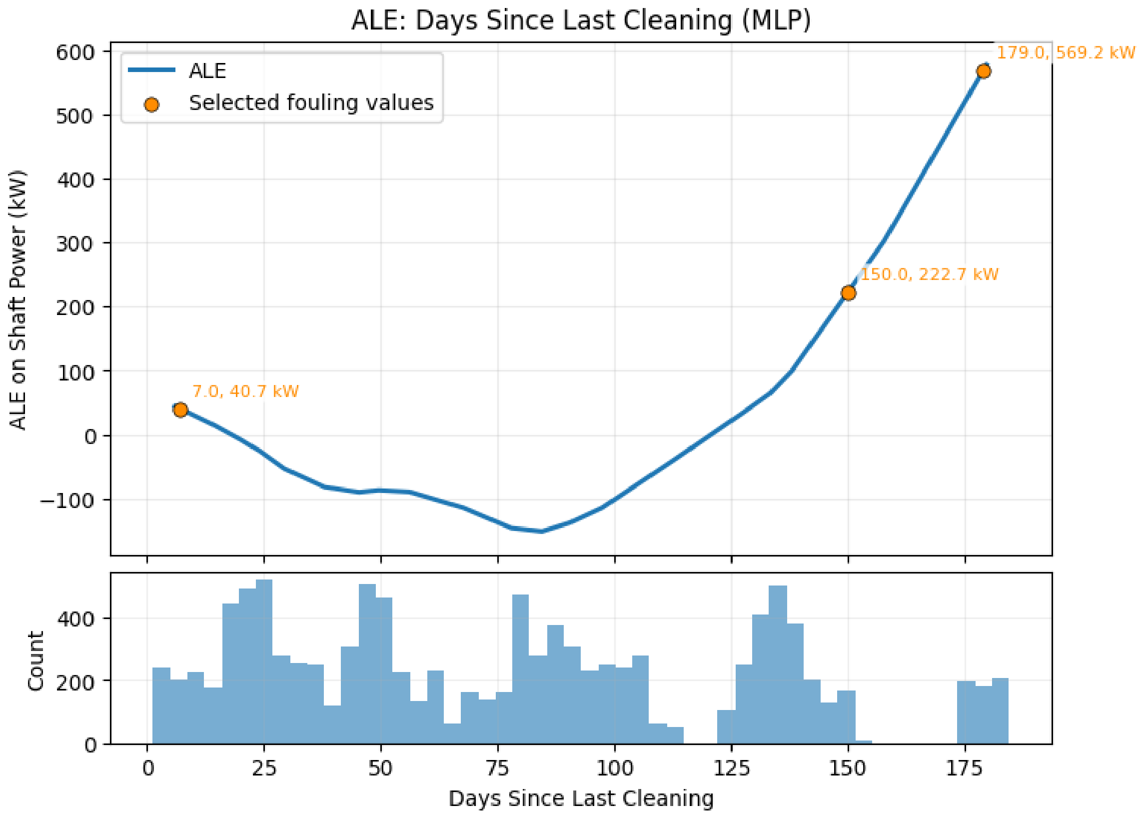

Figure 6.15: *DSC ALE curve for the MLP using the DNV-recommended threshold for rolling std for STW and SOG. Orange dots show the accumulated local effect at selected fouling levels. The histogram indicates the distribution of DSC observations along the curve.*

7 Discussion

The following four sections discuss the answers to the research questions based on the results, while assessing their robustness and validity. Section 7.1 discusses RQ1, i.e., what the results suggest about the development of fouling during the first 6 months following a cleaning event, with a focus on this particular vessel. Section 7.2 discusses RQ2, i.e., whether the results support ML as a reliable alternative to traditional fouling estimation methods. Section 7.3 discusses RQ3, i.e., whether the MLP makes up for its complexity by providing better fouling estimates. Lastly, Section 7.4 puts the findings into perspective and discusses their implications for academia and practitioners in the shipping industry.

7.1 Fouling Development

This section discusses what the estimated fouling effects suggest about the most plausible underlying development of fouling during the first 6 months after a cleaning event. Since the fouling state is not observed directly, this interpretation is based on the learned effects and is evaluated along three dimensions: the time pattern of the estimated development, the magnitude of the implied added power, and the credibility of the estimated, i.e., 150 days and 180 days of fouling, condition of the vessel, represented by speed-power curves. The credibility depends mainly on whether the results are consistent with theory and the literature.

The GAM PE curve (Figure 6.6) and the MLP ALE (Figure 6.7) curve both suggest similar fouling patterns that become progressively worse with time since the last cleaning event, which is consistent with both theory (Lewis, 1988) and literature (Gupta et al., 2022; Schultz, 2007; Schultz et al., 2011). For both models, the results indicate a negligible impact of fouling on power needs during the first 2.5 months after cleaning, after which the additional power increases sharply. The fact that the models yield similar results, despite the fundamental methodological differences between PE and ALE, could indicate that this pattern reflects a real physical signal. However, it cannot be ruled out that they both fall victim to the same data artifacts, such as collinearity between DSC and draft.

It should be noted that although the vessel was cleaned twice, DSC exceeded 150 days only once in the filtered dataset (July 16th to 26th; see Appendix V). This gap in data coverage means that the models' learned effect beyond this point is largely based on this period. The kink in the curve could indicate that calcareous fouling began developing around this time, which would be plausible if the vessel was in very warm water and/or operated at low speed during this period (Davidson et al., 2020; Van Mooy et al., 2014). The surrounding waters were generally warm (above 20 degrees, see Figure 5.4), and the vessel had an entire day where it was almost idle, operating at 2-3 knots. A reasonable hypothesis for the steep increase is light calcareous fouling, which may have benefited from even more favorable conditions due to existing biofilm formation on the hull in the first 2.5 months (Schultz et al., 2011; Van Mooy et al., 2014). However, this once again requires the assumption that the observed relationship is due to a real physical signal and not a common data artifact.

There are several notable reasons to treat these interpretations with caution. Firstly, the results give little sense of the uncertainty behind the results. While the GAM PE does provide confidence intervals, they rest on the violated assumption of independence among observations, so they should not be trusted. Meanwhile, the MLP ALE does not produce any uncertainty indicators. Secondly, the low data support for between 150 and 180 DSC suggests that the steep increase in this period is likely more representative of the particular operating conditions during the late July voyage, making it less generalizable to other ships or even voyages.

The timing of the kink is also somewhat later than expected, based on the stationary-surface study by Davidson et al. (2020), who found that even calcareous fouling can develop within weeks under favorable conditions. Lastly, the negative correlation between DSC and draft may have affected the shape of the curves, since the vessel tends to sail with higher loads shortly after cleaning events (as seen in Figure 5.11), which could make the model prone to underestimating the effect of fouling at low values of DSC, and vice versa. However, the overall development implied by the patterns learned by both models is relatively credible, given their cumulative nature and the magnitude of the added power.

While the kink appears consistently around 2.5 months in both models, there are notable differences in how the added power is distributed over time. The pattern learned by the MLP implies that a substantially larger share of the added power accumulates before the 150-day mark, as it estimates a 16.0% increase in required power at that point compared with 9.7% in the GAM. The GAM, by contrast, attributes more of the development to the final month of the period, thereby implying a more abrupt late-stage increase in deterioration. The final power penalties are, however, relatively close to each other by the end of the observed period, with the GAM estimating a 36.3% increase after 180 days and the MLP estimating 31.6% after 179 days.

One possible explanation for this difference is that the MLP may be capturing genuinely condition-dependent fouling effects that the additive GAM cannot represent as easily. Whereas the GAM imposes one common DSC curve across all operating contexts, the MLP can allow the relationship to vary with speed, draft, and other features, which may lead it to allocate a larger share of the added power earlier in the period. However, the opposite interpretation cannot be ruled out, namely that the added flexibility of the MLP allows it to absorb variation that is not actually attributable to fouling but to other developments such as loading conditions, engine deterioration, or environmental patterns. Some of the discrepancy may also stem from the interpretation methods themselves, since the GAM curve is read directly from a global curve while the MLP curve is derived post hoc using ALE, which introduces additional assumptions and limitations.

In magnitude, the estimated fouling penalties fall within the range already reported in the literature. The upper estimates are close to CFD-based studies such as Farkas et al. (2020), who report 1.4-36.3% added power for biofilm, and Song et al. (2019), who find roughly 18-60% and 22-73% for calcareous fouling at 24 and 19 knots, respectively. They are also broadly consistent with the more empirically oriented studies by Schultz (2007) and Schultz et al. (2011), where light slime is associated with about 9-11% added power and small calcareous fouling with roughly 18-31%. This makes the relatively steep increase from around five months to near six months broadly plausible in light of previous work. However, because most comparable studies are conducted at speeds above 15 knots, a higher percentage of added power is expected in this dataset at a given level of fouling. Therefore, the higher percentage of added power at lower speeds, as observed in the estimated speed-power curves, is arguably feasible.

The fouling state at DSC=150 and DSC=180 under design load is only partially credible when evaluated using the estimated speed-power curves. The clean-hull baselines are flatter and lower than the sea-trials, with the GAM remaining closer to the reference magnitude than the MLP, but both deviating more strongly around 13-14 kn, where data support is low. The resulting added power figures are generally of similar magnitude to those inferred from the PE and ALE curves, but become very large at lower speeds, reaching 69.5% in the GAM and 86.1% in the MLP at 9 knots after 180 days. Given that the data-support heatmaps (see Appendix Q) indicate that these design-draft curves are based entirely on extrapolation, they are therefore better interpreted as a rough sanity check on the final fouling state than as an actual representation of the speed-power curve at that point.

In terms of shape, the end state still behaves plausibly, in that higher DSC yields higher power at any given speed in both models. Beyond that overall pattern, however, the detailed shapes are less reassuring. The GAM produces an almost linear collection of speed-power curves with constant spacing across speed. In contrast, the MLP produces more pronounced curvature and shrinking vertical distance between curves as speed increases, which is inconsistent with both theoretical CFD and empirical studies (Farkas et al., 2020; Gupta et al., 2022; Laurie et al., 2021; Schultz, 2007; Song et al., 2019). In addition, the conclusion reverses for the MLP if one runs the ALE on high- and low-speed subsets of the data, i.e., the added power increases both in absolute and relative terms, indicating that its estimates are not very robust. Therefore, the speed-specific added power estimates derived from the speed-power curves are not considered very credible.

Overall, the evidence suggests that this vessel did experience a meaningful buildup of fouling during the first six months after cleaning, on average across the two observed post-cleaning periods. The added power estimates appear sufficiently consistent with theory and literature to be interpreted, at least partly, as a real fouling signal. The most plausible physical reading is that the ship generally experienced light fouling in the first 2.5 months of each cleaning cycle, and light calcareous fouling accumulated by the end of July. However, some underestimation remains possible because DSC is negatively correlated with draft. At the same time, the estimated speed-power curves are too inconsistent to be considered a genuine representation of the vessel in a fouled state. While the estimates are sensible for this particular vessel, the results also confirm the empirical consensus that fouling development is highly context-dependent, meaning that it is not very generalizable.

7.2 Machine Learning for Fouling Estimation

The previous section showed that all model classes recover a fouling-related signal and that this signal is, at least to some extent, consistent with physical expectations. The key question, however, is whether that signal is reliable enough to support the claim that ML is a viable methodology compared to traditional methods. Ideally, this would be done by comparing ML to other methodologies in a controlled experiment where the ground truth is known, e.g., on a vessel with a known fouling state. However, conducting such a study is too expensive and cumbersome to carry out in practice. Therefore, the reliability of the methodology is instead assessed qualitatively through four criteria: whether inclusion of the fouling proxy improves predictive performance, whether the estimated effects survive residual diagnostics and robustness checks, whether they remain stable in the presence of voyage-specific and time-specific variation, and whether the models can recover a plausible speed-power relationship despite the discrete nature of ship data.

A first argument in favor of ML is that all three model classes improve when DSC is included as a feature. In the main analysis, the one-sided Wilcoxon signed-rank tests are significant for the ridge regression, GAM, and MLP, which suggests that the reduction in paired absolute prediction errors is systematic rather than incidental. Moreover, the nonlinear GAM and MLP achieve markedly lower test errors than the ridge, indicating that the relationship between operating conditions and shaft power is to some extent nonlinear. This should, at least in theory, give ML an advantage over traditional performance estimation methods such as ISO 19030, which often rely on coarse corrections (DNV, 2023; ISO, 2016). It should be noted that the significance of the Wilcoxon signed-rank tests does not necessarily mean that the DSC effect is large in every case, since the test statistic only accounts for the magnitude of errors in an ordinal sense. Nonetheless, the results support the claim that including a proxy for fouling contributes valuable information to the ML models.

The models, particularly the MLP, also deal relatively well with some of the voyage-specific variance. For the GAM, the residual diagnostics (Section 6.5) show some systematic clustering over time, a somewhat larger spread across parts of the DSC range, and visible diagonal banding, whereas for the MLP only the latter pattern remains, and in a weaker form. The fact that some of the patterns persist for the MLP could also suggest that some voyage-specific variation is not represented in the data. While it is conceivable that there could be some omitted voyage-specific variable, a much more likely explanation is that it is due to the repetitive nature of the shaft power variable seen in Figure 6.1. Therefore, the results suggest that ML can, at least with a certain level of model complexity, accommodate the voyage-specific variance that characterizes operational ship data.

The robustness checks, however, point to important weaknesses. The estimated fouling magnitudes are sensitive to preprocessing choices, most clearly when the stricter speed-variability filter is applied. The observation that predictive performance improves while the estimated added power decreases substantially (35% for GAM and 64% for the MLP) is too big to disregard. While it could theoretically be possible that the approximately 10% of observations that are removed carried most of the fouling signal, a more likely explanation is that the higher variance worsens the identification problem, i.e., it introduces more noise that the model can mistake for fouling and therefore use the DSC feature to model instead. This instability is relatively large considering the low amount of data that is removed, and is therefore a strong argument against the reliability of ML models. However, it should be noted that this problem will not necessarily persist when applying ML to larger and/or more balanced datasets.

The infeasible fouling curve generated by the GAM without the monotonicity constraint on DSC, despite its better predictive performance, gives an additional reason for caution. This discrepancy supports Shmueli (2010)'s notion that a good predictive model is not necessarily good for interpretation. If considered in isolation, it could also support the argument that the sensible fouling relationship observed in the PE curve is spurious. However, the fact that the MLP fits a reasonable relationship without any monotonicity constraints points towards a real signal being present. Therefore, a more plausible explanation is that the DSC relationship contains interactions, most likely with speed or draft, which prevents the GAM from fitting a sensible relationship in the absence of explicit constraints (ISO, 2016; Lewis, 1988). This robustness check, therefore, suggests that failing to include domain knowledge can indeed cause ML models to produce very unreliable and physically wrong fouling estimates.

The estimated speed-power curves also give reason to doubt the ability of ML models to reliably estimate fouling. Since this is most likely due to data sparsity, this is a weakness compared with traditional naval engineering approaches, as they are not dependent on training data to learn the physical relationship. Instead, the physical relationships are already considered in the applied formulas, albeit in a simplified form, which means that these models don't make uninformed extrapolations in the same sense that ML models do. The issue would likely be smaller for ML with more balanced data across drafts and fouling stages, but that is difficult to obtain in real ship operations because loading conditions are effectively discrete. While a more balanced dataset than the one at hand is likely obtainable, finding one that completely removes this issue is unlikely. Therefore, limited reliability in speed-specific fouling estimates is likely a general limitation of ML-based fouling estimation, not just a weakness of the present analysis.

While a case can be made for ML as a fouling estimation methodology, there is no support for making the unambiguous claim that ML is universally useful or better than traditional methods. The GAM and MLP learn broadly similar fouling patterns and capture the relationship between operating conditions and shaft power reasonably well, where data support is high. However, the estimated fouling magnitudes are too sensitive to preprocessing and modeling choices to support strong claims about their exact size. The estimated speed-power curves are also questionable, mainly because data sparsity forces the models to extrapolate across combinations of speed, draft, and DSC that are not present in the training data.

7.3 Interpretability Assessment

As discussed in Section 3.3, the present modeling task is primarily one of inference rather than prediction. Therefore, predictive performance cannot by itself determine which model is most useful, as a model may reduce error while still learning relationships that are implausible. The trade-off between the GAM and the MLP must therefore be evaluated using both formalized and unformalized criteria: on the one hand, predictive accuracy and the ability of DSC to improve fit, and on the other hand, interpretability and consistency with domain knowledge. In this setting, the lower interpretability of the MLP is only justified if the gains in both types of criteria are meaningful.

Judged on predictive performance alone, the MLP clearly outperforms the GAM by a substantial margin. The error magnitudes are similar to previous research, which found MAPEs of 1-3% and a slight, but existing performance gap between white and black-box models (Bakka et al., 2022; Lang et al., 2022; Laurie et al., 2021). This gap is also consistent with LeCun et al. (2015)'s view that more complex, and, by extension, less interpretable, models can achieve higher accuracy when the underlying relationships are nonlinear. Because the data consist of structured and human-interpretable tabular data, one might have expected the performance gap to be smaller based on Rudin (2019)'s argument that black-box advantages are often more limited in such settings. When looking solely at formalized criteria, the results therefore suggest that the complexity is worth the trade-off for the MLP. However, looking at the unformalized criteria adds important nuance.

The main strength of the GAM is not its predictive accuracy, but the fact that its fouling estimate can be inspected directly through the PE curve, which makes the learned role of DSC substantially more transparent than in the MLP, where interpretation requires an additional post-hoc step. In addition, domain knowledge can be incorporated directly into the model specification through shape constraints, which is particularly valuable in this case because the robustness check showed that removing the monotonicity constraint improves predictive error but produces a highly non-monotonic and physically implausible fouling curve. The GAM also appears more robust to preprocessing choices, as it retained a significant Wilcoxon signed-rank test under the stricter filtering, and its estimated added power at 150 days changed by about 35%. In contrast, the corresponding MLP estimate changed by about 64% and no longer yielded a significant Wilcoxon signed-rank test. As expected, the GAM is thus preferred when looking solely at its transparency.

This transparency advantage comes with an important limitation. In its present specification, the GAM imposes additivity and therefore cannot represent the interaction effects that are very likely present between fouling, speed, and loading condition. Both propulsion theory (ISO, 2015; Lewis, 1988) and the existing literature (Gupta et al., 2022; Laurie et al., 2021) give strong reason to expect such interactions, and several of the results already discussed point in the same direction, most notably the GAM's residual structure. It also offers a plausible explanation for why the unconstrained GAM loses physical credibility when the monotonicity restriction on DSC is removed. Once the model is given more freedom, it cannot express the DSC effect through interactions, and instead uses a single global curve to absorb time-related variation in a way that quickly becomes physically incoherent. However, it should be stressed that this is a weakness that only applies to the strictly additive GAM adopted in this thesis.

The main advantage of the MLP in this case is that it can model the interactions that the GAM misses. This likely explains why its residual plots are healthier than those of the GAM, since the additive GAM appears to leave more voyage-specific relationships unmodeled. Because the unconstrained GAM overestimates power at low values of DSC, where draft tends to be high (see Figure 5.11), the MLP fits a more reasonable smooth curve, which could suggest that this unmodelled voyage-specific relationship is the interaction between speed and draft. Once again, this suggests that the GAM is inferior to the MLP in terms of consistency with domain knowledge, at least without constraints and interaction terms.

The ALE reveals that the MLP fits a relatively sensible overall fouling curve without needing an explicit monotonicity constraint, although this strength becomes weaker once the analysis moves to the high/low speed scenarios and speed-power curves. Finally, ALE has an advantage over the PE curve in the sense that it estimates effects from local changes in prediction within the observed data, rather than fitting a global curve, which makes it more robust when predictors are correlated, and data support is low. Therefore, the results suggest that the MLP is only superior with regard to consistency with domain knowledge when looking at the overall relationship.

That same point also reveals an important weakness. Although ALE is a defensible way to interpret the MLP, it is still a post-hoc layer placed on top of the fitted model rather than a relationship read directly from the model itself, so it adds another layer of analytical uncertainty to the fouling estimate. Most importantly, the flexibility of the MLP becomes problematic when analyses are broken down into subsets of the data, e.g., in the high/low speed scenarios or the speed-power curves, where the MLP tends to produce implausible relationships. The best example of this is the speed-power curve for the ballast condition (Appendix R), where the MLP appears to have locally overfitted the lower end of the speed range.

Comparing the models thus points towards a clear tension between the two. The GAM is easier to audit, but it imposes a simplified additive fouling curve by design, which makes it less useful when the true effect is likely to vary with operating context. The MLP avoids that restriction and predicts much more accurately overall, but in sparse regions, it appears to use that extra flexibility to fit relationships that are physically hard to defend, most clearly in the estimated speed-power curves. In that sense, the weakness of the GAM is a structural simplification, while the weakness of the MLP is its flexibility, becoming local overfitting under weak data support.

The most important point, therefore, is not that one model clearly succeeds where the other fails, but that neither model produces fully credible speed-specific fouling estimates in this dataset. Although the GAM curves are somewhat more stable, they still rest on a fundamentally wrong assumption by applying the same absolute added power across speeds, while the MLP often becomes unreliable precisely where flexibility is most needed. Based on the latter, it can be argued that the MLP does not substantially outperform the GAM with regard to the unformalized criteria.

A strong argument can also be made that data sparsity, especially across loading conditions and at the edges of the speed range, is the main reason neither approach fully resolves the problem. More broadly, this suggests that the best model for this use case is likely one that sits somewhere between rigid additivity and unconstrained black-box flexibility, e.g., a GAM with tensor products. Alternatively, using hybrid approaches such as that proposed by de Haas et al. (2023) could also improve domain knowledge consistency by incorporating more complex physical relationships where they are well understood.

7.4 Implications

This section assesses the findings in relation to the broader perspective of current research and industry practices. Section 7.4.1 discusses the findings of the thesis in the context of the existing body of academic knowledge, while Section 7.4.2 assesses the implications of the findings for practitioners, such as shipowners, operators, and data scientists in the maritime industry.

7.4.1 Implications for Research

The findings are consistent with previous research in terms of both the magnitude and general development of fouling effects. However, one notable difference is the extended initial period with little apparent fouling. This pattern may indicate that fouling can remain negligible, even for some time, when operating and environmental conditions are favorable. The results also suggest that fouling is not best represented as a simple linear function of DSC, but rather as a more nonlinear and stage-like process, which supports the view that improved fouling proxies should account for factors such as idle time, water temperature, and salinity.

The findings also support further research on physics-informed ML in this domain. Although the results do not justify the conclusion that ML models are unambiguously superior to traditional fouling-estimation methods, they suggest that such models can capture a substantial part of the underlying physical system and therefore represent a promising alternative. The findings also highlight the need to evaluate fouling models using both predictive metrics and physical plausibility, as predictive performance alone may be misleading, and they underline that both white-box and black-box models depend heavily on domain-informed model design and feature selection.

The results do not support the claim that black-box models are inherently superior to white-box approaches for this particular task. This conclusion strengthens the broader research argument that interpretable and semi-interpretable models remain highly relevant in structured physical systems. In particular, the results point to the value of more flexible but still interpretable approaches, such as extended GAM formulations or hybrid models, with tensor-product GAMs appearing especially promising as a way to balance predictive flexibility with interpretability.

7.4.2 Implications for Industry

The findings suggest that operators should not assume that fouling develops linearly with time between cleaning events. This conclusion implies that hull cleaning should not necessarily follow a fixed and uniform schedule. Still, it should instead account for the operating and environmental conditions the vessel has been exposed to, with particular attention to the later months after cleaning or following idle periods in warm waters, when the economic consequences tend to accumulate rapidly.

The results indicate that ML-based fouling estimation can be a useful alternative when standard reference-condition methods, such as ISO 19030, are difficult or impossible to apply, for example, when sea trial data is not comparable or available. At the same time, ML appears to be more sensitive to data sparsity, meaning that they depend more heavily on the availability of sufficiently balanced data across multiple cleaning cycles. It can therefore be argued that ML models are currently better suited to retrospective analysis of average fouling development than to real-time assessment of how fouled a vessel is at a given moment. This does not rule out real-time applications using ML, but it does suggest that such use cases would likely need a slightly different approach, perhaps one including improved temporal representations or hybrid models.

The findings also suggest that a slightly more accurate black-box model may not be the best choice if its fouling estimates are harder to explain, validate, or trust. In practice, this makes white-box models such as GAMs attractive, while black-box models should be used for more predictive use cases or, at the very least, combined with domain-informed modeling and XAI methods. However, the results do not support the argument that added power estimates produced by the black-box model are better to an extent that justifies the added effort and uncertainty involved with applying XAI methods.

8 Limitations

Several important limitations need to be taken into account when interpreting the presented findings. Firstly, this study is limited by its post-positivist and bounded case-study design. Marine fouling is treated as a real phenomenon. Still, it is only observed indirectly through sensor measurements, fouling proxies, modeling assumptions, and analytical choices, so the estimated effects should be understood as approximations rather than direct measurements. Moreover, because the analysis is based on a single vessel in a delimited operational setting, the findings primarily support conclusions about this particular vessel for the year 2024. They may still indicate broader strengths and weaknesses of ML-based fouling estimation, but they do not justify definitive generalizations about the concrete development of fouling or the applicability of different methodologies.

Perhaps the most important limitation is the identification problem. All of the results presented in this thesis rest on the assumption that DSC is a reliable quantification of the fouling state of the ship. While the findings do support the claim that DSC is a decent proxy, it is undoubtedly a rather coarse one. Other developments, including engine deterioration, voyage heterogeneity, omitted variables, changing environmental patterns, or idiosyncratic noise, may in principle generate signals that happen to decrease performance over time and may therefore be partly absorbed by the models through the DSC feature. In addition, DSC appears correlated to some features, particularly draft, which also makes attribution less certain.

In an ideal setting, the identification problem would be addressed either through a (near-)perfect reference measurement of the fouling state or with a tightly controlled experimental design in which the fouling state and all other relevant determinants of shaft power could be observed or held constant. However, such an experiment is at best highly impractical and at worst impossible. Therefore, some degree of identification uncertainty must be tolerated to study fouling from operational data at all, even if more detailed proxies than DSC are applied.

A related limitation concerns measurement quality. The dataset contains several known imperfections that can only be partially addressed, including missing time-zone information in the noon reports, the use of hourly forecast metocean data rather than more frequent nowcasts or hindcasts, and a torquemeter recalibration for which the exact methodology is unknown. The dataset complexity and heterogeneity necessitate synchronization, interpolation, and tolerance-based matching, all of which build on assumptions and approximations. Thus, some uncertainty will inevitably remain, even after cleaning and validation. In addition, further unknown imperfections cannot be ruled out. In an ideal setting, each key signal would be validated against an independently measured ground truth, but this is not feasible here. Hence, the analysis necessarily relies on qualitative plausibility assessments and partial validation between signals instead.

The most important data-related limitation is arguably data coverage, or rather, lack thereof. As already noted in the literature review, ML models in this domain are sensitive to poor and uneven data coverage because they must learn the speed-power relationship from the observed operating range rather than from a fully specified physical structure, and this challenge is also present in the current dataset. In particular, draft observations are unevenly distributed across voyages and loading conditions, which weakens the basis for estimating reliable interactions between draft, speed, and fouling across the full operating range. This issue is compounded by collinearity between draft and DSC, meaning that part of the apparent fouling pattern may reflect the structure of the observed loading conditions rather than fouling alone. Moreover, the collinearity diagnostics applied in the thesis only assess linear dependence. Since concurvity may also distort effect estimates, especially in additive models, the severity of this issue is likely understated by the VIFs.

A limitation of the modeling and training setup is that the 80/20 random split assumes approximate independence between observations, even though the dataset is temporally structured. Observations from the same voyage are likely more similar than the split treats them as, which means that test performance may be somewhat optimistic relative to a genuinely forward-looking evaluation on unseen future periods. This limitation does not invalidate the analysis, since the study's models are used primarily as inference rather than forecasting, but it does make claims about predictive performance less generalizable.

A related limitation is the lack of reliable uncertainty quantification around the fouling estimates. The GAM does provide confidence intervals, but these are not trustworthy given the temporal dependence in the data, while the MLP provides no comparable uncertainty estimates at all. The results should therefore be interpreted primarily in terms of overall patterns and approximate magnitudes rather than precise probabilistic ranges.

A further limitation is that the model comparison is relatively narrow. The analysis only compares three specific model classes, and the conclusions therefore apply to these particular modeling choices rather than to white-box and black-box methods in general. Put differently, some interpretable models may be too restrictive to recover a meaningful fouling signal. At the same time, other black-box models[8] may capture the underlying relationships better than the MLP considered here.

Overall, the results support the interpretation of a time-dependent deterioration signal that is likely, but not guaranteed, to reflect fouling-related performance loss. The strongest conclusions are therefore specific to this vessel and to the observed period, while more granular conditional interpretations become less stable once the analysis is narrowed to particular drafts or speeds. At a broader level, the study suggests that ML models can recover a plausible fouling-related pattern from noisy operational data. Still, it does not establish that the same methods will perform equally well in other

[8]During the exploratory phase of the project, a random forest was tested as an alternative. However, it fared significantly worse than the MLP (see Appendices X and Y for ALE and estimated speed-power curves)

vessel contexts. More specifically, the findings support a retrospective average interpretation of how fouling-related deterioration appears to develop across varying environmental conditions and voyage differences.

9 Conclusion and Future Work

This thesis examined whether ML can be used to quantify the energy efficiency impact of marine fouling through a bounded case study on a bulk carrier. Two models, a GAM and an MLP, were fitted to predict shaft power with DSC as a proxy for the fouling state of the hull and propeller.

The results suggest that fouling appears to accumulate very little during the first 2.5 months after cleaning and then increase sharply after five months, with the GAM and MLP estimating 9.7% and 16.0% added power after five months and 31.6% to 36.3% after six months, respectively. This development is most consistent with a progression from slime or biofilm toward light calcareous fouling. The findings give some support to using ML for quantifying fouling-related efficiency loss in this setting, since including the fouling proxy improves model performance and both the GAM and MLP recover a similar and sensible fouling-related signal. However, sparse data coverage limits robustness and prevents more granular conclusions and the construction of speed-power curves. The results do not support the claim that black-box models outperform white-box models to an extent that justifies their complexity. Although the MLP includes interactions and achieves better predictive accuracy than the GAM, the gain is not large enough to clearly outweigh the loss of interpretability and local overfitting exhibited by the MLP.

Three avenues for future work are identified. Firstly, more controlled research designs that move closer to a known ground-truth fouling state would allow a cleaner and more direct comparison between ML-based estimation and traditional methods. Secondly, future work should aim to increase data coverage, especially with regard to loading conditions, possibly by sourcing data across sister vessels or over longer time periods. Such studies will clarify the severity of the extrapolation issue and whether the fouling and speed-power curves become more robust and physically credible under better data coverage. Lastly, future work should explore models that remain interpretable while capturing more of the underlying physics, e.g., GAMs with interactions or hybrid models with more detailed fouling proxies.

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Appendices

Appendix A: Prompts for Generating Front Page Image

Please generate an image that I can use as background in my thesis.

The motive itself should be a ship with noticable fouling. It should be a somewhat simplified image without to many colors, without just being an icon.

More specific requirements:
- Use a white background and only light colors. it should be pretty discrete and not make the text on the front page harder to read.
- The motive itself should be horizontal in the middle in sensible dimensions. However, it should contain white padding on top and bottom so that it fits the dimensions on an A4 page (the front page of my thesis). I have given you a screenshot of the text i want to fit on the page

Figure A1: *First prompt in the conversation*

Good start. I like the shading in the water and dimensions of the ship. Few adjustments:

- It should be less detailed
- The ship should be a bulk carrier, not a container ship
- It should be less like a drawing and more like a crisp design
- It should be more 2-dimensional
- The part of the hull sitting underneath the surface should be visible
- The fouling should be more noticable
- I would actually like the ship motive to be in the bottom 30% of the page, so please move it down

Figure A2: *Second prompt in the conversation*

That is great. Now add a version of the logo for copenhagen busines school (attached) in the top left corner, but with the color profile adjusted to fit the ship. It should be even more discrete than the ship, only differing slightly in color from the white background, and the letters "CBS" should be to the right of the logo instead of underneath. It is very important that the logo otherwise stays true and representative to the original

Figure A3: *Third prompt in the conversation*

Appendix B: Full List of Available Variables

B.1 Vessel Sensor Variables

Table B1: Operational monitoring variables from sensors onboard the vessel.

Source	Value name	Unit	Sampling frequency	Comment
Transducer	Main Engine Fuel Load	%	15s	
Control Alarm Monitoring System	Main Engine Rotational Speed	rpm	15s	
Transducer	Main Engine Scavenging Air Pressure	bar	15s	
RPM Indicator	Main Engine Turbocharger Rotational Speed	rpm	15s	
Gyrocompass	Vessel Hull Heading True Angle	degrees	15s	
Gyrocompass	Vessel Hull Heading Turn Rate	deg/min	15s	
GPS 1	Vessel Hull Over Ground Speed	knots	15s	
Echosounder	Vessel Hull Relative To Transducer Water Depth	meters	15s	
Speedlog	Vessel Hull Through Water Longitudinal Speed	knots	15s	
Torquemeter	Vessel Propeller Shaft Mechanical Energy	KWh	15s	Cumulative over lifetime of the vessel
Torquemeter	Vessel Propeller Shaft Revolutions (cumulative)	Revolutions	15s	Cumulative over lifetime of the vessel
Torquemeter	Vessel Propeller Shaft Mechanical Power	KW	15s	
Torquemeter	Vessel Propeller Shaft Rotational Speed	rpm	15s	
Torquemeter	Vessel Propeller Shaft Thrust Force	KN	15s	
Torquemeter	Vessel Propeller Shaft Torque	N*meter	15s	

B.2 Metocean Variables

Table B2: *Weather variables, including onboard anemometer measurements and externally provided forecast fields. Note that "MB" and "S" are anonymized metocean data providers.*

Source	Value name	Unit	Sampling frequency	Comment
Onboard anemometer	Wind Relative Angle	degrees	15 s	Higher sampling rate due to direct measurement.
Onboard anemometer	Wind Relative Speed	knots	15 s	Higher sampling rate due to direct measurement.
Forecast (Provider MB)	Eastward Sea Water Velocity	ms	1h	
Forecast (Provider MB)	Northward Sea Water Velocity	ms	1h	
Forecast (Provider MB)	Swell Significant Height	m	1h	
Forecast (Provider MB)	Wave Significant Height	m	1h	
Forecast (Provider MB)	Wind True Angle	degrees	1h	
Forecast (Provider MB)	Wind True Speed	knots	1h	
Forecast (Provider S)	Eastward Sea Water Velocity	ms	1h	
Forecast (Provider S)	Eastward Wind Velocity	ms	1h	
Forecast (Provider S)	Northward Sea Water Velocity	ms	1h	
Forecast (Provider S)	Northward Wind Velocity	ms	1h	
Forecast (Provider S)	Sea Water Temperature	C	1h	
Forecast (Provider S)	Swell Significant Height	m	1h	
Forecast (Provider S)	Wave Period	sec	1h	
Forecast (Provider S)	Wave Significant Height	m	1h	
Forecast (Provider S)	Wave True Angle	degrees	1h	

B.3 Noon Report Variables

Table B3: Overview of available noon report variables and units.

Source	Value name	Unit	Sampling frequency	Comment
Manual Recording	Slip	%	Daily	
Manual Recording	Aft Draft	m	Daily	
Manual Recording	Fwd Draft	m	Daily	
Manual Recording	Mid Draft	m	Daily	
Manual Recording	Displacement	t	Daily	
Manual Recording	Consumption for Propulsion	MT/day	Daily	Exact timing of measurement cannot be determined since the position, and therefore timezone, of the ship is unknown.
Manual Recording	Fuel	categorical	Daily	
Manual Recording	Air Temp	°C	Daily	
Manual Recording	Bar Pressure	Millibars	Daily	
Manual Recording	Sea Direction	°	Daily	
Manual Recording	Sea State	Douglas scale	Daily	
Manual Recording	Sea Temp	°C	Daily	
Manual Recording	Wind Direction	°	Daily	
Manual Recording	Wind Force	Beaufort	Daily	

Appendix C: Sea Trial Curves

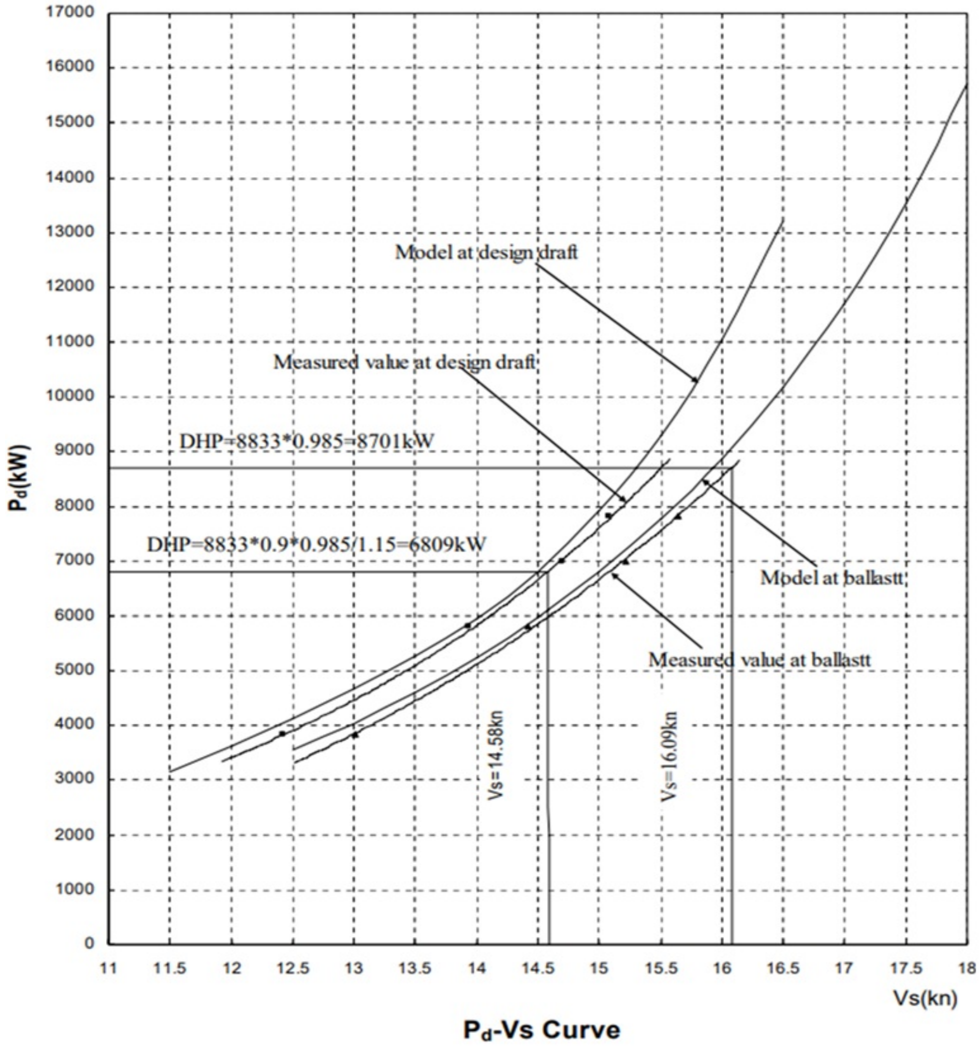


Figure C1: *Sea trial curve showing the relationship between shaft power (P_d) and STW (V_s) under different loading conditions. Source: F. Lehn, Personal Communication, 2026*

Appendix D: Correlation Comparison with Wave Height

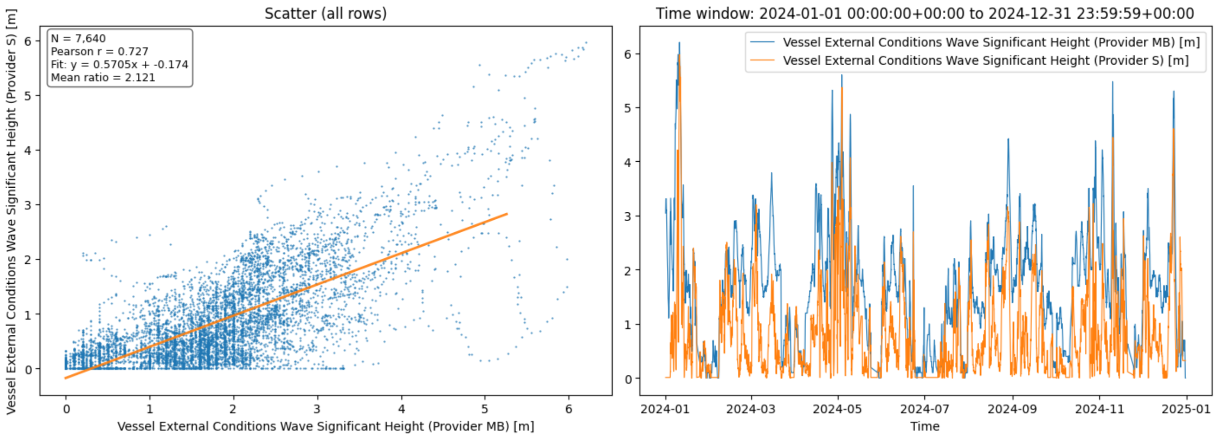


Figure D1: *Correlation between wave height measurements from Provider S and Provider MB.*

Table D1: *Correlation comparison by interval.*

	interval	speed_min	speed_max	n_samples	corr_MB	corr_S	winner
0	[0.53, 9.40]	0.533333	9.400000	507	0.558725	0.300950	Provider MB
1	[9.42, 10.44]	9.420000	10.442857	464	0.410659	0.412214	Provider S
2	[10.46, 10.90]	10.460000	10.900000	559	0.396306	0.355567	Provider MB
3	[10.91, 11.20]	10.914286	11.200000	501	0.366178	0.288116	Provider MB
4	[11.22, 11.60]	11.216667	11.600000	420	0.449933	0.460151	Provider S
5	[11.62, 12.10]	11.620000	12.100000	491	0.726837	0.698113	Provider MB
6	[12.11, 12.50]	12.111111	12.500000	503	0.503916	0.308768	Provider MB
7	[12.51, 12.90]	12.514286	12.900000	494	0.169906	0.022824	Provider MB
8	[12.92, 13.25]	12.916667	13.250000	428	0.062303	-0.133044	Provider S
9	[13.26, 14.20]	13.260000	14.200000	485	0.061968	0.019611	Provider MB
10	OVERALL	NaN	NaN	4852	0.038781	0.015351	Provider MB

To reach the results above, the raw data were filtered for rows with reasonable wave heights (< 15) and positive shaft power, then divided into speed-through-water quantile buckets with at least 100 samples each. Within each speed bucket, the Pearson correlations between each provider's significant wave height and shaft power were computed, and the label " winner " is assigned to the provider whose wave height had the strongest absolute correlation with shaft power in that speed range.

Appendix E: Distribution of Valid Time Segment Lengths

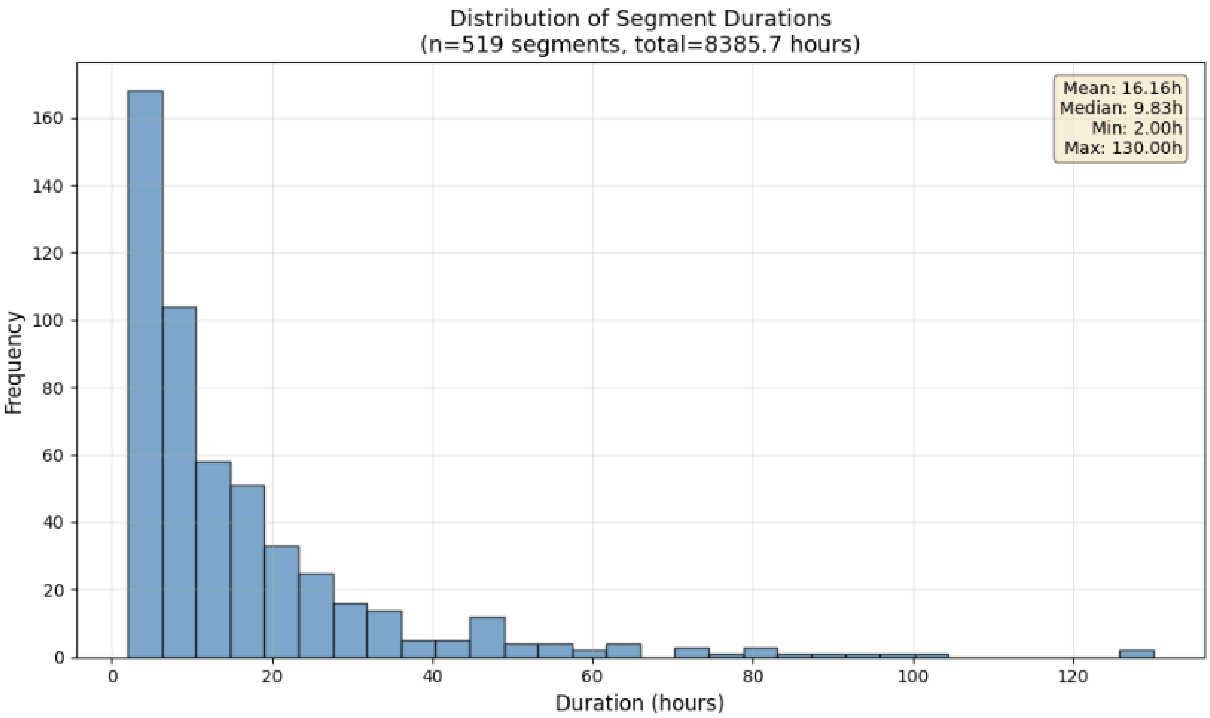


Figure E1: *Distribution of segment lengths after time jump identification.*

Appendix F: Example of Time Alignment of Onboard Sensors

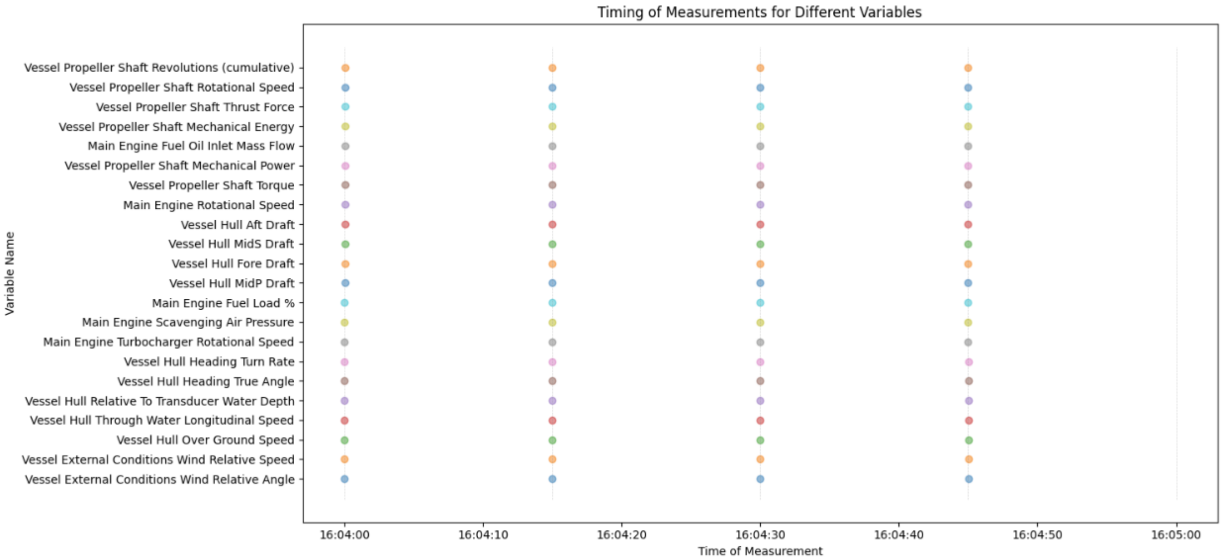


Figure F1: *Example of timing in sensor measurements across a 1-minute window. Note that the observations are almost perfectly aligned to the intended 15-second sampling interval.*

Appendix G: Example of Spike in Shaft Power Measurements

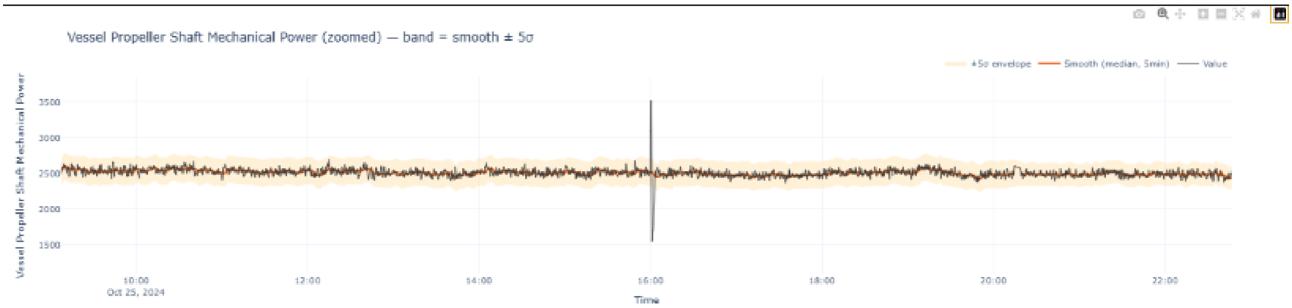


Figure G1: *Example of a spike in shaft power measurements. Orange bands indicate the ±5 MAD threshold.*

Appendix H: Standard Deviation Thresholds as Suggested by DNV

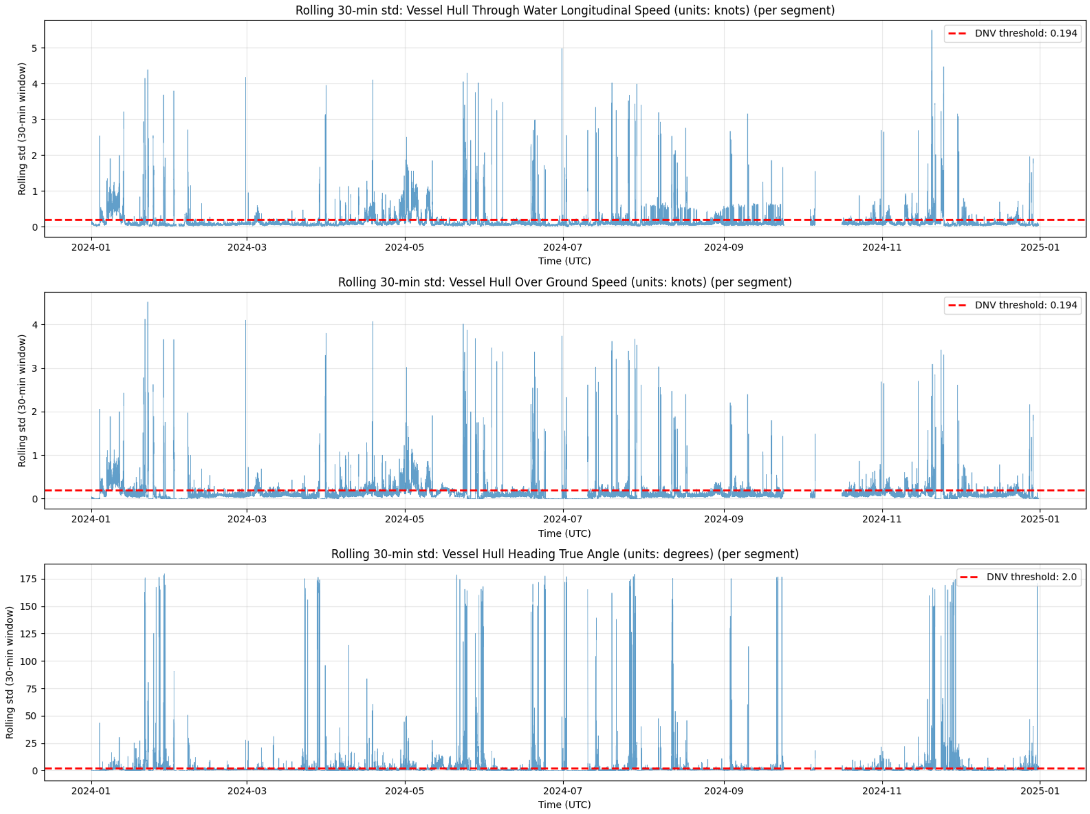
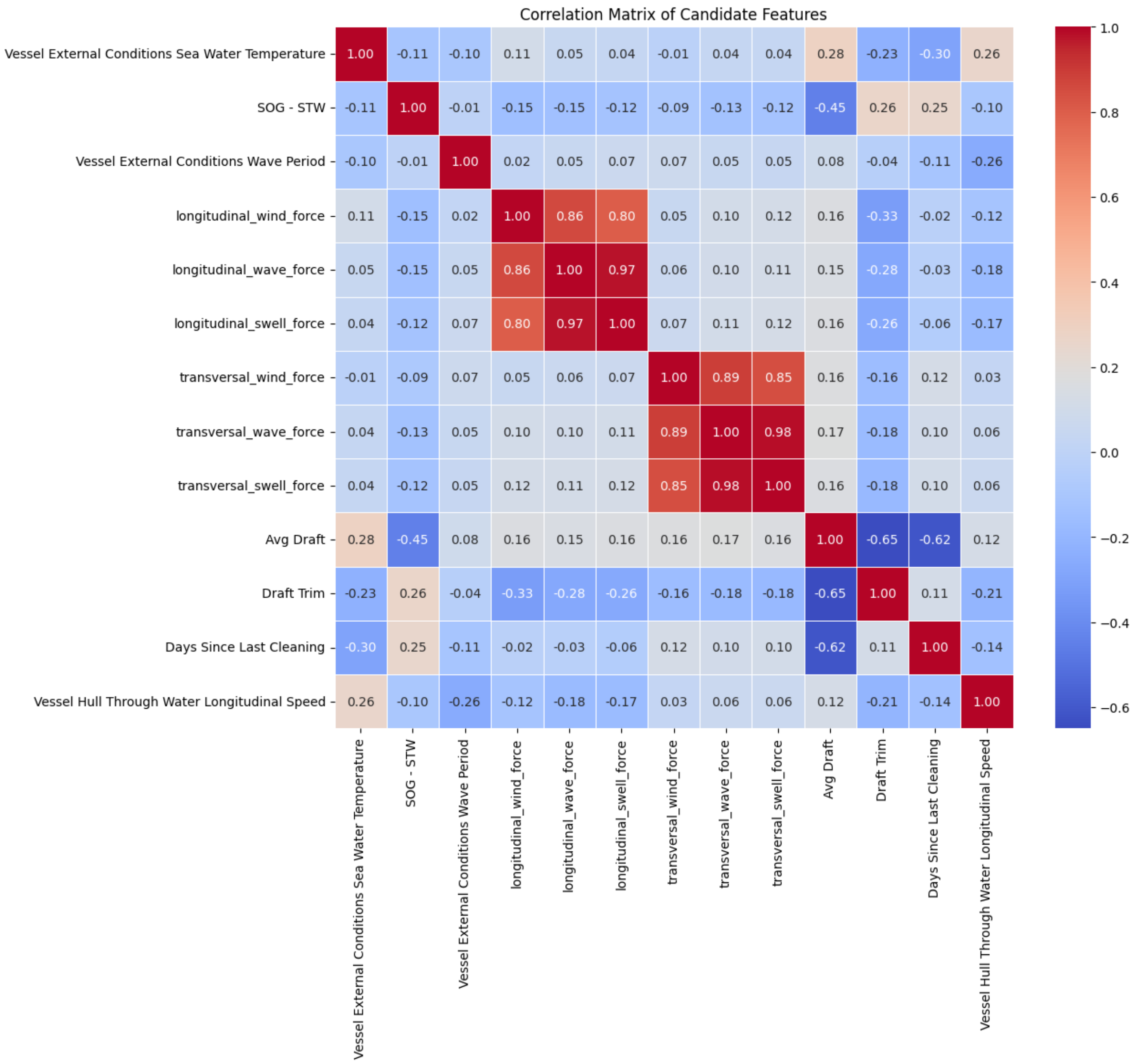


Figure H1: *Line graphs of STW (top), SOG (middle), and Vessel heading (bottom) on the raw data. Red lines represent the rolling standard deviation thresholds as suggested by DNV.*

Appendix I: Correlation Matrix



	Vessel External Conditions Sea Water Temperature	SOG - STW	Vessel External Conditions Wave Period	longitudinal_wind_force	longitudinal_wave_force	longitudinal_swell_force	transversal_wind_force	transversal_wave_force	transversal_swell_force	Avg Draft	Draft Trim	Days Since Last Cleaning	Vessel Hull Through Water Longitudinal Speed
Vessel External Conditions Sea Water Temperature	1.00	-0.11	-0.10	0.11	0.05	0.04	-0.01	0.04	0.04	0.28	-0.23	-0.30	0.26
SOG - STW	-0.11	1.00	-0.01	-0.15	-0.15	-0.12	-0.09	-0.13	-0.12	-0.45	0.26	0.25	-0.10
Vessel External Conditions Wave Period	-0.10	-0.01	1.00	0.02	0.05	0.07	0.07	0.05	0.05	0.08	-0.04	-0.11	-0.26
longitudinal_wind_force	0.11	-0.15	0.02	1.00	0.86	0.80	0.05	0.10	0.12	0.16	-0.33	-0.02	-0.12
longitudinal_wave_force	0.05	-0.15	0.05	0.86	1.00	0.97	0.06	0.10	0.11	0.15	-0.28	-0.03	-0.18
longitudinal_swell_force	0.04	-0.12	0.07	0.80	0.97	1.00	0.07	0.11	0.12	0.16	-0.26	-0.06	-0.17
transversal_wind_force	-0.01	-0.09	0.07	0.05	0.06	0.07	1.00	0.89	0.85	0.16	-0.16	0.12	0.03
transversal_wave_force	0.04	-0.13	0.05	0.10	0.10	0.11	0.89	1.00	0.98	0.17	-0.18	0.10	0.06
transversal_swell_force	0.04	-0.12	0.05	0.12	0.11	0.12	0.85	0.98	1.00	0.16	-0.18	0.10	0.06
Avg Draft	0.28	-0.45	0.08	0.16	0.15	0.16	0.16	0.17	0.16	1.00	-0.65	-0.62	0.12
Draft Trim	-0.23	0.26	-0.04	-0.33	-0.28	-0.26	-0.16	-0.18	-0.18	-0.65	1.00	0.11	-0.21
Days Since Last Cleaning	-0.30	0.25	-0.11	-0.02	-0.03	-0.06	0.12	0.10	0.10	-0.62	0.11	1.00	-0.14
Vessel Hull Through Water Longitudinal Speed	0.26	-0.10	-0.26	-0.12	-0.18	-0.17	0.03	0.06	0.06	0.12	-0.21	-0.14	1.00

Figure I1: *Correlation matrix of the features. The figure displays pairwise Pearson correlations between the features and is included to supplement the VIF assessment.*

Appendix J: Feature and Power Histograms

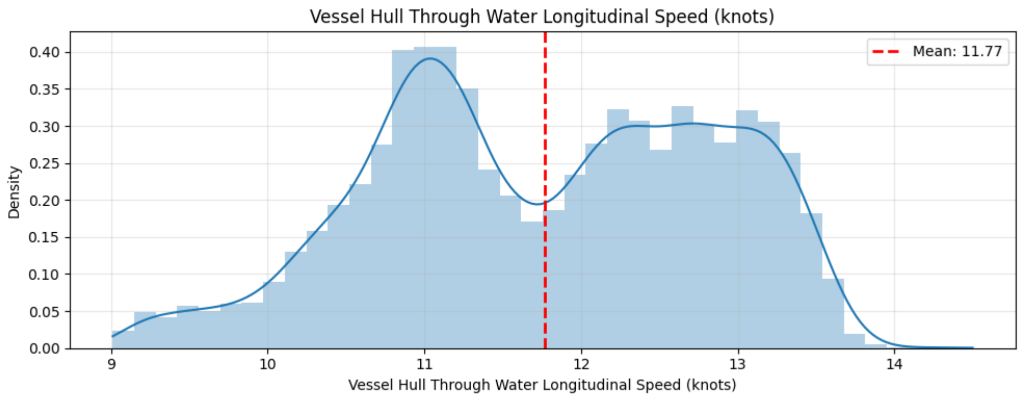


Figure J1: *Distribution of Speed Through Water (kn)* $(n = 15{,}061)$

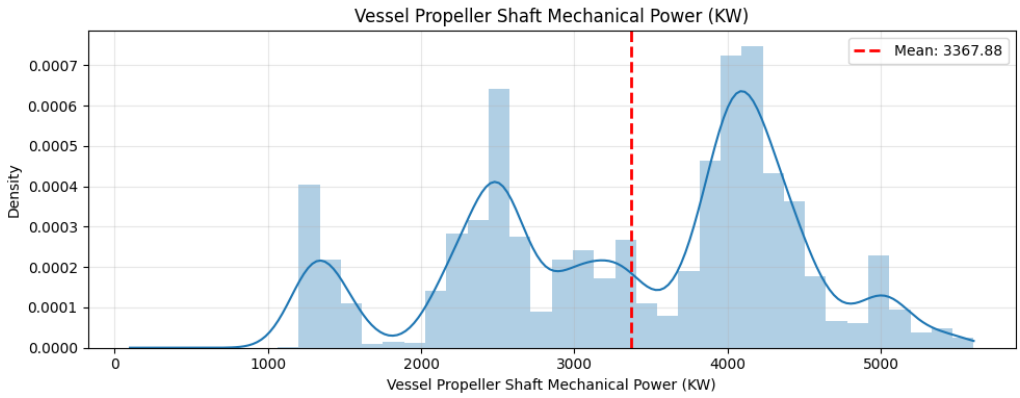


Figure J2: *Distribution of Shaft Power (kW)* $(n = 15{,}061)$.

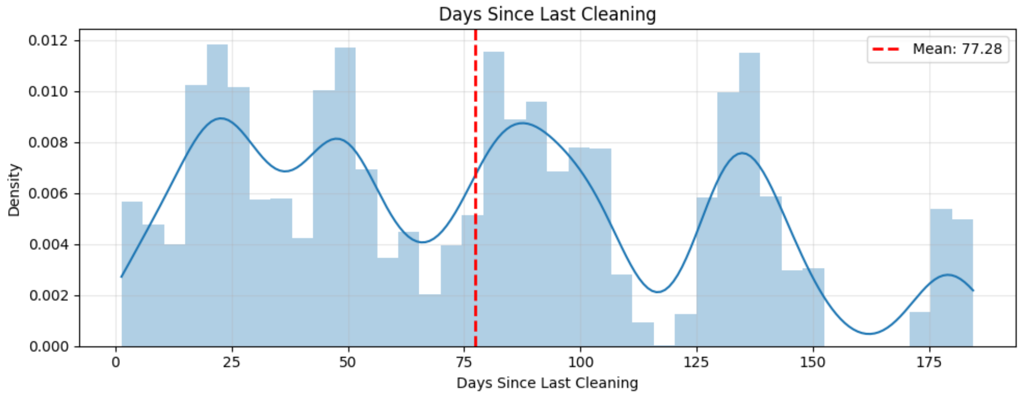


Figure J3: *Distribution of DSC ($n = 15{,}061$). The lower count reflects the exclusion of observations prior to the first recorded cleaning event.*

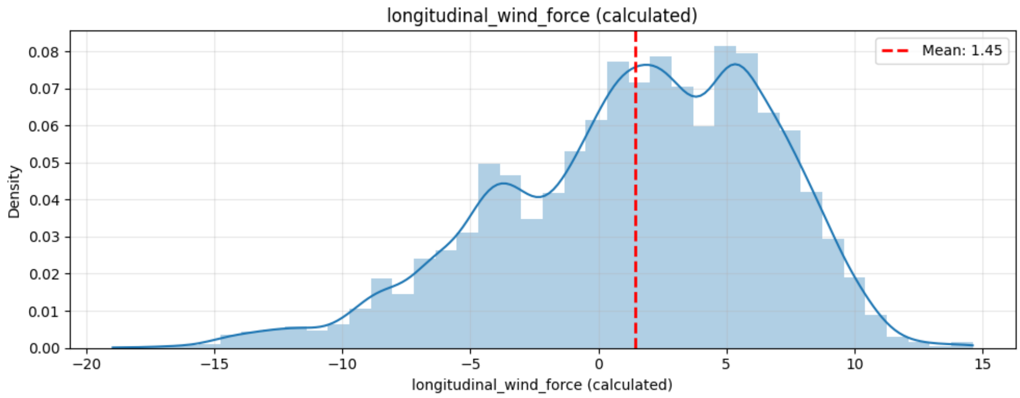


Figure J4: *Distribution of longitudinal wind force (m/s) ($n = 15{,}061$). Positive values indicate head conditions opposing forward motion.*

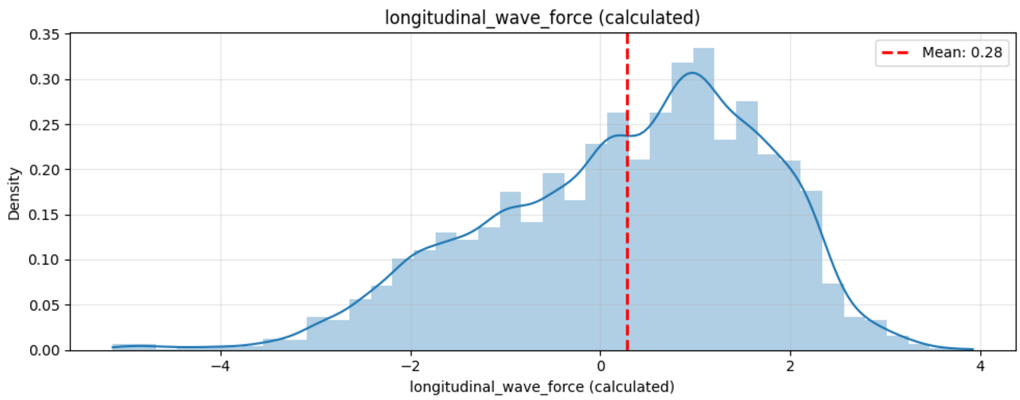


Figure J5: *Distribution of longitudinal wave force (m of wave height) ($n = 15{,}061$). Positive values indicate head conditions opposing forward motion.*

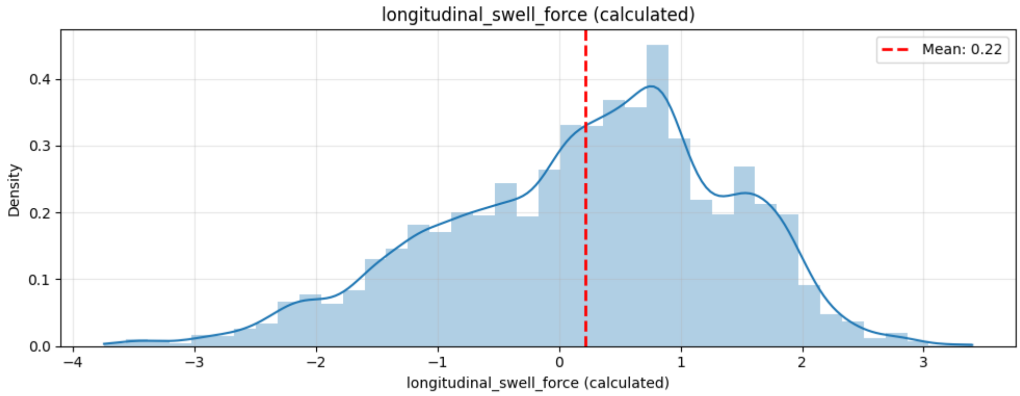


Figure J6: *Distribution of longitudinal swell force (m of swell height) (n = 15,061). Positive values indicate head conditions opposing forward motion.*

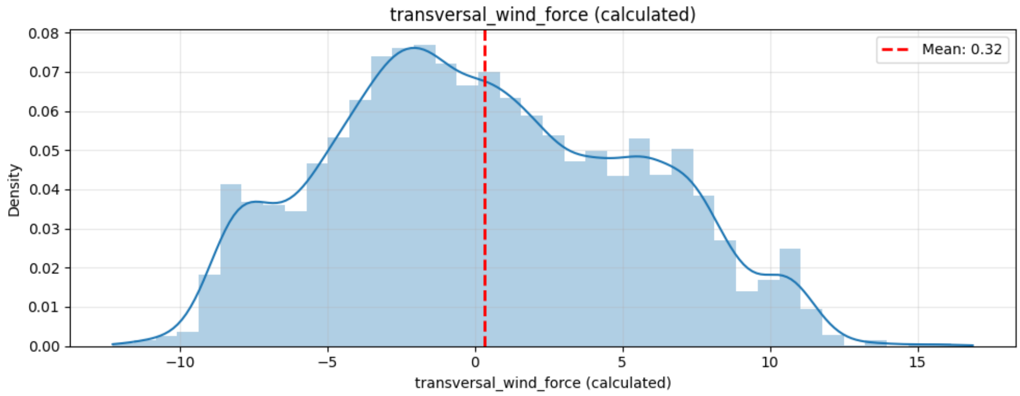


Figure J7: *Distribution of transversal wind force (m/s) (n = 15,061). Positive values indicate force from starboard, negative from port.*

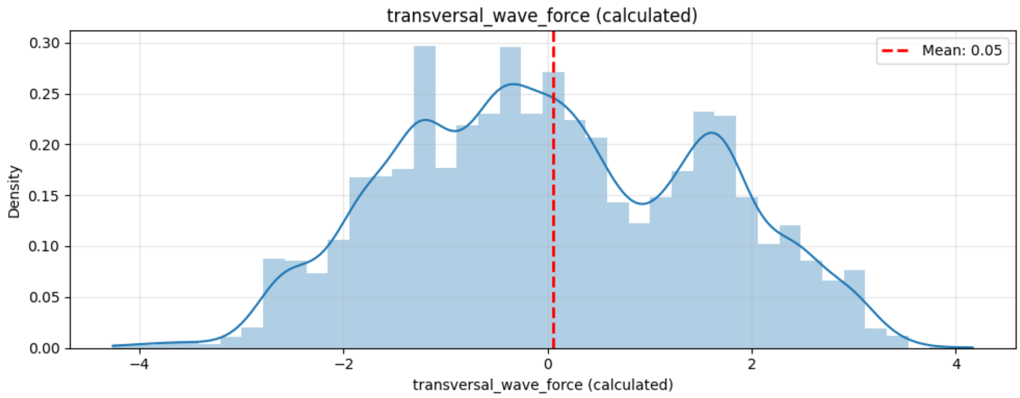


Figure J8: *Distribution of transversal wave force (m of wave height) (n = 15,061). Positive values indicate force from starboard, negative from port.*

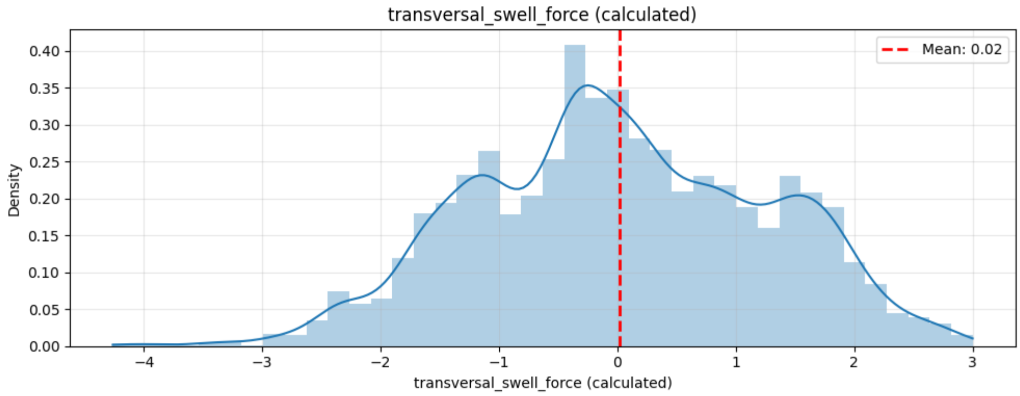


Figure J9: *Distribution of transversal swell force (m of wave height) ($n = 15{,}061$). Positive values indicate force from starboard, negative from port.*

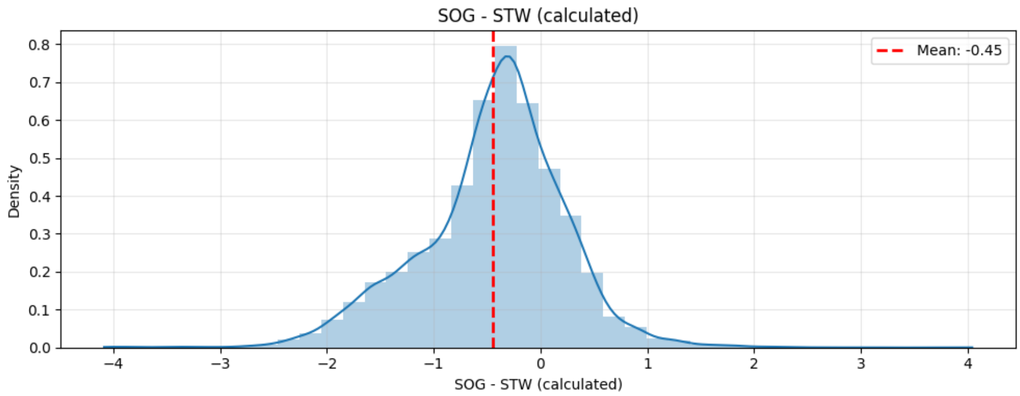


Figure J10: *Distribution of $SOG - STW$ (kn) ($n = 15{,}061$). Negative values indicate a current opposing the vessel's heading.*

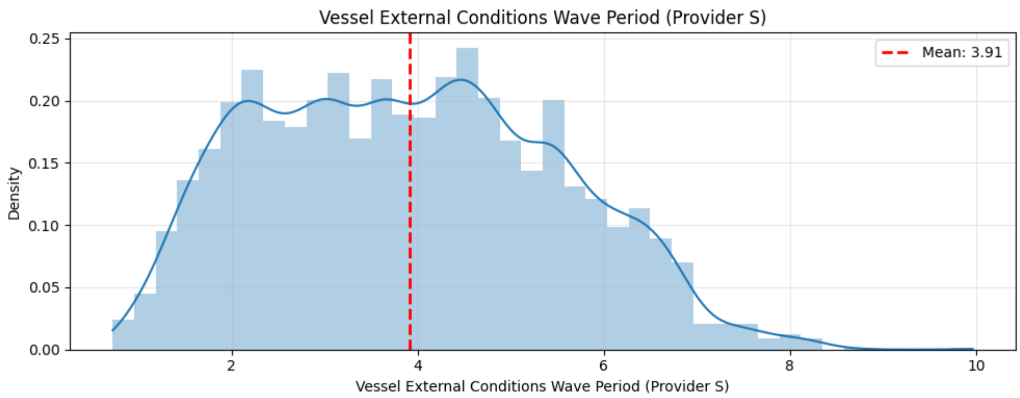
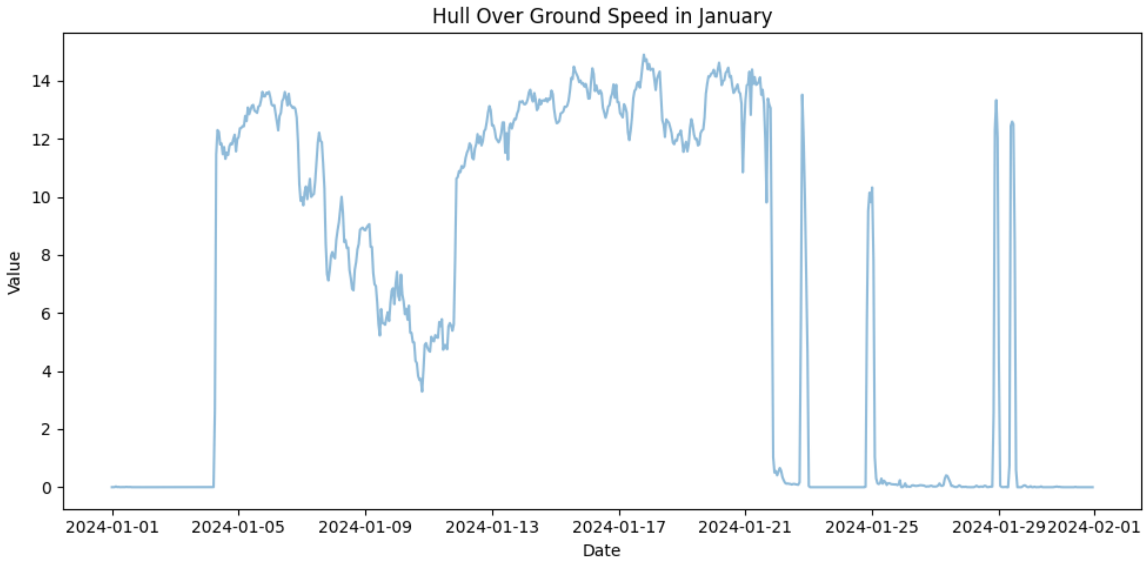
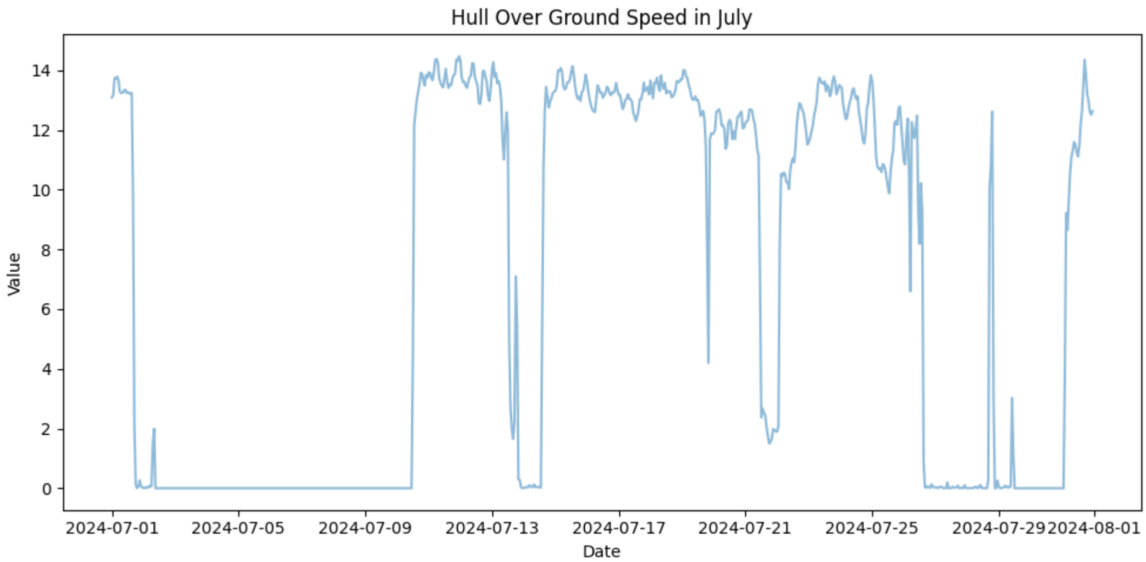


Figure J11: *Distribution of wave period (s) after preprocessing and filtering to open-water sailing conditions ($n = 15{,}061$).*

Appendix K: Speed Over Ground Time Series for January and July



(a) *SOG series for January before filtering for open water observations.*



(b) *SOG series for July before filtering for open water observations.*

Figure K1: *Speed over ground around the two inferred cleaning events. The assumed cleaning dates are the final days of the two port stays (Jan 24 and Jul 30).*

Appendix L: Common Activation Functions

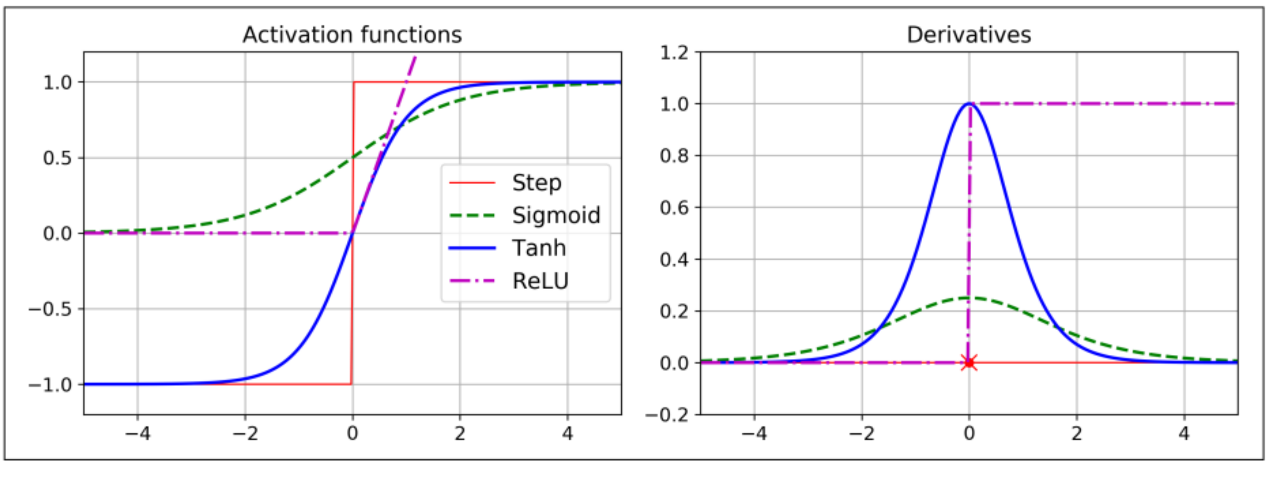


Figure L1: *Four common activation functions applied in neural networks. Source (Géron, 2019, Section 10)*

Appendix M: Comparison of ALE Curves with Partial Dependence and Marginal Curves

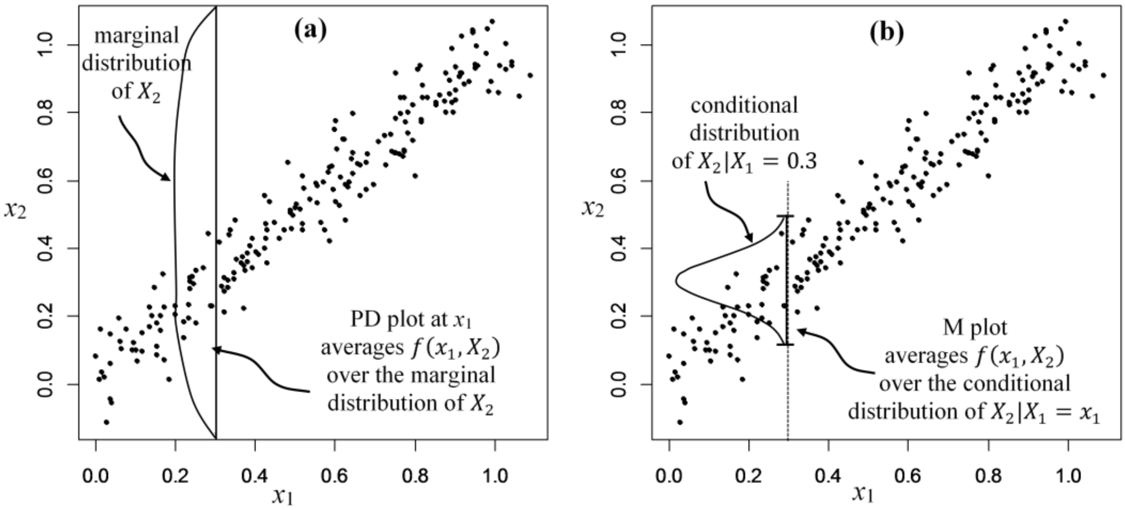


Figure M1: *Illustration of the difference between PD and marginal (M) plots for correlated predictors, adapted from Apley and Zhu (2020). In panel (a), the PD plot at a fixed value of x_1 averages predictions over the marginal distribution of X_2, which can require evaluation in regions with little or no observed data and therefore lead to extrapolation. In panel (b), the marginal plot instead averages over the conditional distribution of $X_2 \mid X_1 = x_1$, which avoids this extrapolation but may confound the effect of x_1 with the effect of correlated predictors.*

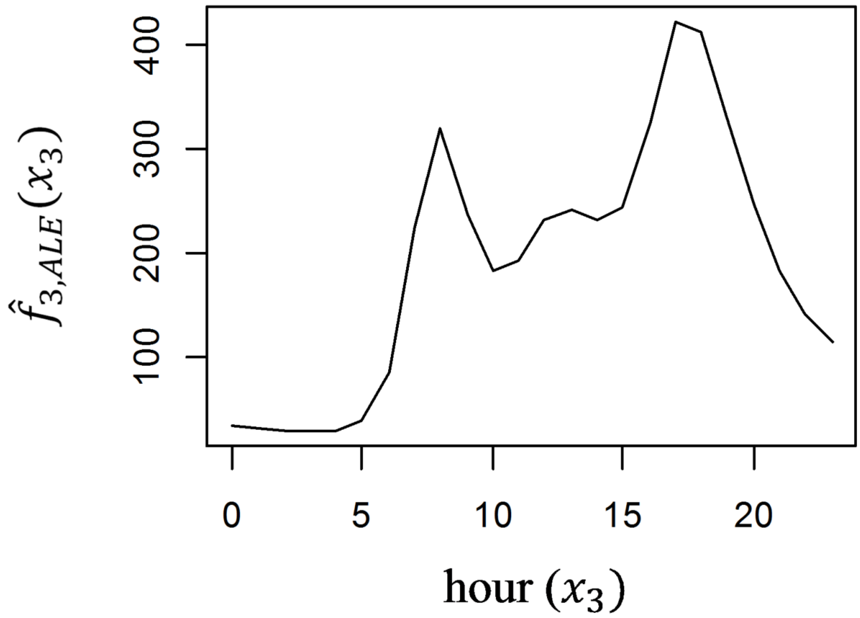


Figure M2: *Comparison of ALE, partial dependence, and marginal plots for the tree model fitted to the synthetic example of Apley and Zhu (2020). Panel (a) shows the estimated effect of x_1, whose true effect is linear, and panel (b) shows the estimated effect of x_2, whose true effect is quadratic. In both panels, the ALE estimate (blue line with markers) tracks the true effect (black solid line) more closely than the partial dependence plot (red dotted line) and the marginal plot (black dashed line). Source: Apley and Zhu (2020).*

Appendix N: Value Counts of Observed Loading Conditions Adjacent to Standard Conditions

Table N1: *Number of observations within a pm 1m draft and trim of each standard loading condition. Note that n=0 for heavy ballast, since the vessel did not operate at any loading conditions similar to it. S*

Scenario	Target Draft	Target Trim	Points in ±1m window	Avg Draft (Calculated)	Draft Trim (Calculated)	Count
Ballast	6.40	2.8	5536	6.10	2.60	**2132**
Ballast	6.40	2.8	5536	6.42	2.24	1419
Ballast	6.40	2.8	5536	6.14	2.29	1006
Ballast	6.40	2.8	5536	6.28	1.84	969
Ballast	6.40	2.8	5536	6.42	3.64	9
Design	12.20	0.0	3848	11.62	0.81	**2718**
Design	12.20	0.0	3848	11.84	0.00	666
Design	12.20	0.0	3848	11.47	0.32	445
Design	12.20	0.0	3848	11.50	0.09	19
Scantling	14.25	0.0	2011	14.20	1.00	**1078**
Scantling	14.25	0.0	2011	14.25	0.70	822
Scantling	14.25	0.0	2011	14.28	0.01	82
Scantling	14.25	0.0	2011	14.22	0.99	29
Heavy Ballast	6.90	5.0	0	–	–	0

Appendix O: MLP Architecture Without DSC

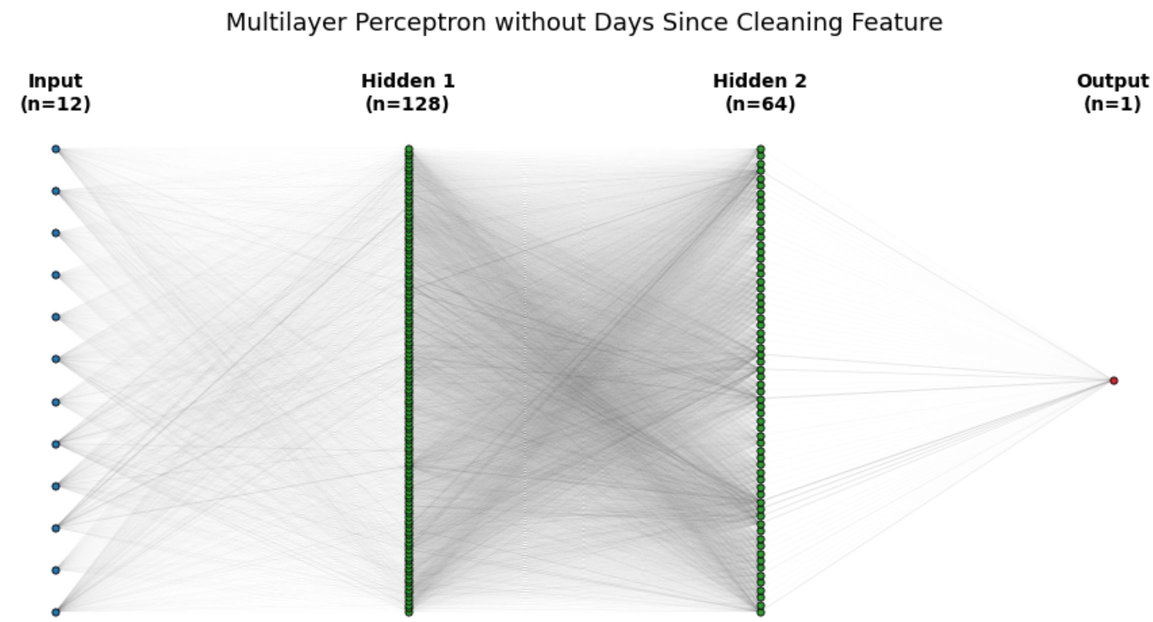
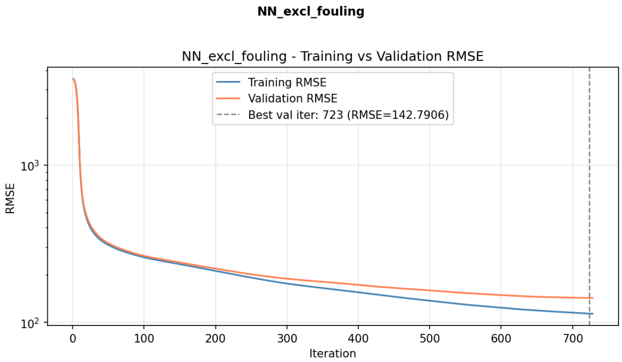
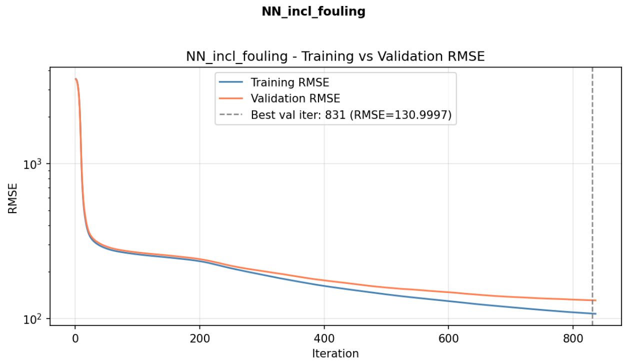


Figure O1: *Architecture of the Multilayer Perceptron excluding DSC. Each blue dot represents an input variable, and each green dot represents a neuron. The thickness of lines indicates the absolute value of weights.*

Appendix P: MLP Training Curves



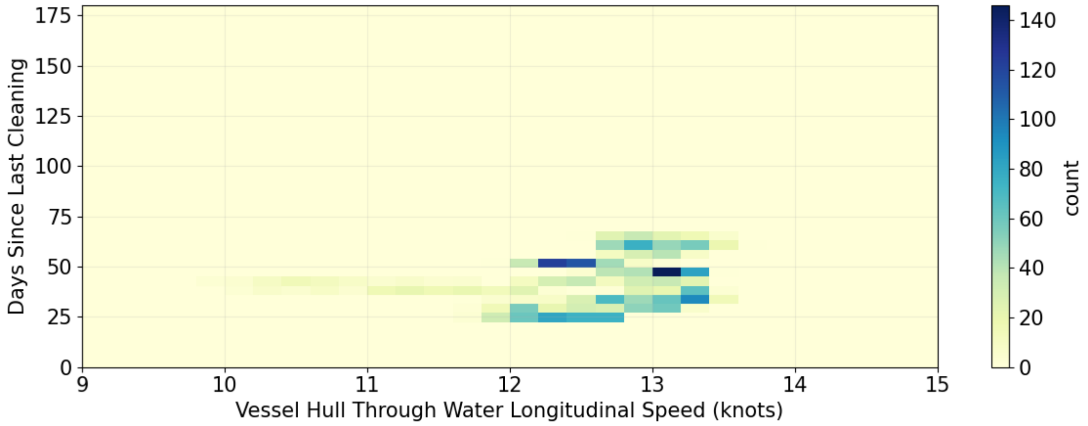
(a) *MLP excluding DSC. Best validation RMSE at iteration 723 (RMSE = 142.79).*



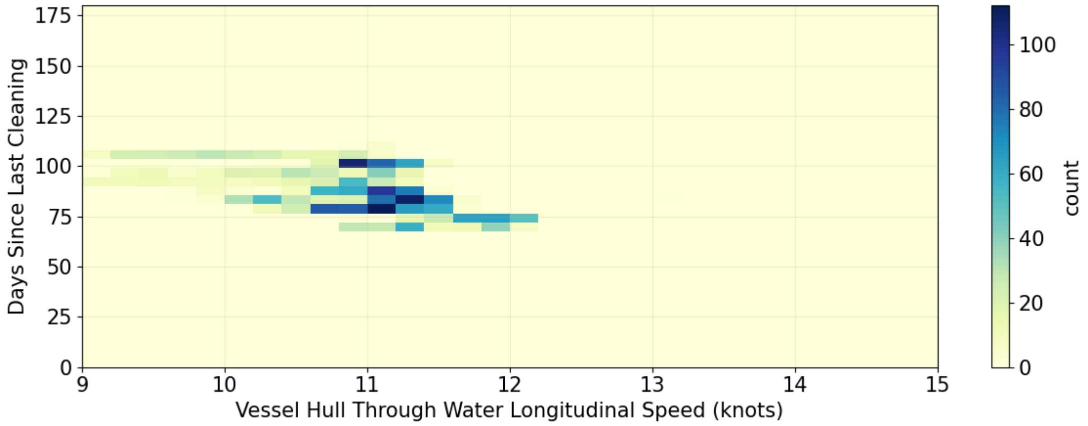
(b) *MLP including fouling features. Best validation RMSE at iteration 831 (RMSE = 131.00).*

Figure P1: *Training and validation RMSE curves for MLPs with and without fouling-related input features. Note that the number of epochs differs despite being trained with the same hyperparameters. This is due to the different feature spaces.*

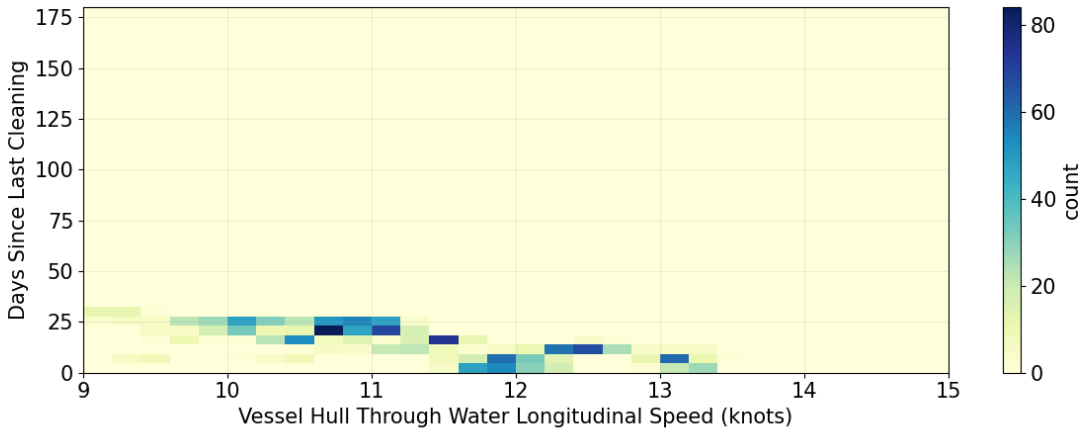
Appendix Q: Data Support Heatmaps by Speed, DSC, and Draft



(a) Design draft



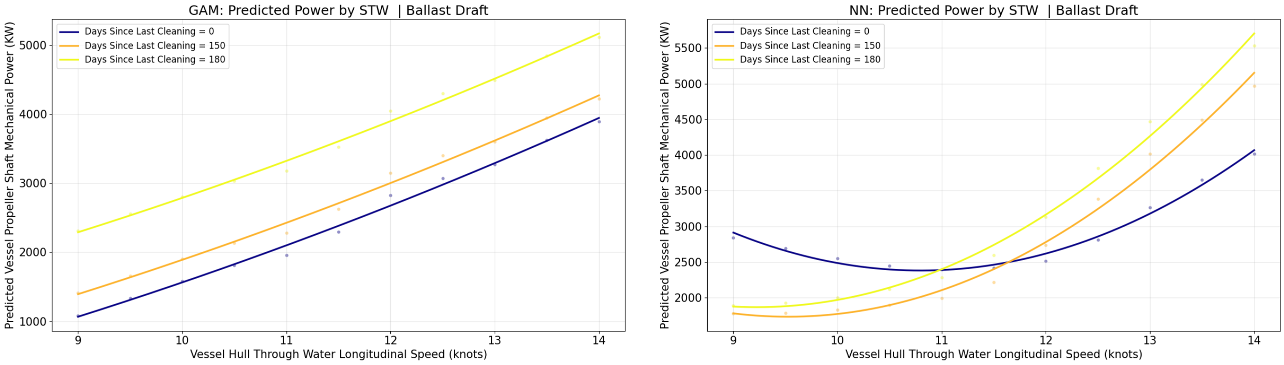
(b) Ballast draft



(c) Scantling draft

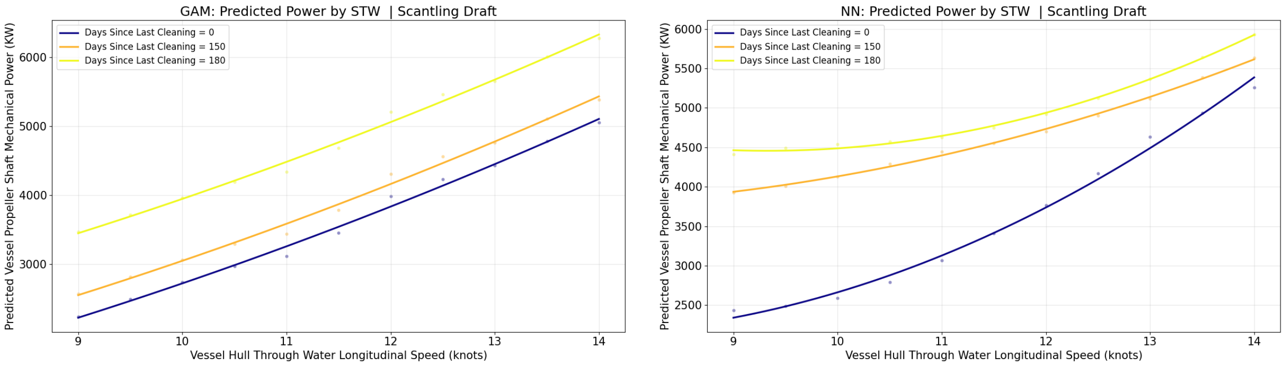
Figure Q1: *Data support heatmaps (observational coverage) for different loading conditions. Based on data points within* $\pm 1\,\mathrm{m}$ *avg draft,* $\pm 1\,\mathrm{m}$ *trim, and* ± 10 *DSC.*

Appendix R: Estimated Speed-power Curves for Ballast & Scantling Drafts



(a) *GAM* ***(b)*** *MLP*

Figure R1: *Speed–power curves at ballast draft for different fouling levels.*



(a) *GAM* ***(b)*** *MLP*

Figure R2: *Speed–power curves at scantling draft for different fouling levels.*

Appendix S: Feature and Residual Scatterplots

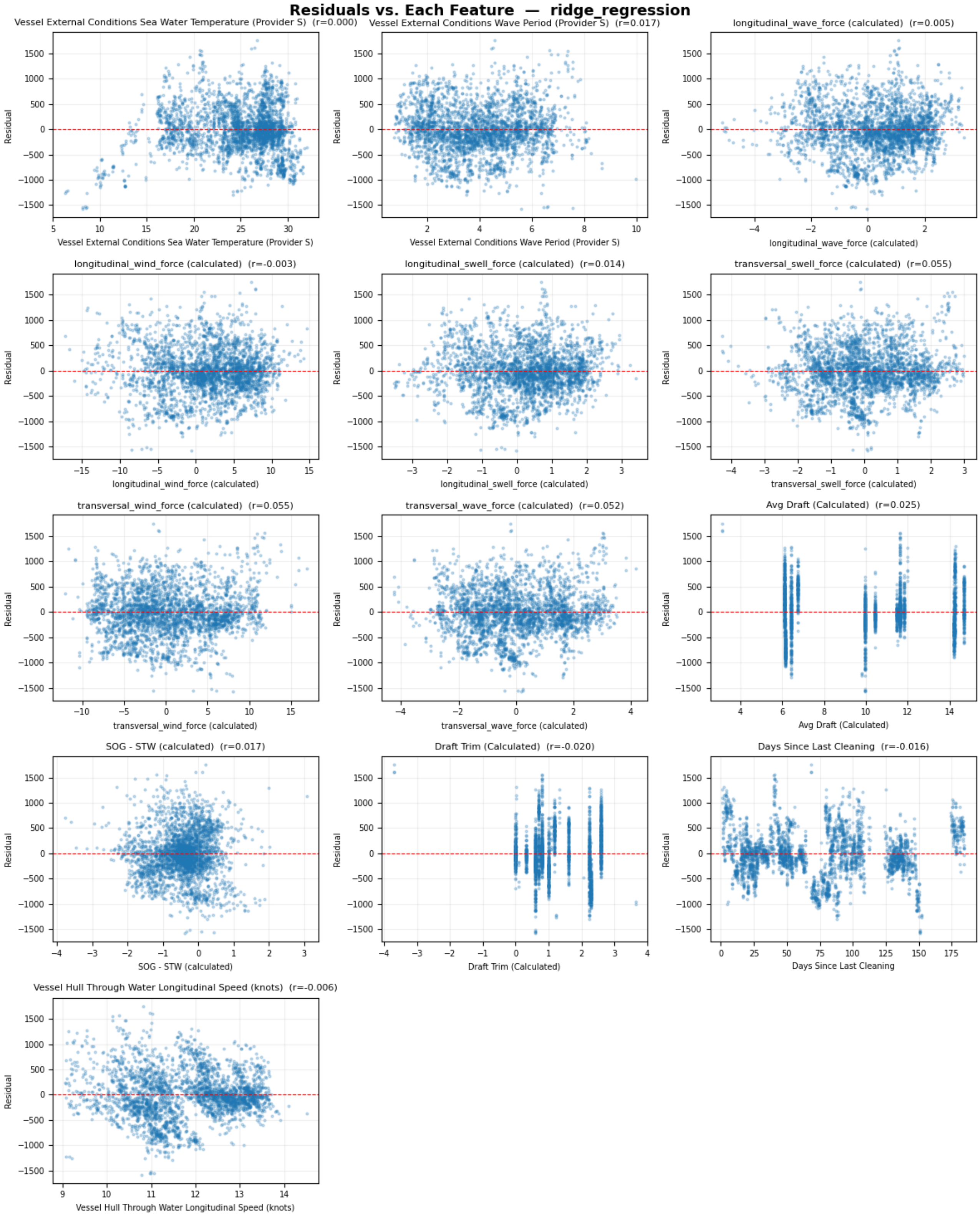


Figure S1: *Residuals vs. features for the Ridge model.*

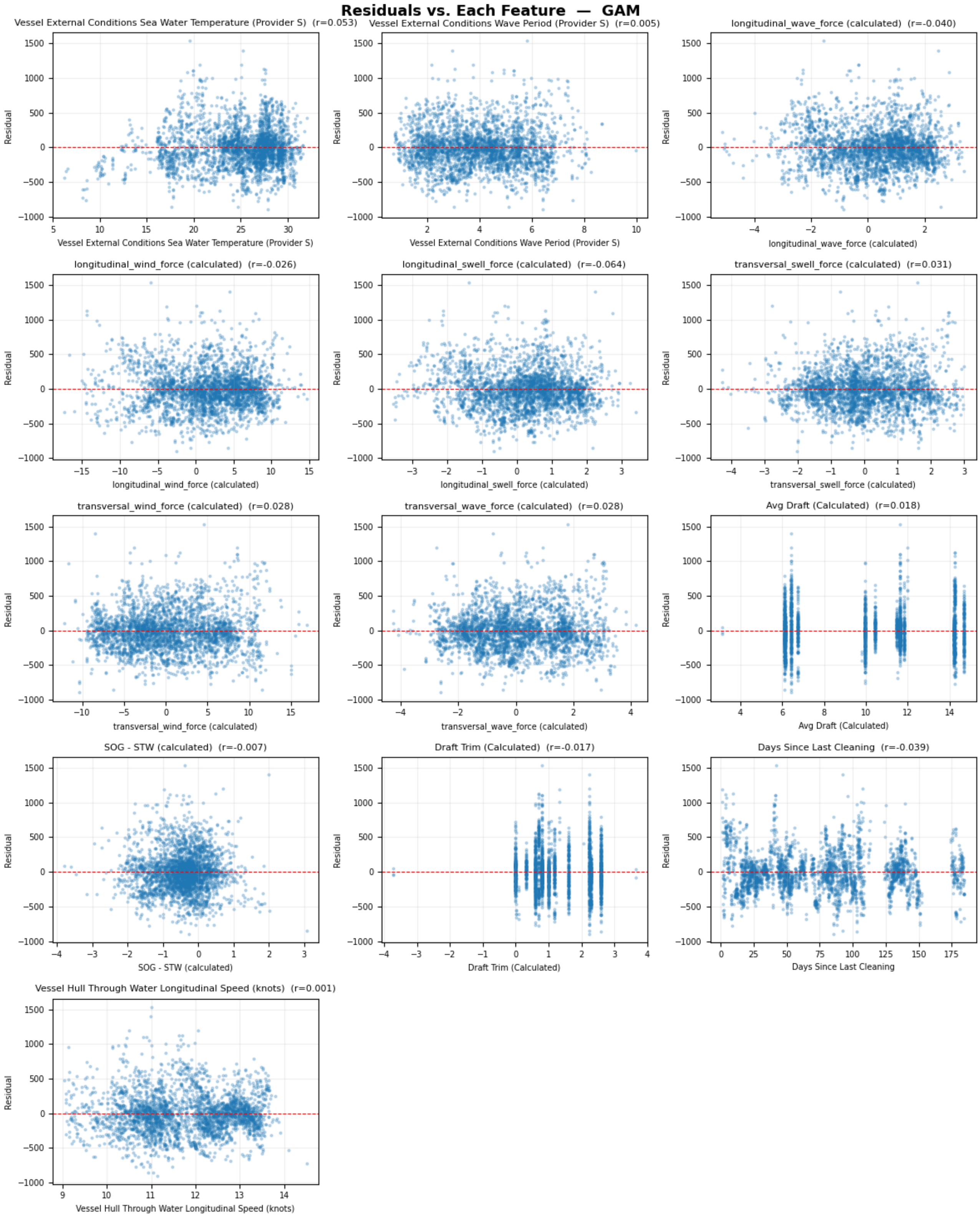


Figure S2: *Residuals vs. features for the GAM model.*

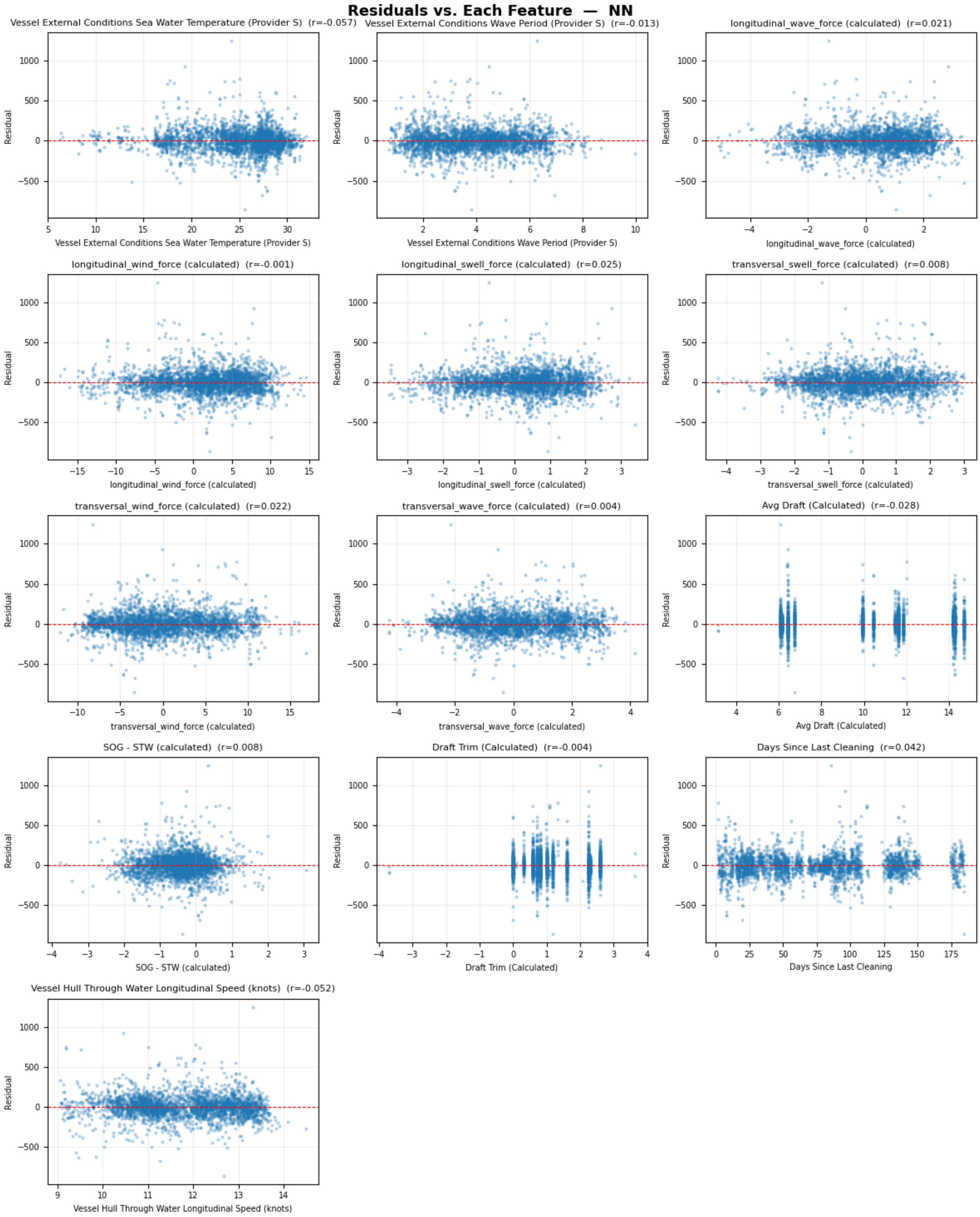


Figure S3: *Residuals vs. features for the MLP model.*

Appendix T: Feature and Residual Correlations

Table T1: *Correlation (r) between feature values and model residuals for each model*

Feature	Ridge Regression	GAM	NN
Sea Water Temperature	0.000	0.053	0.057
Wave Period	-0.017	-0.005	-0.013
Longitudinal Wave Force	0.005	-0.040	-0.021
Longitudinal Wind Force	-0.003	-0.026	-0.001
Longitudinal Swell Force	0.014	-0.064	0.025
Transversal Swell Force	0.055	0.031	0.008
Transversal Wind Force	0.055	0.028	0.022
Transversal Wave Force	0.052	0.028	0.004
Avg Draft	0.025	0.018	-0.028
SOG - STW	0.017	-0.007	0.008
Trim	-0.020	-0.017	-0.004
Days Since Last Cleaning	-0.016	-0.039	-0.042
STW	-0.006	0.001	-0.052

Appendix U: QQ Plots

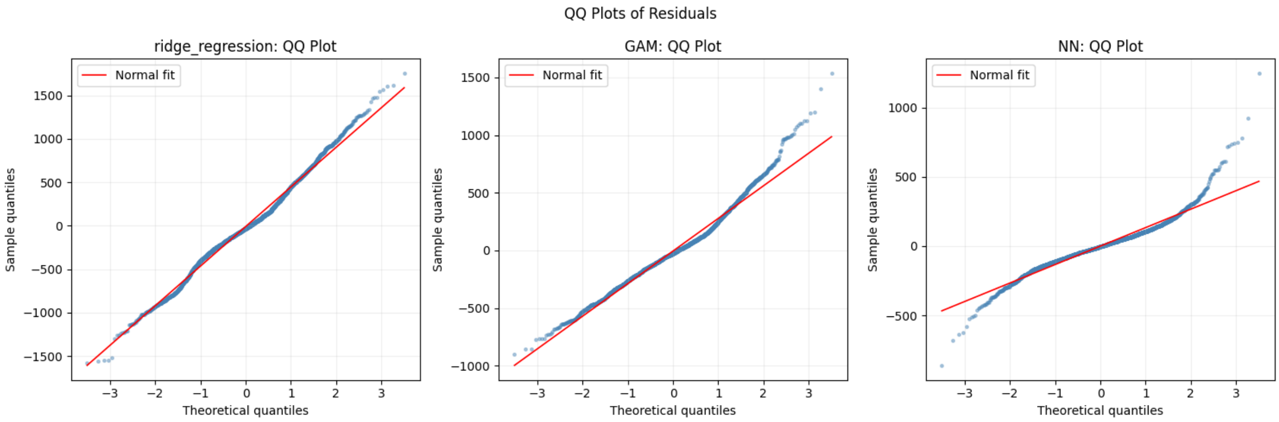


Figure U1: *QQ plots of residuals for the evaluated models.*

Appendix V: DSC Over Time

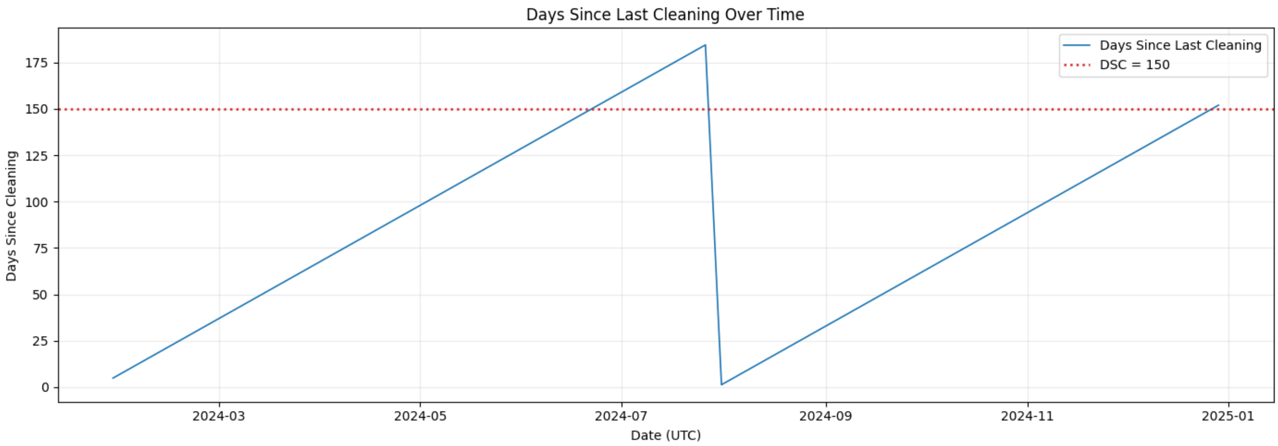


Figure V1: *DSC over time.*

Appendix W: Added Fuel Cost Methodology

For each observation i, the exact DSC value is extracted from the aggregated dataset. Since the estimated fouling effect curves were defined only at discrete intervals, linear interpolation is used to obtain a continuous estimate of the added shaft power demand, ΔP_i, at the exact cleaning age of each observation, separately for the PE- and ALE-based curves.

The interval-level fuel penalty is then calculated as

$$Fuel_i = \Delta P_i \times \Delta t \times SFOC_i$$

where Δt is the observation length in hours and $SFOC_i$ is the assumed specific fuel oil consumption in tons per kWh. For each 15-minute observation i, $SFOC_i$ is estimated directly from the aggregated fuel mass flow and shaft power as

$$SFOC_i = \frac{\dot{m}_{\text{fuel},i}}{P_{\text{shaft},i}}$$

where $\dot{m}_{\text{fuel},i}$ is the main engine fuel oil inlet mass flow in kg/h and $P_{\text{shaft},i}$ is the shaft power in kW. Because the calculation is applied row by row, unlogged periods such as downtime and port stays were naturally excluded.

The corresponding interval-level cost is then obtained as

$$Cost_i = Fuel_i \times P_{\text{fuel}}$$

where P_{fuel} denotes the assumed fuel price in USD per ton.

Finally, the cumulative added fuel cost curve is constructed as the running sum of the interval-level costs:

$$\text{Cumulative Cost}_n = \sum_{i=1}^{n} Cost_i$$

This yields the total estimated added fuel cost up to and including observation n.

Appendix X: Overall ALE Curve for Random Forest Model

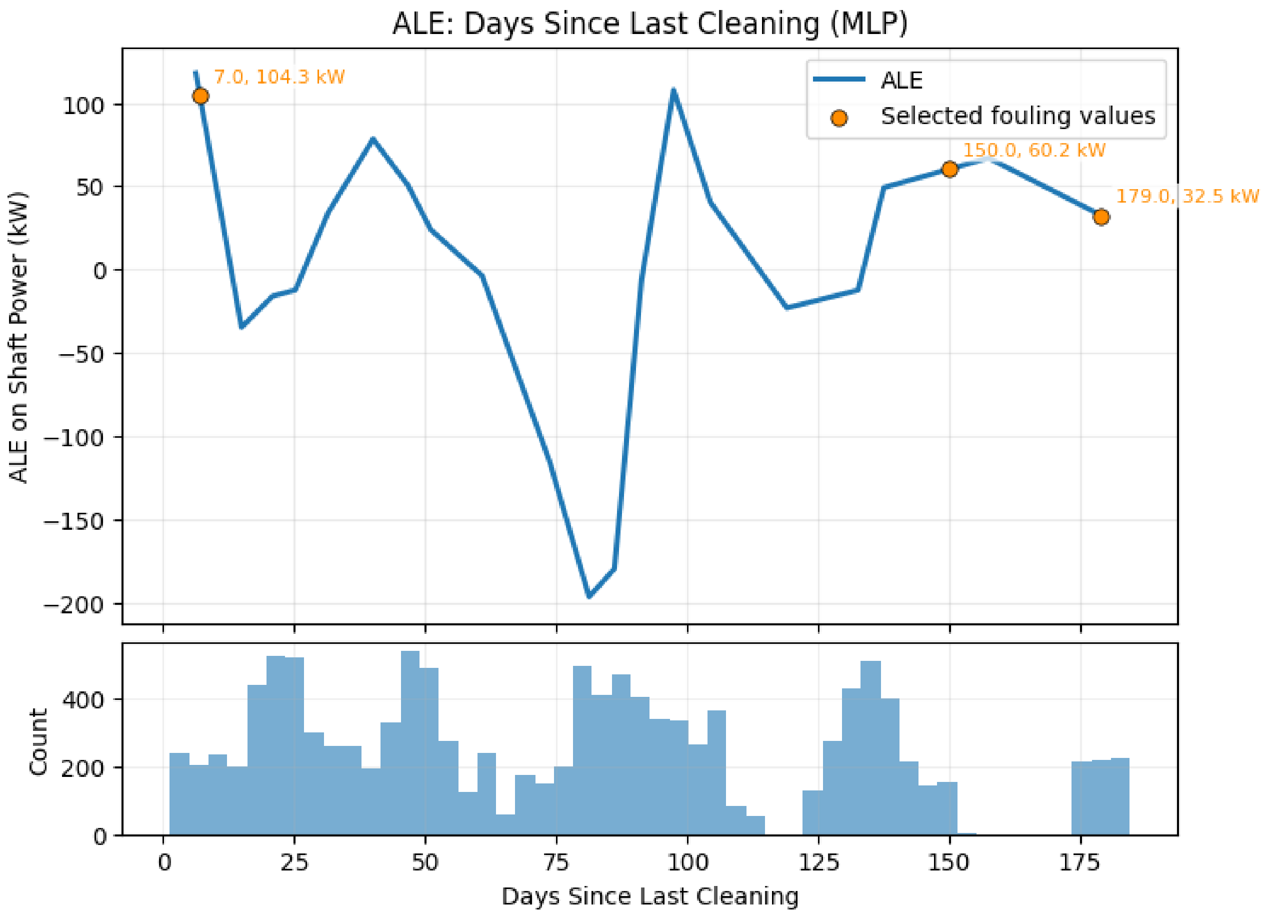


Figure X1: *Overall ALE curve for the random forest model.*

Model hyperparameters: n_estimators=300, max_depth=8, min_samples_split=20, min_samples_leaf=20, max_features=0.85.

Appendix Y: Speed-power Curves for Random Forest Model

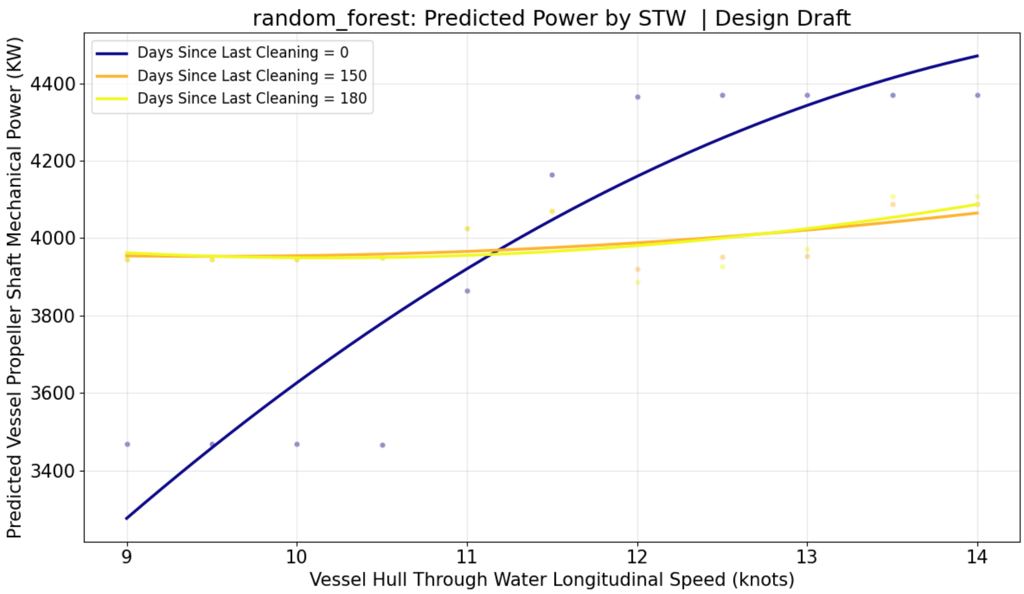


Figure Y1: *Random forest speed-power curve (design loading condition).*

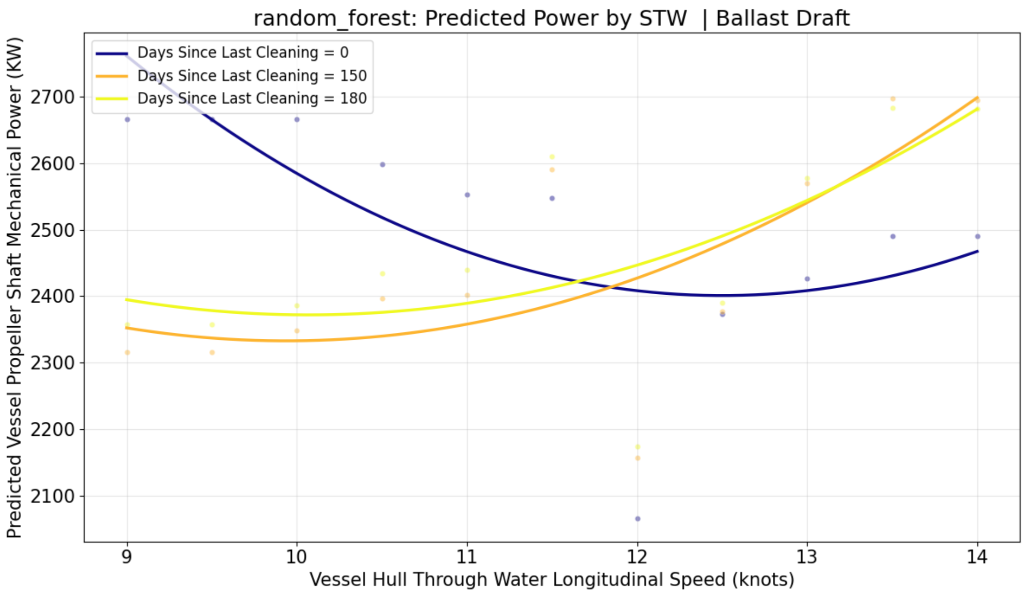


Figure Y2: *Random forest speed-power curve (ballast loading condition).*

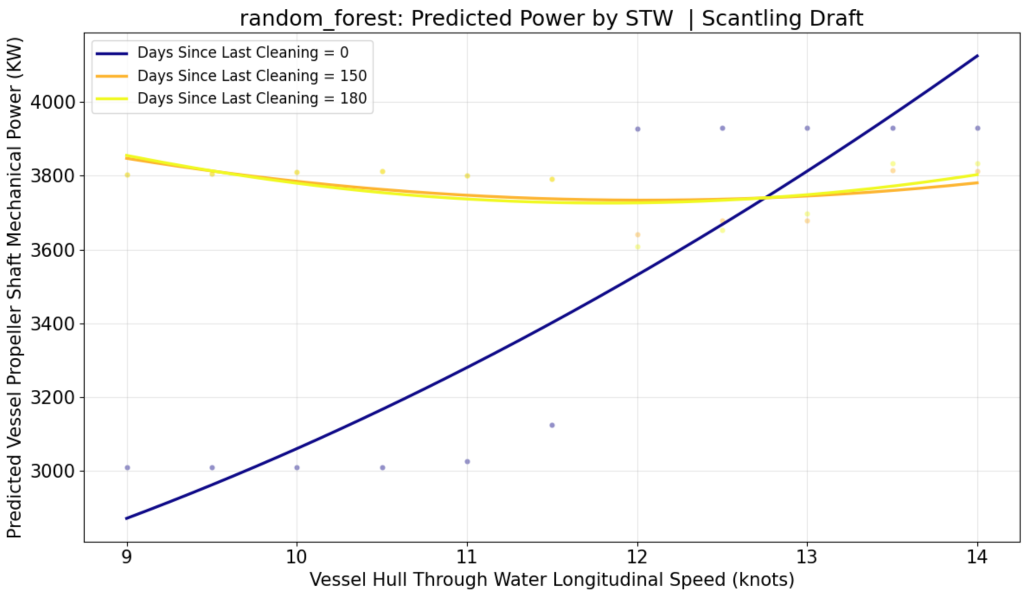


Figure Y3: *Random forest speed-power curve (scantling loading condition).*

Model hyperparameters: n_estimators=300, max_depth=8, min_samples_split=20, min_samples_leaf=20, max_features=0.85.